

# EC22 Lecture 2: Macro Variables

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# 1 Slides

# Economics 22 - Macroeconomic Variables

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## Macroeconomic Data

- GDP
  - Definition
  - Components
  - Real vs. nominal
- Prices
  - GDP Deflator
  - CPI
- Unemployment

## Basic US Economic Numbers

- GDP = \$29.2 Trillion (2024, annual, nominal)
- US Population = 343 million (January 2026, est.)
- World Population = 8.2 billion (January 2026, est.)
- Unemployment Rate = 4.6 (Nov 2025)
- Inflation = 2.7 (CPI-all items, Nov 2024 - Nov 2025)
- Core inflation = 2.6 (CPI-all items less food and energy, Nov 2024 - Nov 2025)
  - Sources: <http://www.census.gov/popclock/>, <http://www.bea.gov/>,  
<http://www.bls.gov/>

## Why it is good to know GDP



**Alexandria Ocasio-Cortez**  

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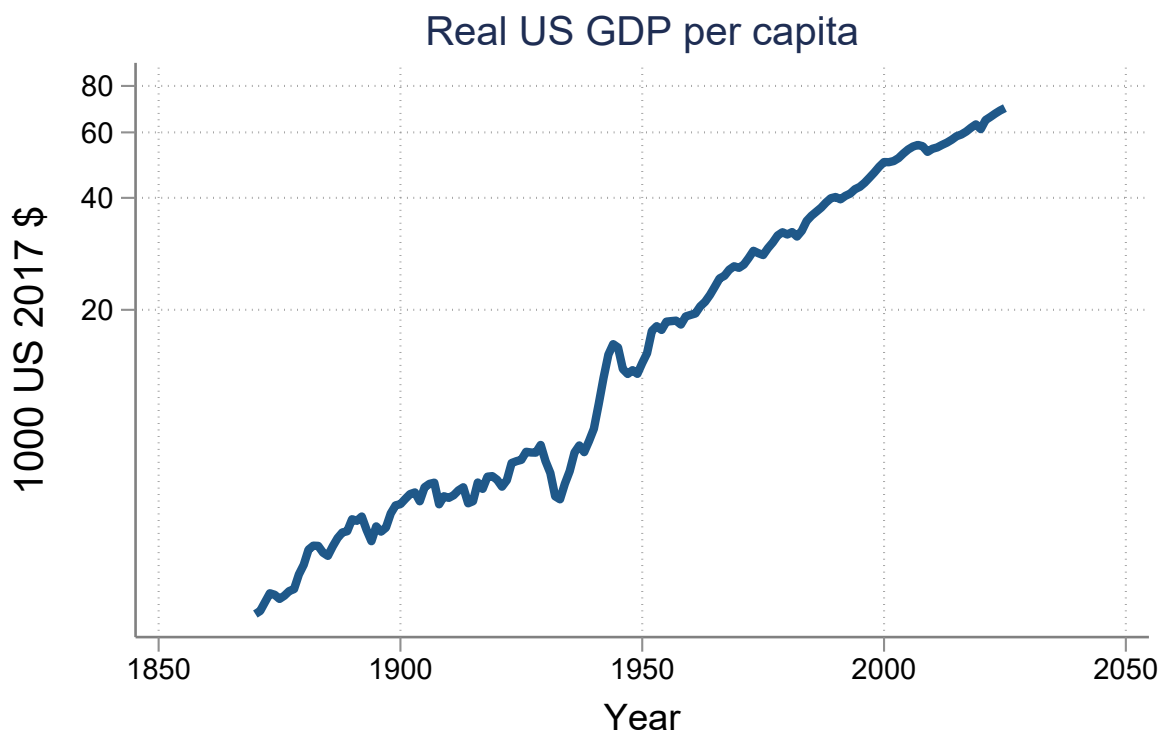


\$21 TRILLION of Pentagon financial transactions “could not be traced, documented, or explained.”

\$21T in Pentagon accounting errors. Medicare for All costs ~\$32T.

That means 66% of Medicare for All could have been funded already by the Pentagon.

And that’s before our premiums.



## Measuring GDP

- Lots of things go into GDP
  - Millions of Chickens
  - Billions of paperclips
  - Thousands of econ lectures
- How do we add Apples and Oranges?



## Money

- The natural metric for adding GDP is money prices.
- Prices provide the relative value of things in an economy.
- This turns out to be one of money's most important functions.

## Measuring GDP

- Suppose that an economy had only two goods.
  - Apples
  - Oranges
- GDP in terms of prices would be

$$GDP = (Q_A \times P_A) + (Q_O \times P_O)$$

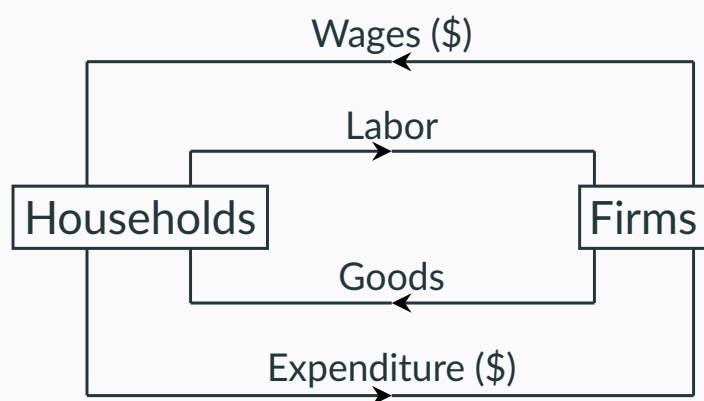
## Measuring GDP

- One way of measuring GDP would be to measure the price and quantity of everything sold in the country.
- There is a second way – we take advantage of circular flows in the economy.

## Measuring GDP

- Imagine an economy with only one good, bread, produced using only labor. There are two types of actors in this economy:
  - Households
    - Provide labor to make bread
    - Get paid wages by the firm (which they use to buy bread)
  - Firms
    - Pay workers to make bread
    - Sell bread to households

## Circular Flow Diagram (simple)



- Households earn wages from firms. All the income is in the form of wages.
- Households spend all their income on goods.
- GDP is both therefore both national income and national expenditure.

## Complications in Counting GDP

- Used Goods
  - How do you count the sale of a used car?
- Intermediate Goods
  - Firm 1 makes Flour – sells it to Firm 2
  - Firm 2 makes bread – sells it to consumers
  - How do we count up GDP?

## Intermediate Goods

- Adding up the value of flour and bread would be double counting.
- We can fix this in two ways
  - Just count final goods – in this case bread.
  - Add up the value added at each step; the value of the output minus the value of the inputs.

## Complications

- Non-market goods
  - Home cooked meals
  - Child care
  - Snow Shoveling
- Imputed value
  - Housing services
  - Government services



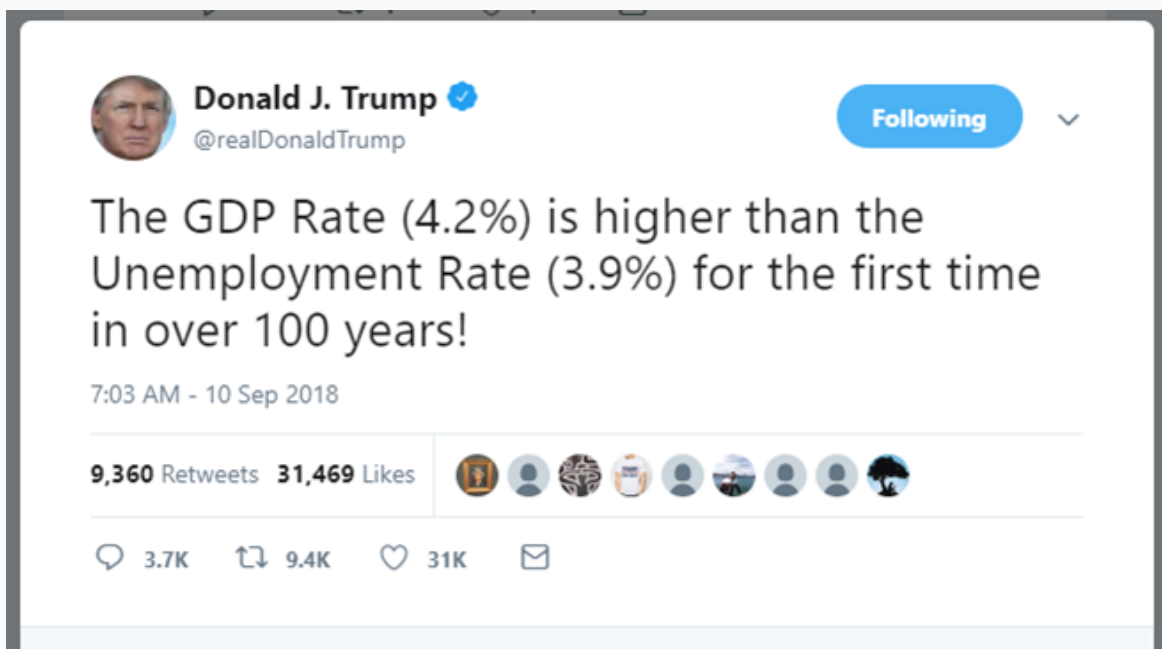
## Stocks vs. Flows

- Think of a bathtub
  - Stock – level of water in the tub at any time
  - Flow – quantity of water flowing in/out of the tub
- Goods aren't produced instantly -- they are produced over time.
  - We don't say a factory makes 100 loaves of bread, we say they make 100 loaves per day.

## Stocks vs. Flows

- Stock – quantity measured at any point in time.
  - Credit Card Balance
  - Capital stock
  - Number of unemployed people
- Flow – Quantity measured per unit of time.
  - Credit card charges/payments
  - Investment
  - Number of people losing their jobs

## Stocks vs flows == rates vs levels



## Components of GDP

- There are four uses to which GDP can be put

$$Y = C + I + G + NX$$

- This is called the national income accounting identity.

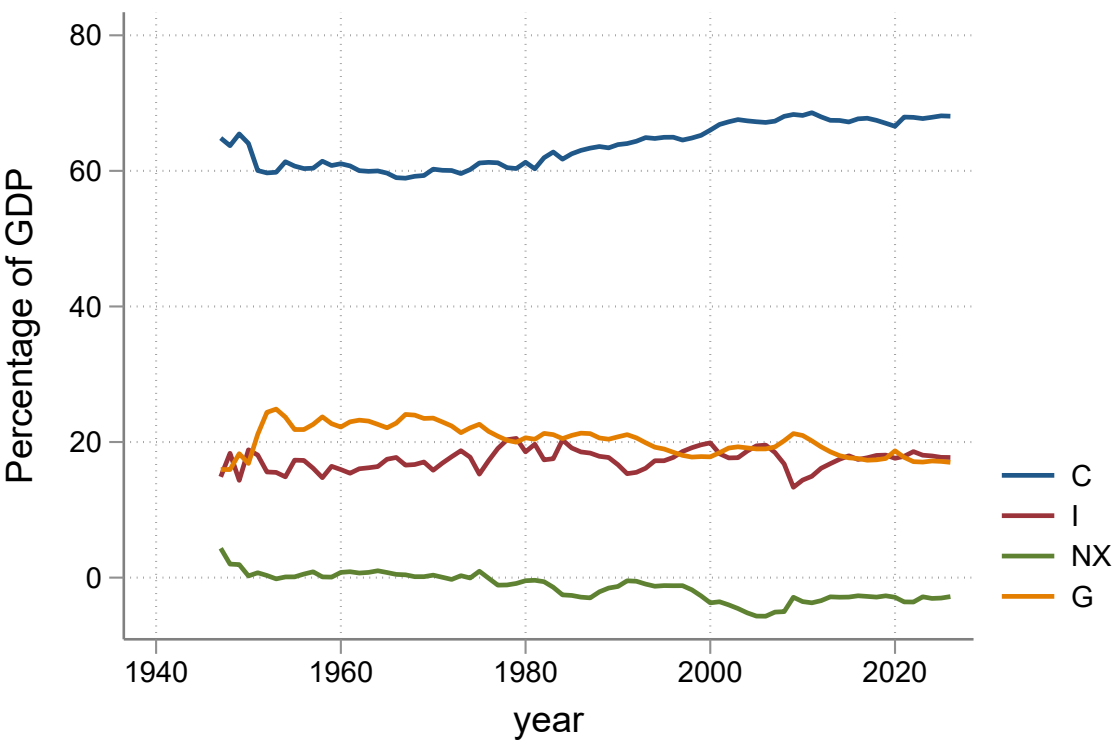
$$Y = C + I + G + NX$$

- C – Consumption of goods and services by households.
- I – Investment. Investment is anytime you buy something for use in the future. Houses, factories.
- G – Government spending on goods and services – like tanks and teachers. Note that this does not include transfer payments.
- NX – Net exports. This is the amount that we export, less the amount that we import.

How large are the components?  
All values are annual nominal values for 2024

|    |         |                |
|----|---------|----------------|
| Y  | \$29.3T | Percent of GDP |
| C  | \$19.9T | 68%            |
| I  | \$5.3T  | 18%            |
| G  | \$5.0T  | 17%            |
| NX | -\$898B | -3%            |

# Components over time - Consumption



## Real GDP

- Q: Given our framework for calculating GDP, what would happen if prices doubled?
- A: Nominal GDP would double
- Would this mean anything real?



## Real GDP

- Since we want a measure of real stuff produced we need to filter out changes in prices.
- We do this by using a uniform set of prices.

## Constructing Real GDP

- Find the price of each good in a given base year.
- Take the quantity of each good produced during the current year and multiply it by the price of that good in the base year.
- Sum the value of all the goods produced.
- Because the only thing changing over time is the quantity of goods, this gives us the changes in the quantity of stuff, irrespective of changes in prices.

## Real GDP - example

- Suppose we want to construct RGDP in 2025 using 2017 as the base year. In our economy with apples and oranges:

$$RGDP_{2025} = (Q_{A,25} \times P_{A,17}) + (Q_{O,25} \times P_{O,17})$$

- This gives us RGDP for 2025 in 2017 prices

## Real GDP

- How would you calculate RGDP for 2017 using 2017 as the base year?

## Some numbers

- Nominal GDP in 2024– \$29.2 T
- Nominal GDP in 2017– \$19.6 T
- Real GDP in 2024– \$23.4 T (in 2017\$)
- Real GDP in 2017– \$19.6 T (in 2017\$)
  
- The nominal increase was \$9.6 T
- The real increase was \$3.8 T
- The increase due to inflation was \$5.8 T

## The GDP deflator

- Suppose that now we want to know how much prices have gone up.
- There are two major price indices that we use. One index is called the GDP deflator (also called the “implicit price deflator for GDP”).

$$GDPD = NGDP / RGDP$$

## The GDP deflator

$$GDPD = NGDP / RGDP$$

- For example, for 2024:

$$GDPD = \frac{(Q_{A,24} \times P_{A,24}) + (Q_{O,24} \times P_{O,24})}{(Q_{A,24} \times P_{A,17}) + (Q_{O,24} \times P_{O,17})}$$

- Note that GDPD is 1 for the base year

## The GDP deflator - example

- $NGDP_{24} = \$29.2 \text{ T}$
- $RGDP_{24} = \$23.4 \text{ T (2017\$)}$
- $GDPD = NGDP / RGDP = 1.248$
- Prices rose by about 24.8% between 2017 and 2024



## Consumer Price Index (CPI)

- The GDPD measures the prices of all the goods and services that are produced in the country.
- What if we wanted to measure the price of a different bundle?
  - Cost of being a Dartmouth student
  - Cost of being caffeinated for your 9am classes.

## Caffeine Price Index

- Suppose you got data on the prices of coffee and tea.
- Q: How much has the price of caffeine risen?

|        | 2024 | 2025 |
|--------|------|------|
| Tea    | 1.00 | 1.20 |
| Coffee | 1.00 | 1.40 |

## Caffeine Price Index

- A: You can't tell because you don't know how much to weight the two components.
- We know that tea has gone up by 20% and that coffee has gone up by 40%.
- Overall, the price of caffeinated beverages has gone up by somewhere between the two. The overall change will be a weighted average. But what weights to use?

## Caffeine Price Index

- Suppose that we did a survey and found that in 2024 (the base year), consumption of the two beverages was:
  - Tea – 1000 cups
  - Coffee – 3000 cups
- We can use these quantities as weights in our price index. Coffee will have three times the weight of tea.

## Caffeine Price Index

- Our caffeine consumption cost is:
  - $1000 * 1.0 + 3000 * 1.0 = \$4,000$  in 2024
  - $1000 * 1.2 + 3000 * 1.4 = \$5,400$  in 2025
- The price of caffeine had increased by 35%.

# PNC Cost of Christmas

## PNC Christmas Price Index - 2008

|                                | Traditional  |              |                     | Internet      |               |                     |
|--------------------------------|--------------|--------------|---------------------|---------------|---------------|---------------------|
|                                | 2007         | 2008         | % Change<br>2008/07 | 2007          | 2008          | % Change<br>2008/07 |
| One Partridge in a Pear Tree   | \$ 164.99    | \$ 219.99    | 33.3%               | \$ 224.94     | \$ 235.00     | 4.5%                |
| Partridge                      | \$ 15.00     | \$ 20.00     | 33.3%               | 75.00         | 85.00         | 13.3%               |
| Pear Tree                      | \$ 149.99    | \$ 199.99    | 33.3%               | 149.94        | 150.00        | 0.0%                |
| Two Turtle Doves               | \$ 40.00     | \$ 55.00     | 37.5%               | 130.00        | 130.00        | 0.0%                |
| Three French Hen               | \$ 45.00     | \$ 30.00     | -33.3%              | 195.00        | 195.00        | 0.0%                |
| Four Calling Birds             | \$ 599.96    | \$ 599.96    | 0.0%                | 455.00        | 475.00        | 4.4%                |
| Five Gold Rings                | \$ 395.00    | \$ 349.95    | -11.4%              | 329.75        | 329.75        | 0.0%                |
| Six Geese-a-Laying             | \$ 360.00    | \$ 240.00    | -33.3%              | 660.00        | 660.00        | 0.0%                |
| Seven Swans-a-Swimming         | \$ 4,200.00  | \$ 5,600.00  | 33.3%               | 7,035.00      | 7,035.00      | 0.0%                |
| Eight Maids-a-Milking          | \$ 46.80     | \$ 52.40     | 12.0%               | 310.08        | 310.08        | 0.0%                |
| Nine Ladies Dancing            | \$ 4,759.19  | \$ 4,759.19  | 0.0%                | 7,106.00      | 7,237.00      | 1.8%                |
| 10 Lords-a-Leaping             | \$ 4,285.06  | \$ 4,413.61  | 3.0%                | 11,283.23     | 11,629.79     | 3.1%                |
| 11 Pipers Piping               | \$ 2,213.40  | \$ 2,284.80  | 3.2%                | 1,900.00      | 2,100.00      | 10.5%               |
| 12 Drummers Drumming           | \$ 2,397.85  | \$ 2,475.20  | 3.2%                | 1,620.00      | 1,620.00      | 0.0%                |
| Total Christmas Price Index    | \$ 19,507.25 | \$ 21,080.10 | 8.1%                | \$ 31,249.00  | \$ 31,956.62  | 2.3%                |
| % change                       |              |              |                     |               |               |                     |
| True cost of Christmas in song | \$ 78,100.10 | \$ 86,608.51 | 10.9%               | \$ 128,886.37 | \$ 131,150.76 | 1.8%                |
| % change                       |              |              |                     |               |               |                     |
| "Core" index, excluding swans  | \$ 15,307.25 | \$ 15,480.10 | 1.1%                | \$ 24,214.00  | \$ 24,921.62  | 2.9%                |

## CPI – Consumer Price Index

- To calculate the CPI, you choose a base year and a basket of goods.
- The CPI compares the cost of the basket today versus the cost in the base year.

$$CPI_t = 100 \times \frac{\sum p_t \times q_{base}}{\sum p_{base} \times q_{base}}$$

- The basket of goods for the US is chosen by the government.
- The government performs a survey to find out what people consume and in what proportions. This determines the weights.



## Problems with the CPI

- Suppose that you wanted to compensate people for the increased cost of their caffeine consumption.
  - This is the idea behind cost of living adjustments (COLAs).
- What if people only slightly prefer coffee to tea?

## CPI problems - Substitution

- If people switch from coffee to tea, the increase in the price index overstates the needed increase in people's caffeine budgets.
- If you increased everyone's caffeine budget by 35% (the amount that the caffeine CPI has risen), they would be better off than they were before, because they are willing to switch to tea.

## CPI problems - Substitution

- The problem with indexes based on a fixed basket of goods is it does not take account of the fact that people will switch from expensive products to cheaper ones.
- This is an unavoidable problem with this sort of index.

## Differences between CPI and GDPD

- CPI

- Basket of goods is just consumer goods
- Goods produced abroad are counted
- The basket of goods is from the base year (Laspeyres index, fixed basket)

- GDPD

- Basket of goods is all goods
- Goods produced abroad are not counted
- The basket of goods is from the current year (Paasche index, changing basket)

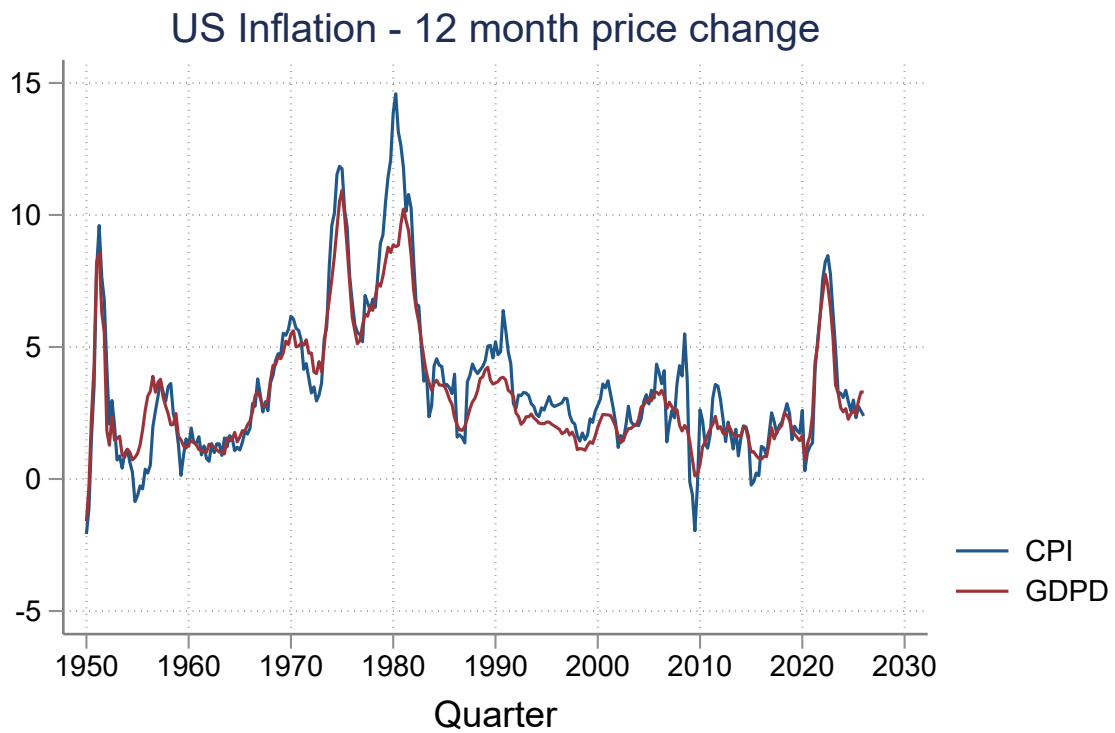
## Problems with choosing baskets

- Suppose that we use a base year consumption basket from 50 years ago
  - the typical basket that the average person got with the average salary.
- Can we consume it now with an average salary?
- Maybe, but some things have gone up tremendously in value, like land.

## Problems with choosing baskets

- On the other hand, suppose that we use today's average consumption basket.
- Could people in the past have consumed it with the average income?
- No. we have things like iPhones, HDTV's and computers that they didn't have then, and other things are much cheaper now, like transatlantic transportation.

## CPI and GDPD

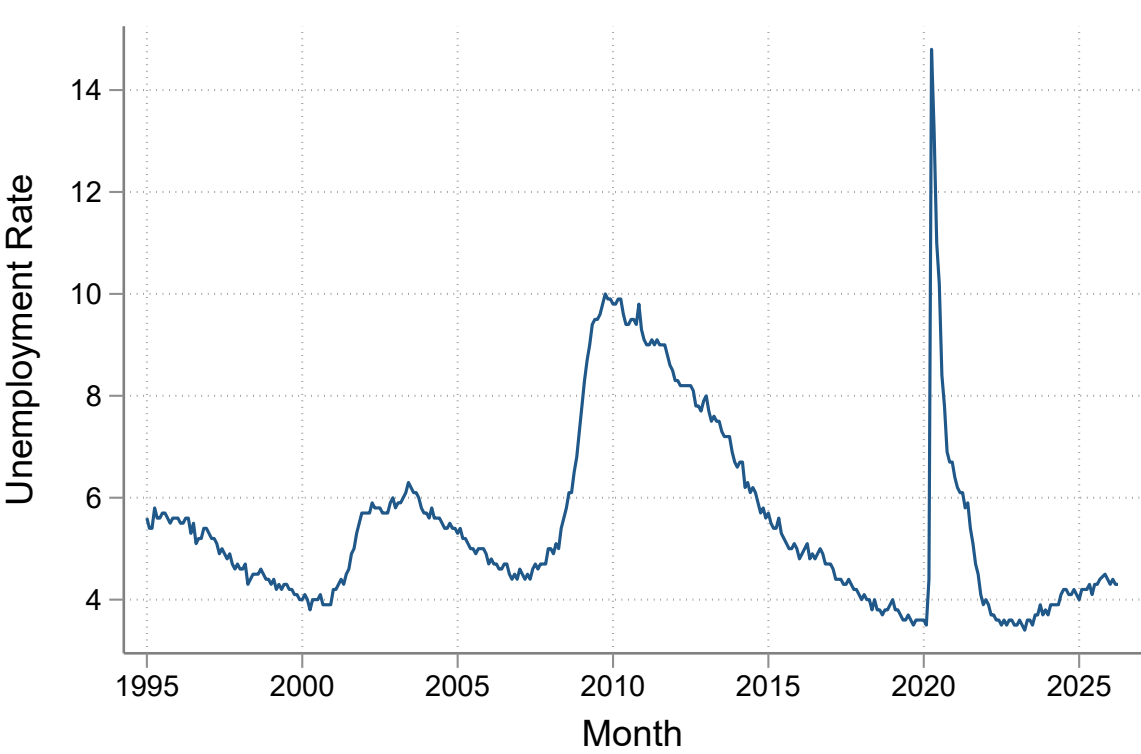


## Unemployment

- Perhaps the most watched economic statistic
- Why?
  - A well functioning economy uses all resources. Unemployment wastes resources.
  - The costs are unevenly distributed



# Unemployment



## Unemployment

- To figure out unemployment, the government does a survey each month.
- They divide the population into three groups.

$$\text{Adult population} = (E + U) + OLF$$

- E – employed
- U – unemployed
- OLF – out of labor force (students, retired, stay at home parents, etc.)

## Unemployment

- The labor force consists of the Employed and the Unemployed

$$LF = E + U$$

- We can then form measures:

- Unemployment Rate

$$UR = U/LF$$

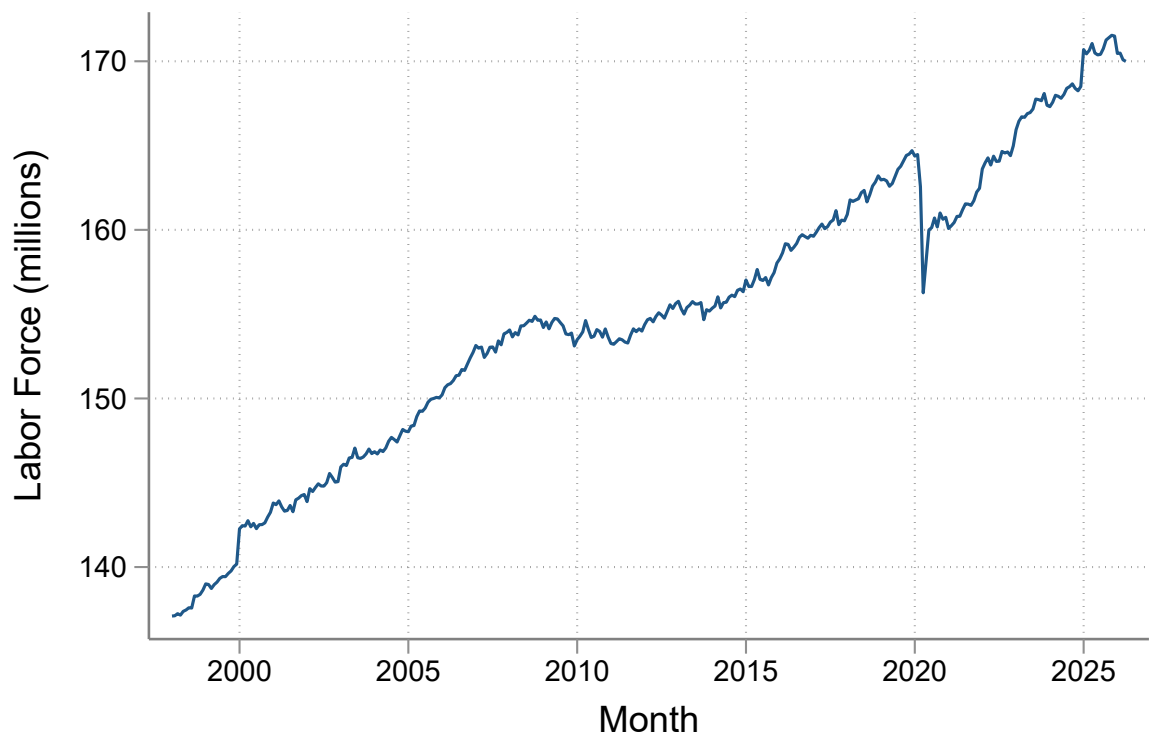
- Labor force Participation Rate

$$LFPR = LF / adult\ pop$$

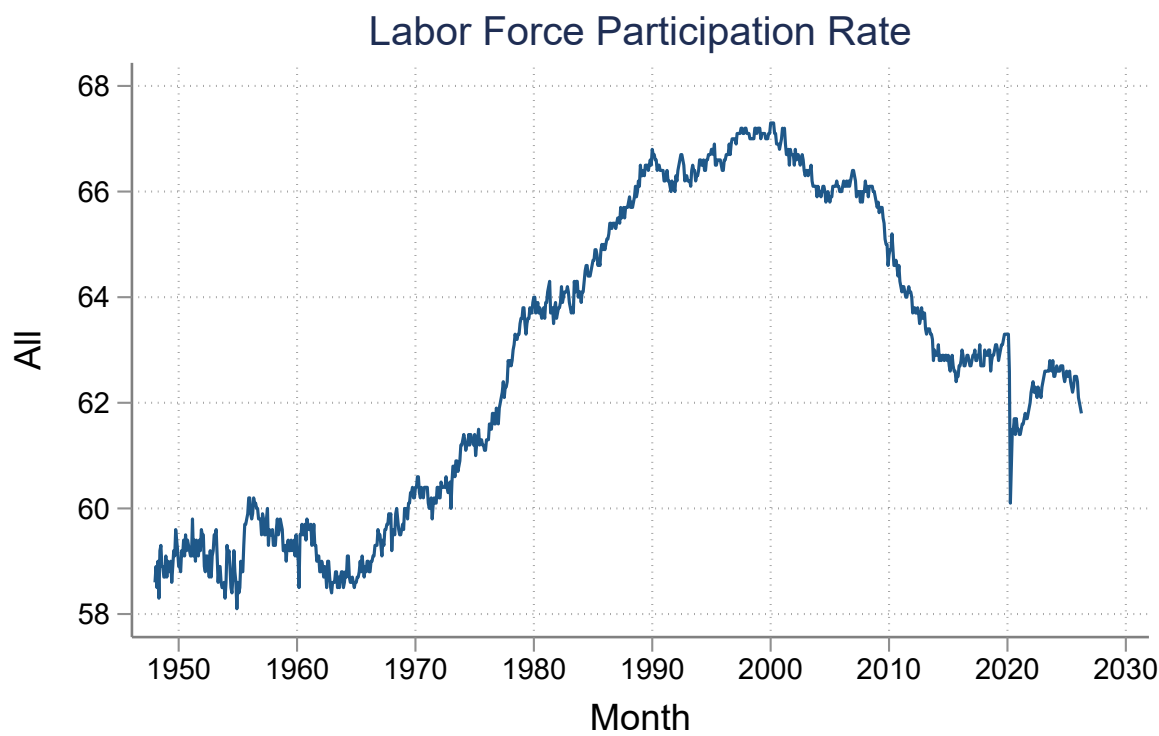
- Employment to Population Ratio

$$EPOP = E / adult\ pop$$

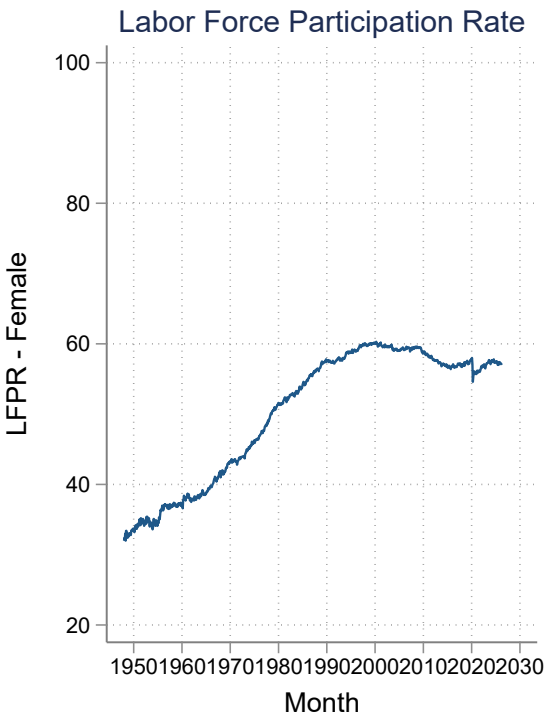
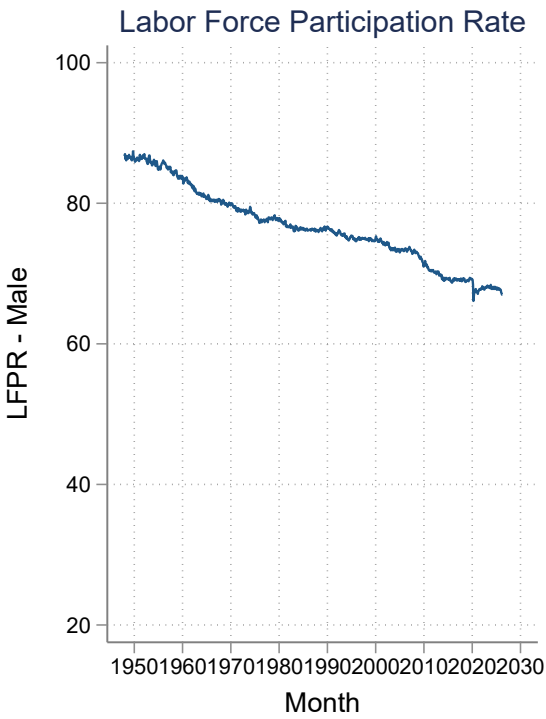
# Labor Force Size



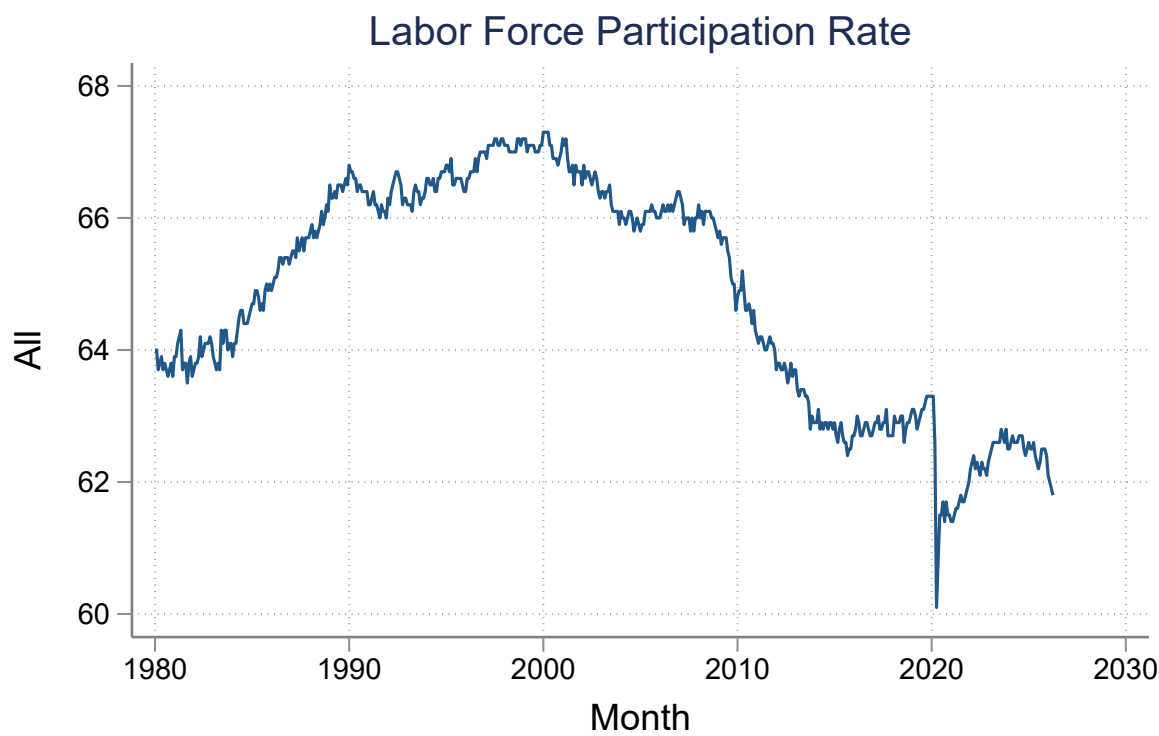
# Labor Force Participation Rate



# LFPR by Gender

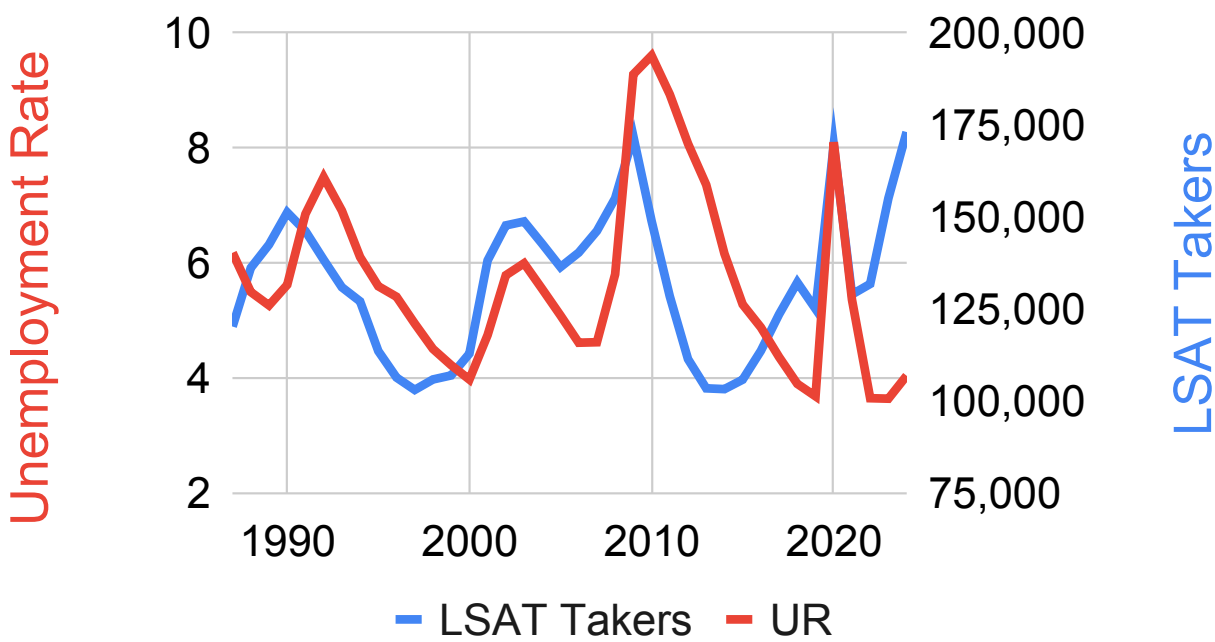


# Labor Force Participation Over the Business Cycle



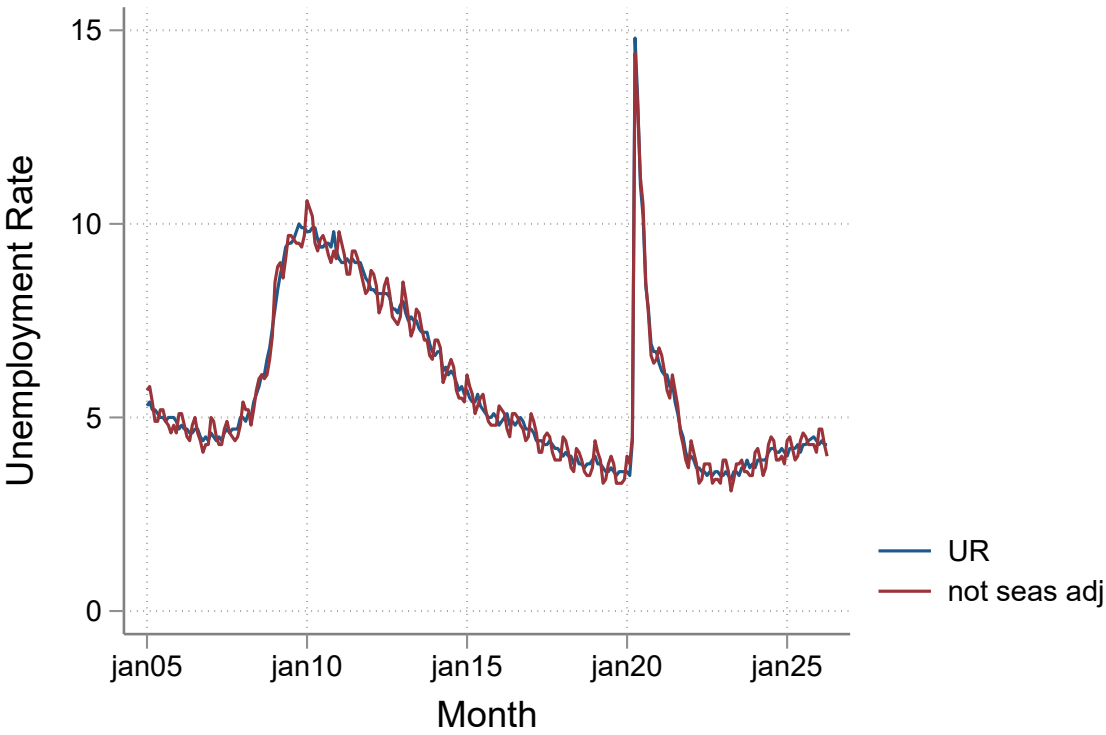
LSAT tests taken vs. Unemployment

LSAT takers Versus the Unemployment Rate

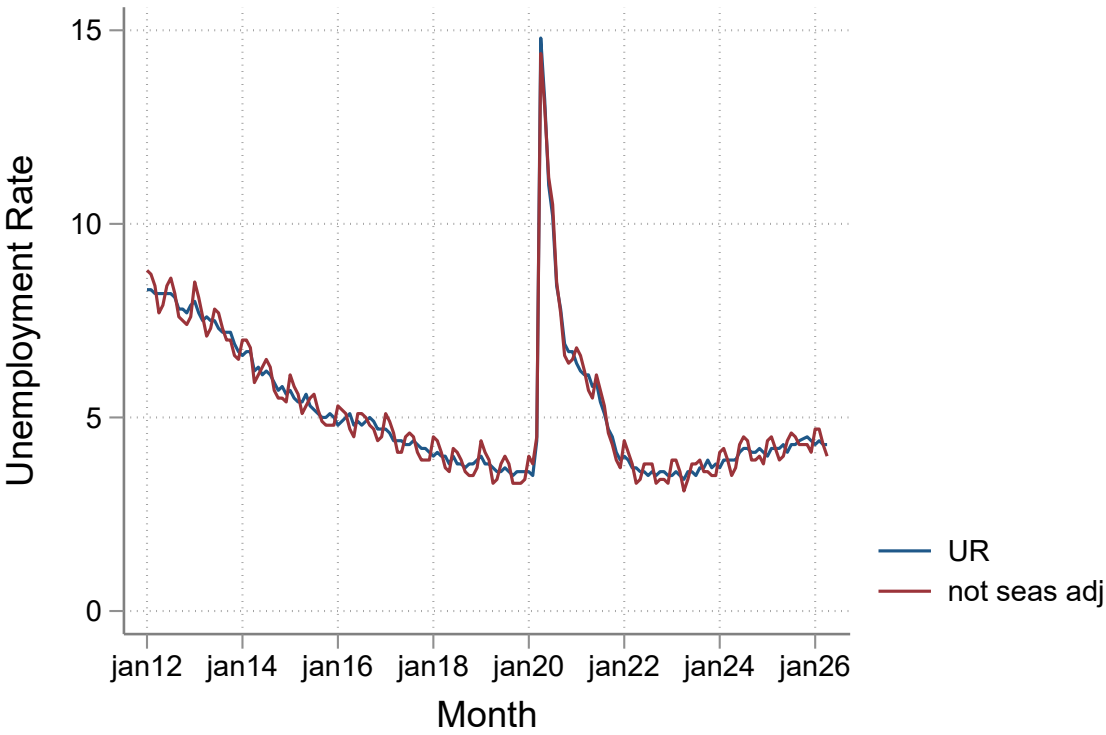




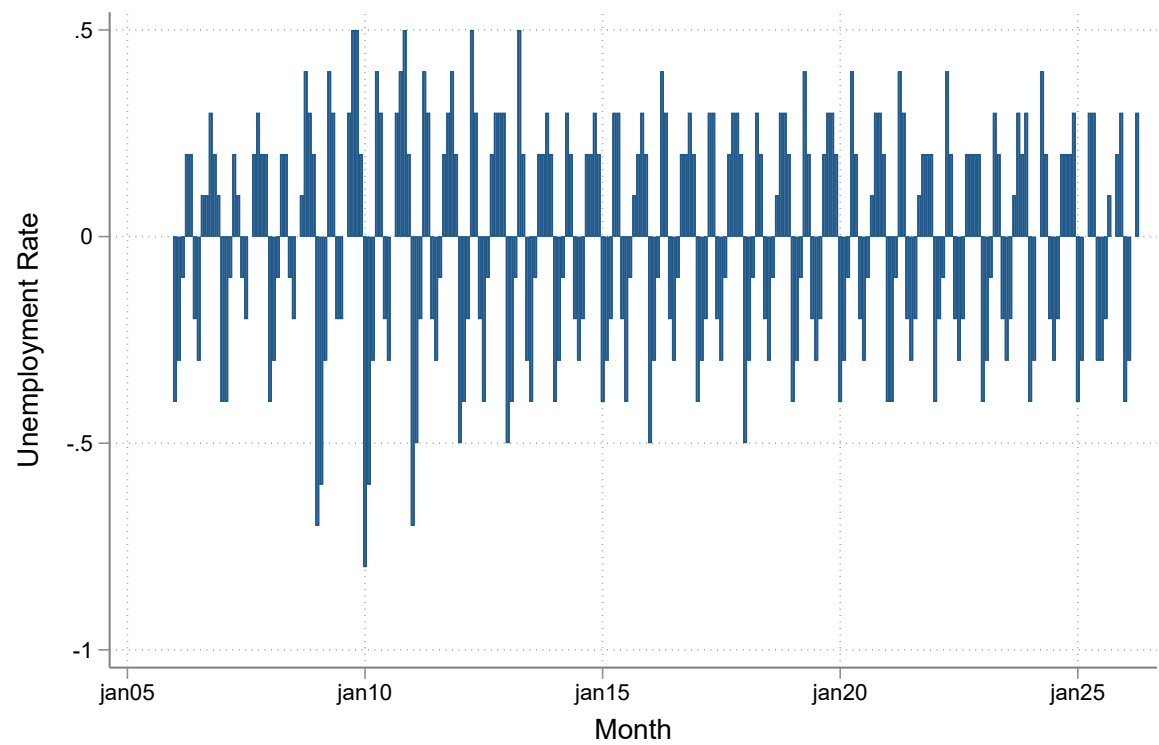
# Seasonal Adjustments



# Seasonal Adjustments



# Size of Adjustments



## Okun's Law

- When unemployment is high, GDP growth tends to be low.

$$\text{Real GDP Growth} = 3\% - 2 * \Delta \text{Unemployment Rate}$$

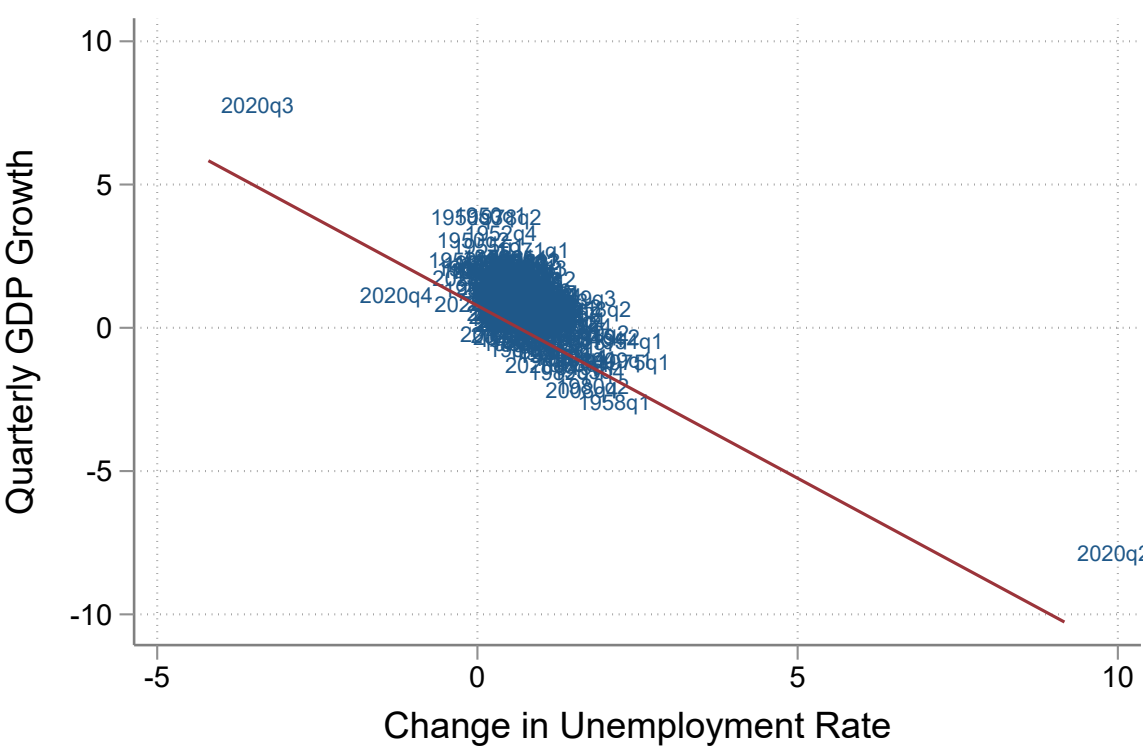
- If the UR isn't changing, Real GDP growth is about 3%.
- For each percentage point increase in unemployment, real GDP growth declines by about 2

## Okun's law

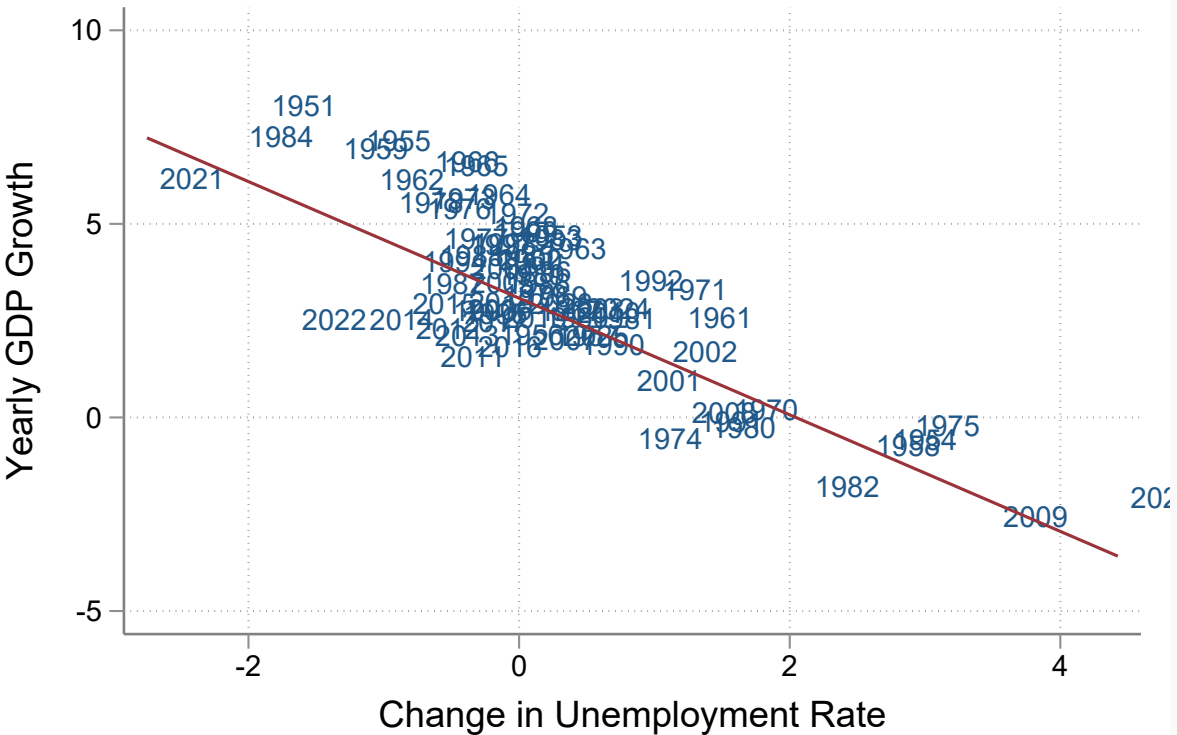
$$\text{Real GDP Growth} = 3\% - 2 * \Delta \text{Unemployment Rate}$$

- To reduce unemployment by two percentage points the US needs to grow at about 7%

Okun's Law on Quarterly Data



Okun's Law on Yearly Data

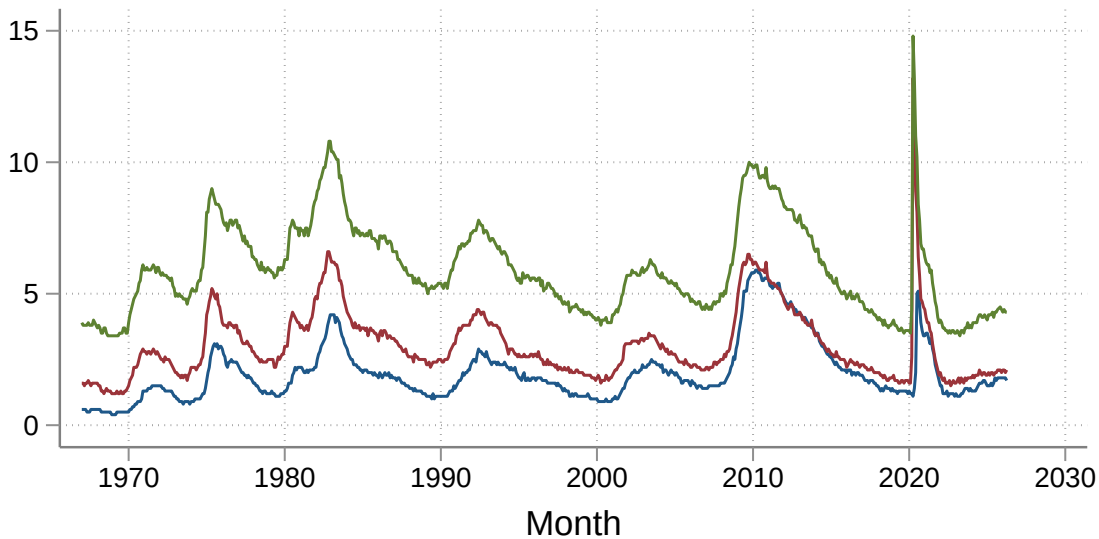


## Okun's Law Regression Results

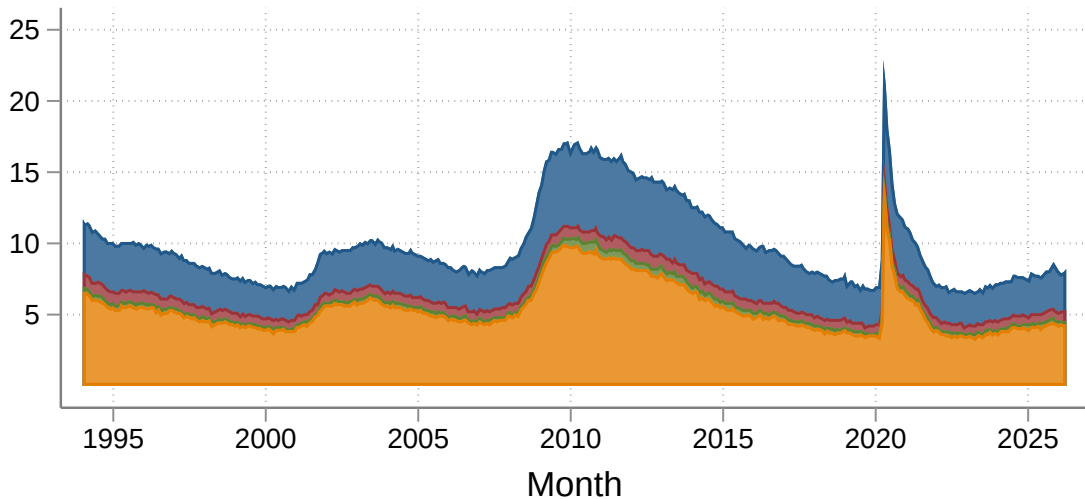
| VARIABLES                   | (1)<br>Yearly      | (2)<br>Quarterly   |
|-----------------------------|--------------------|--------------------|
| Change in Unemployment Rate | -1.50***<br>(0.13) | -1.21***<br>(0.06) |
| Constant                    | 3.09***<br>(0.16)  | 0.78***<br>(0.04)  |
| Observations                | 74                 | 310                |
| R-squared                   | 0.65               | 0.56               |



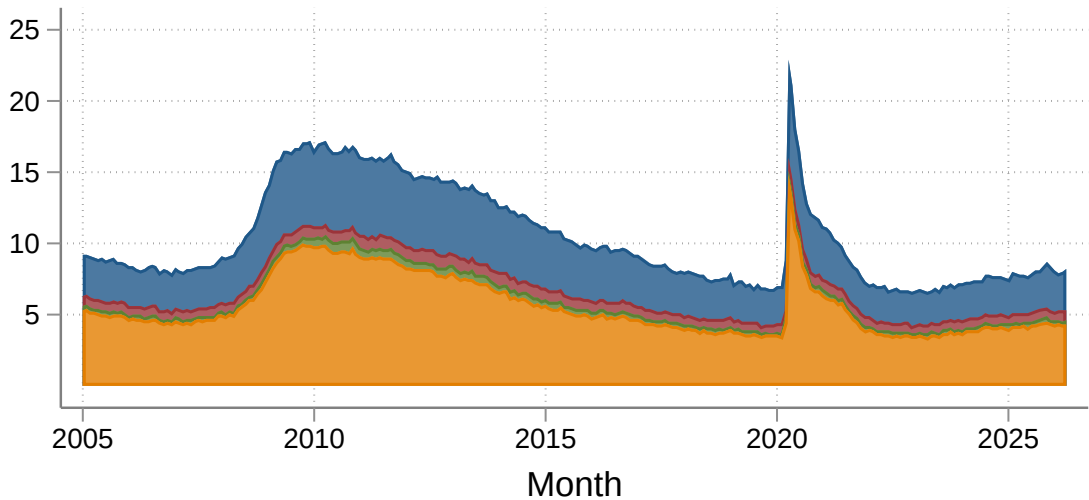
## 2 Generated Graphs



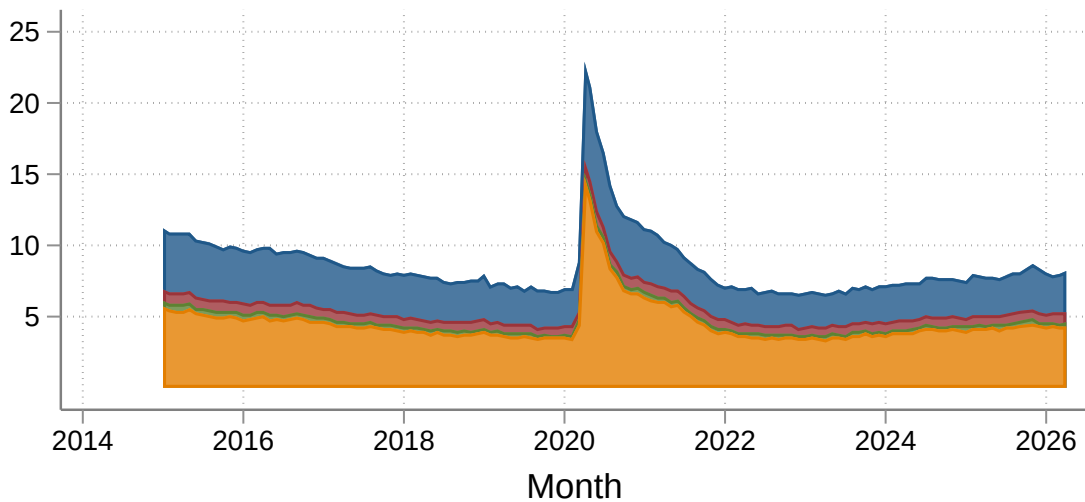
- U1 -- Unemployed 15 weeks or longer
- U2 -- Job losers and persons who completed temporary jobs
- U3 -- Headline Unemployment rate



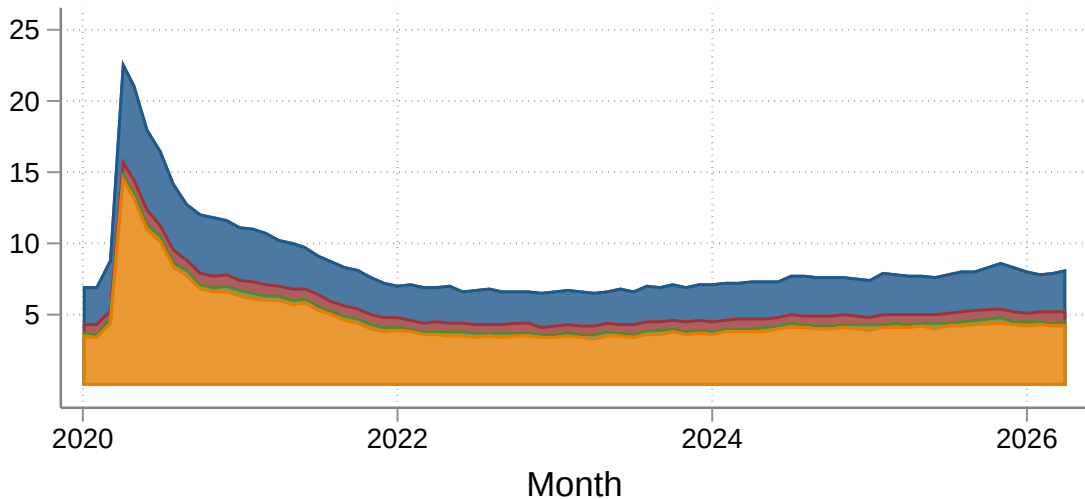
- U6 -- U5 plus part time for economic reasons
- U5 -- U4 plus all other marginally attached workers
- U4 -- U3 plus discouraged workers
- U3 -- Headline Unemployment rate



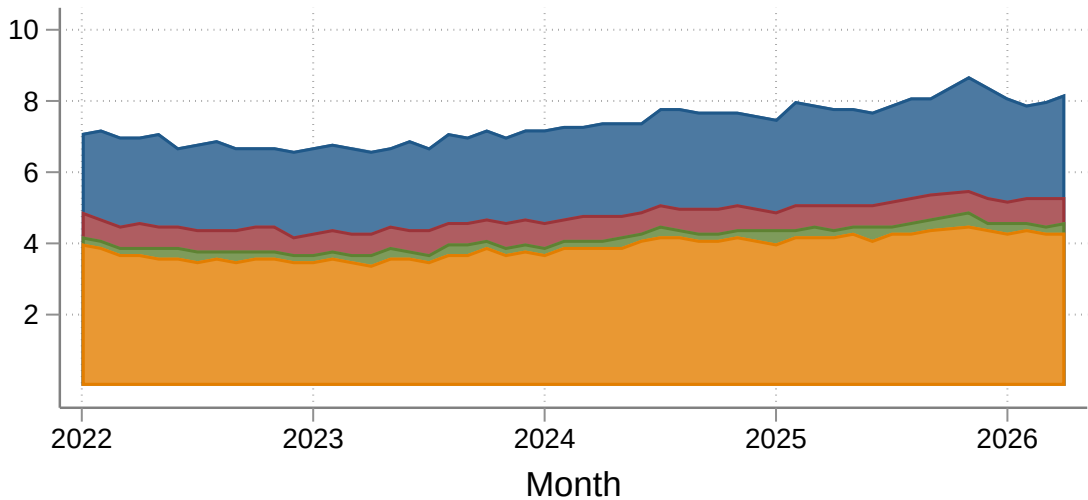
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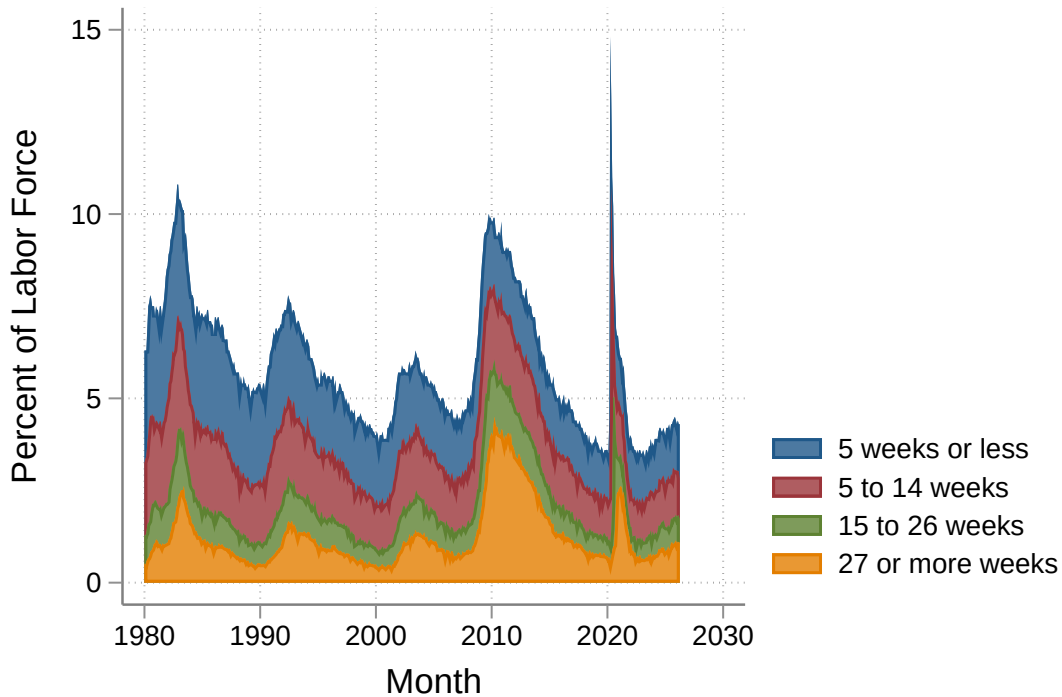
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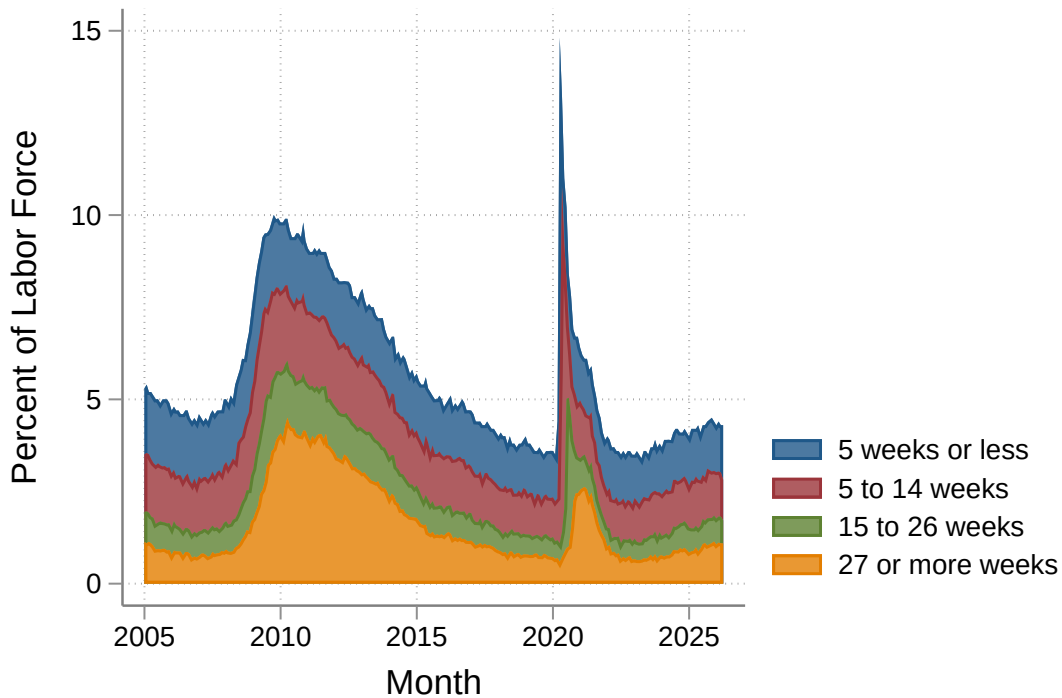
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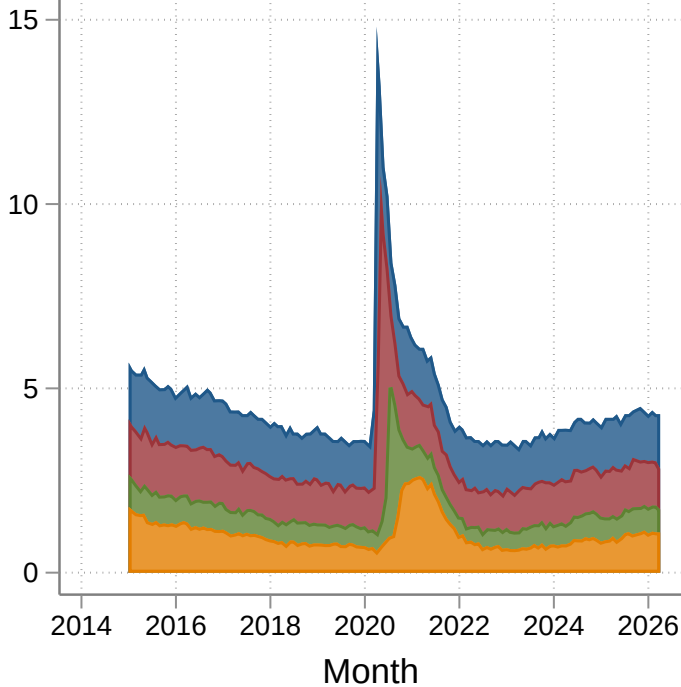
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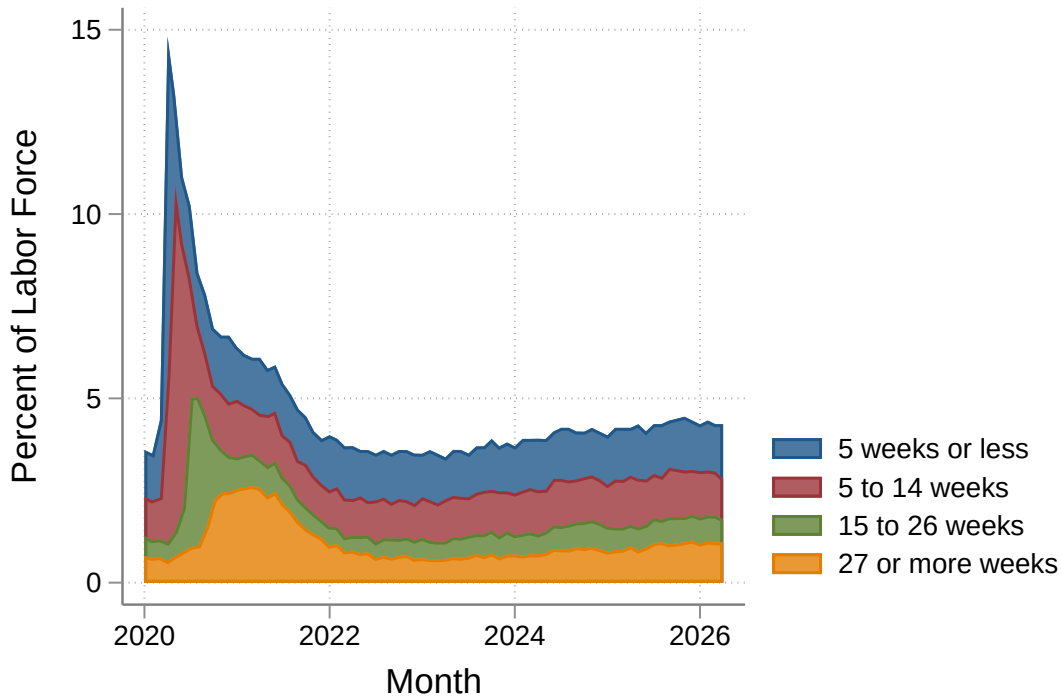


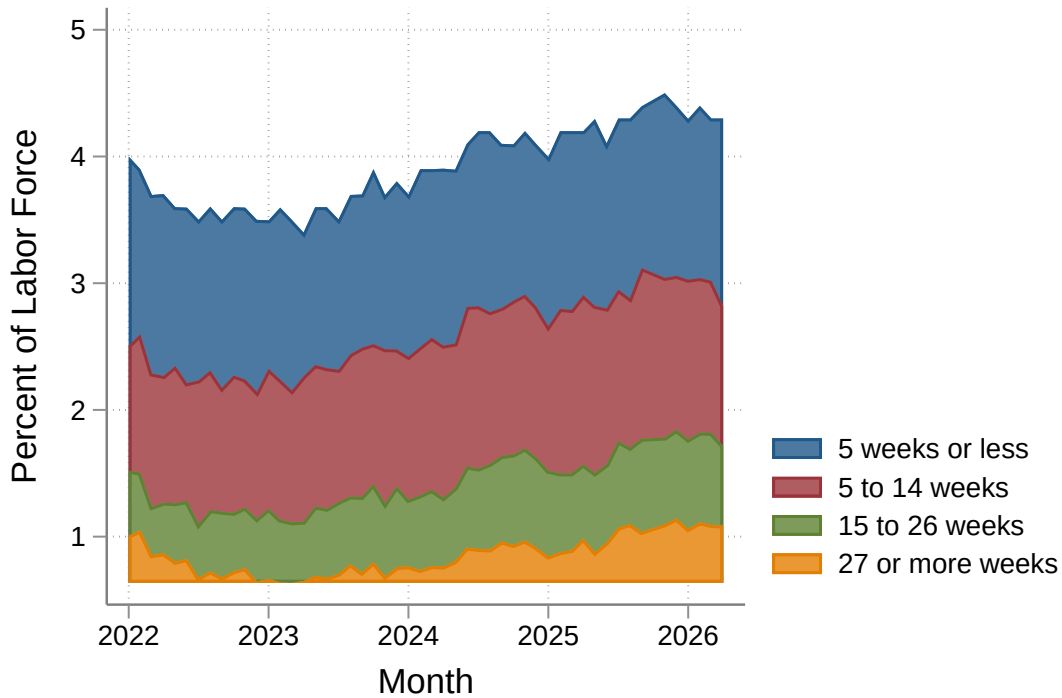


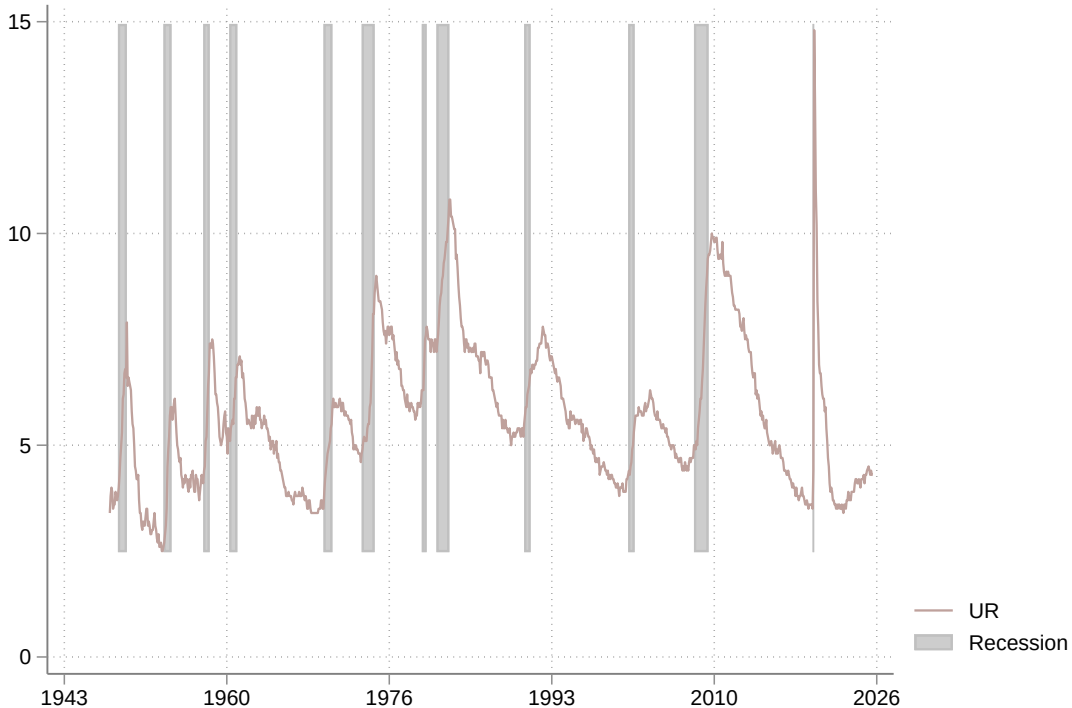
Percent of Labor Force



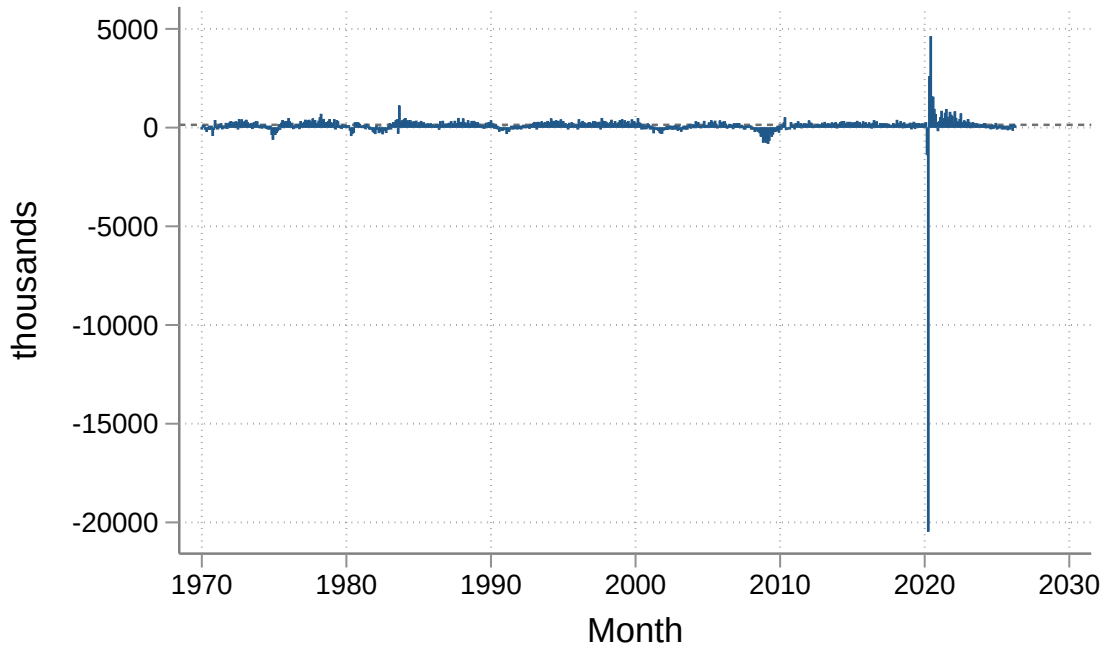
- 5 weeks or less
- 5 to 14 weeks
- 15 to 26 weeks
- 27 or more weeks



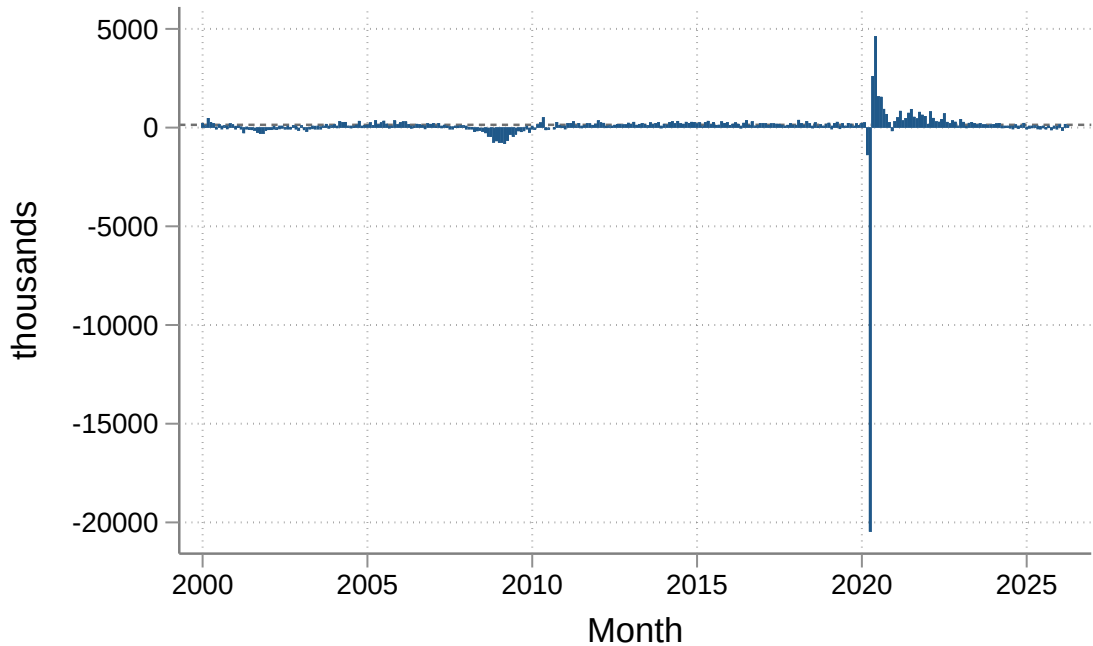




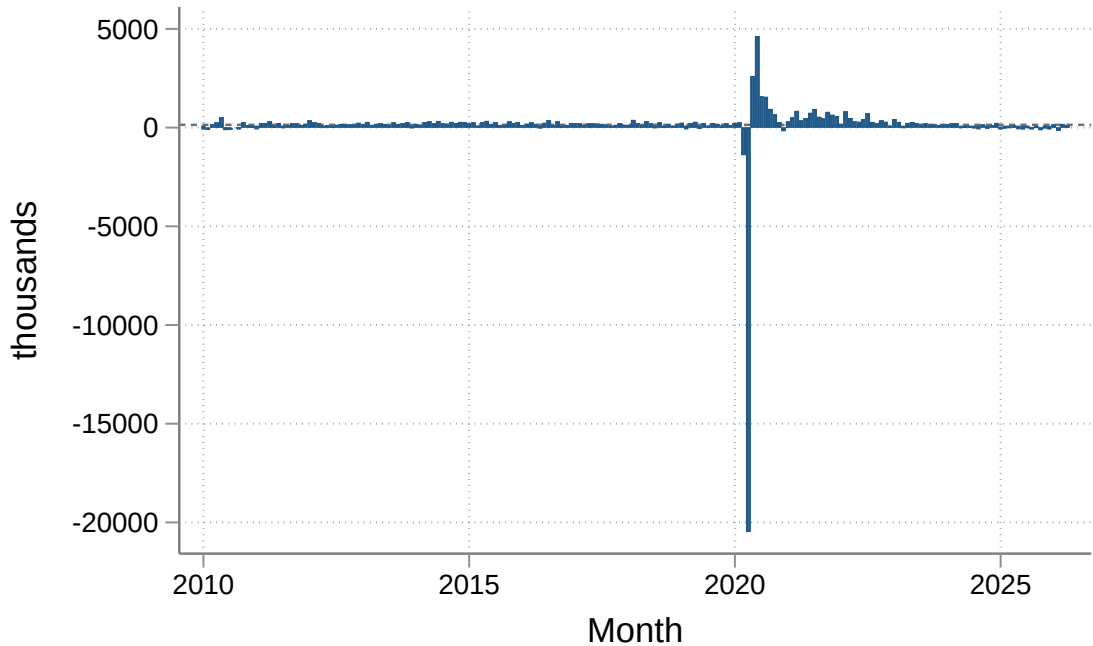
# Monthly Payroll Employment Change



# Monthly Payroll Employment Change

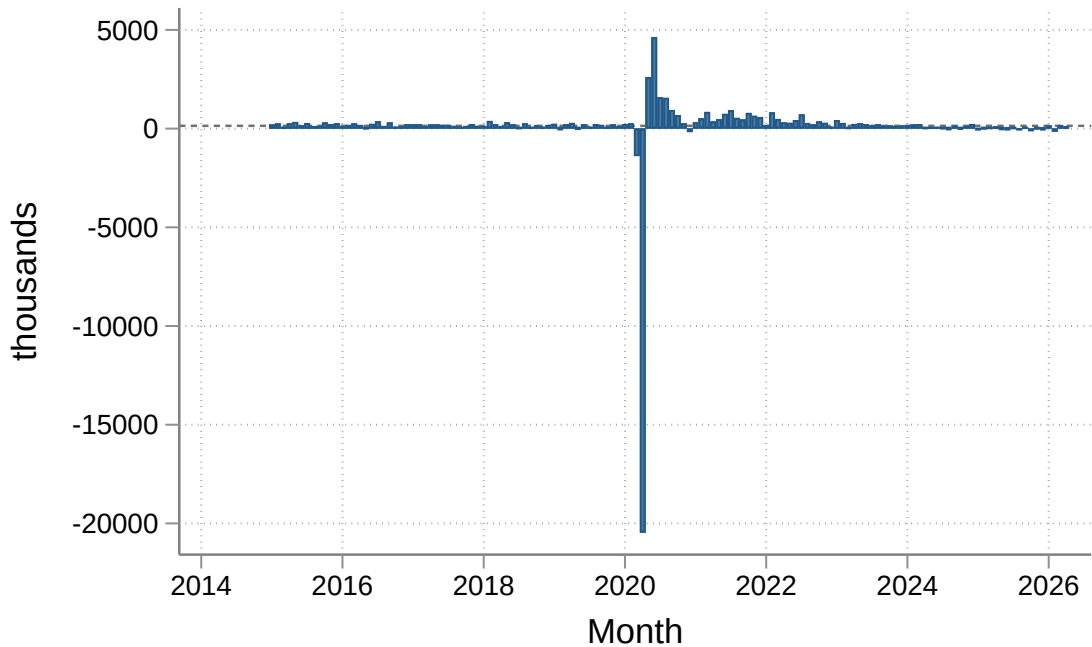


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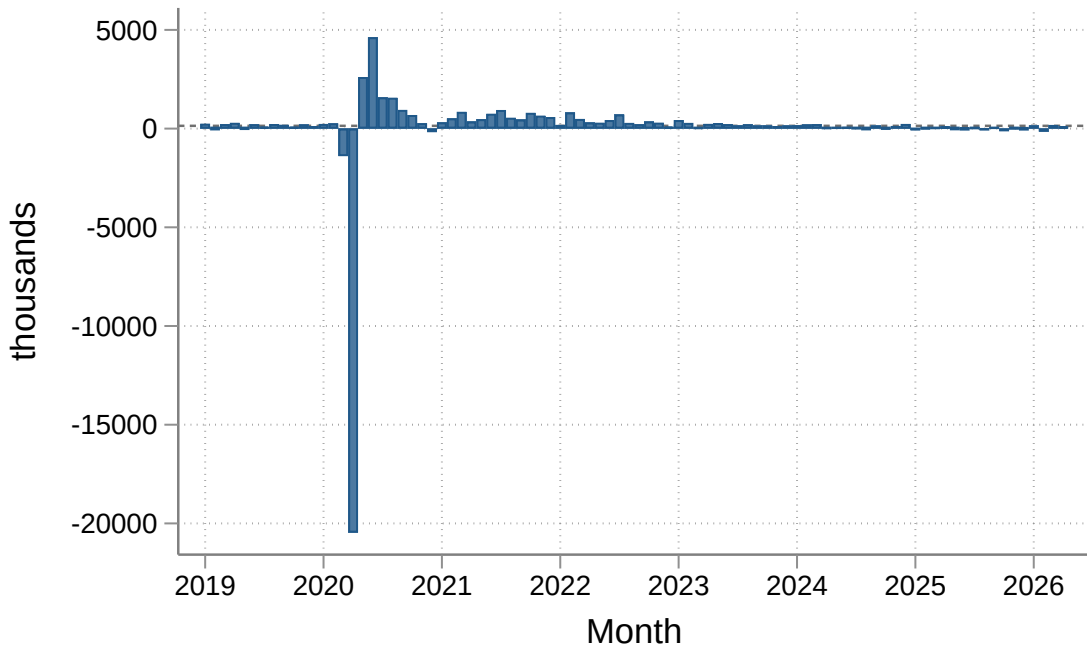




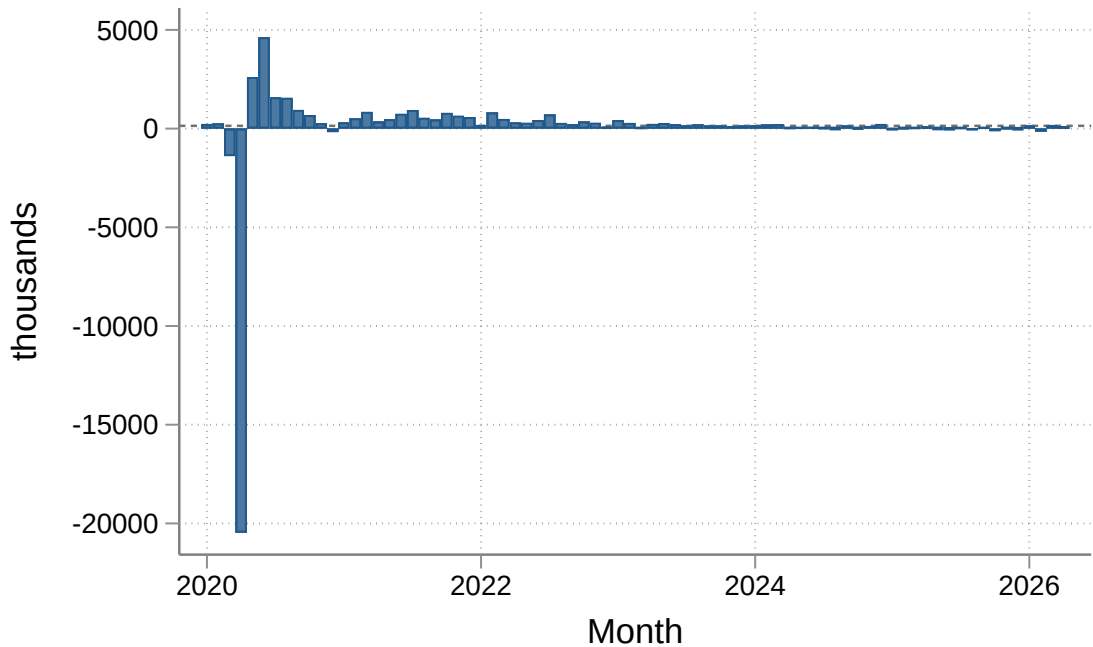
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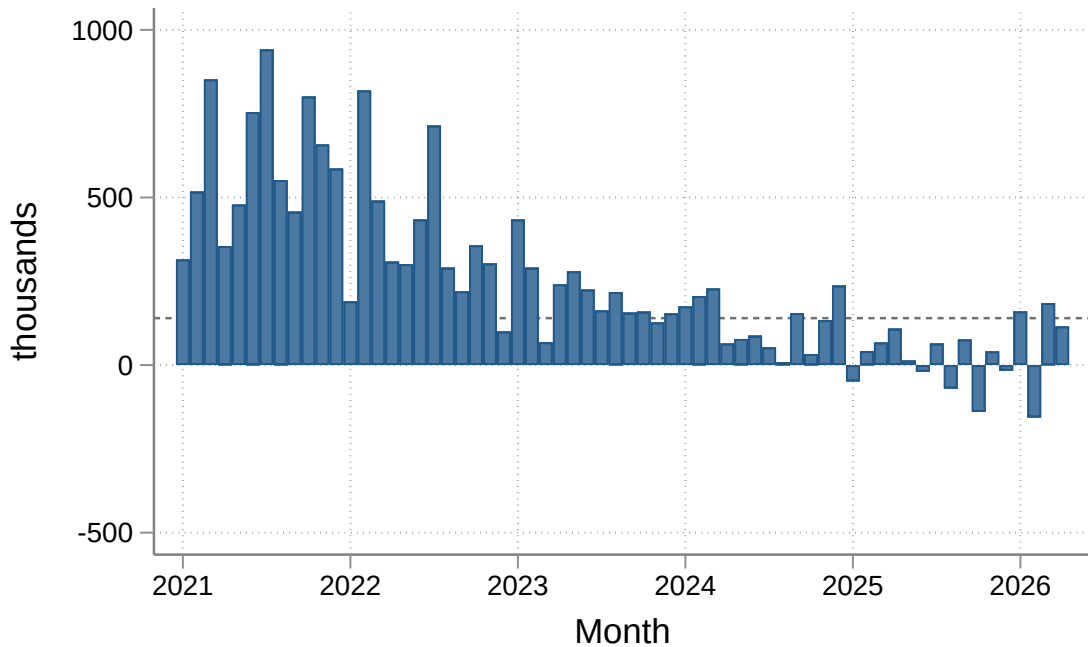
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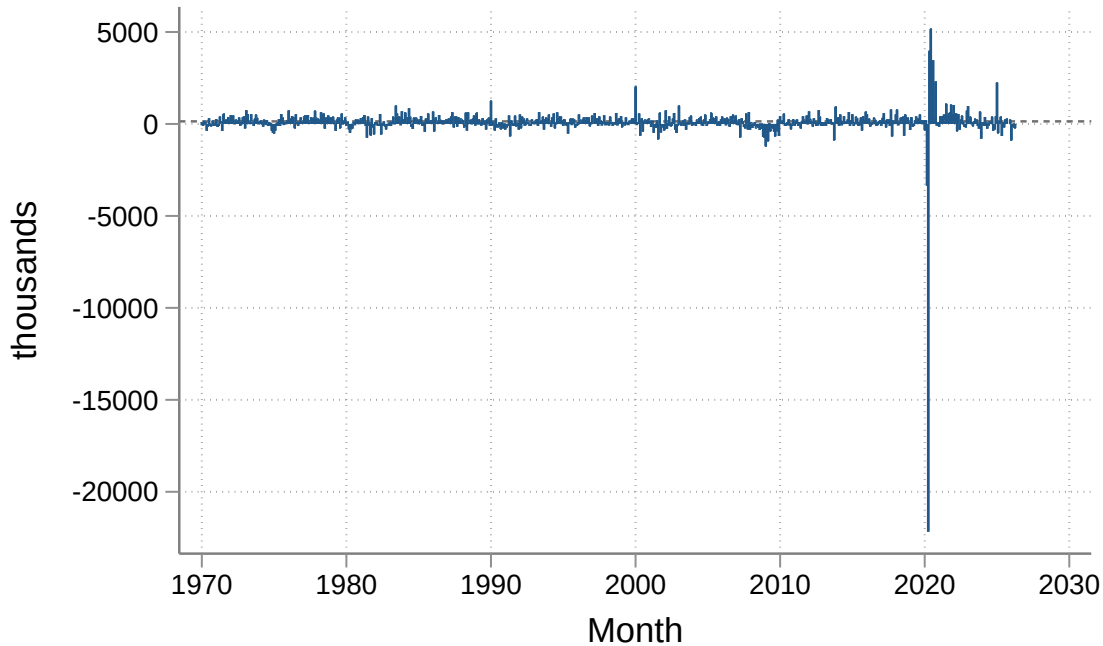
# Monthly Payroll Employment Change



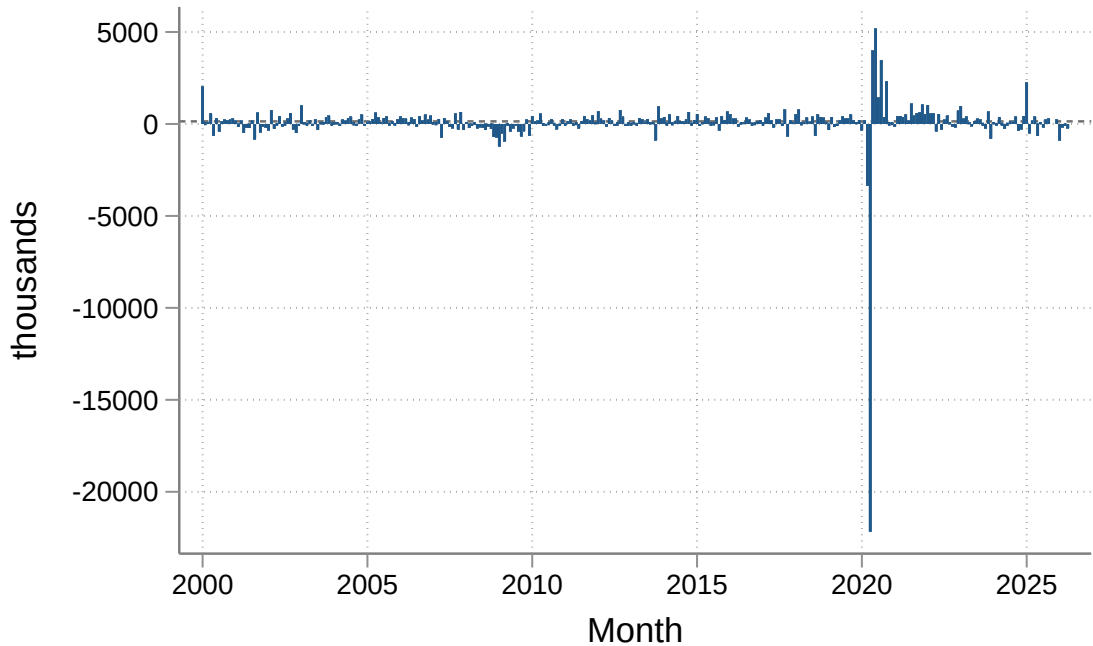
# Monthly Payroll Employment Change



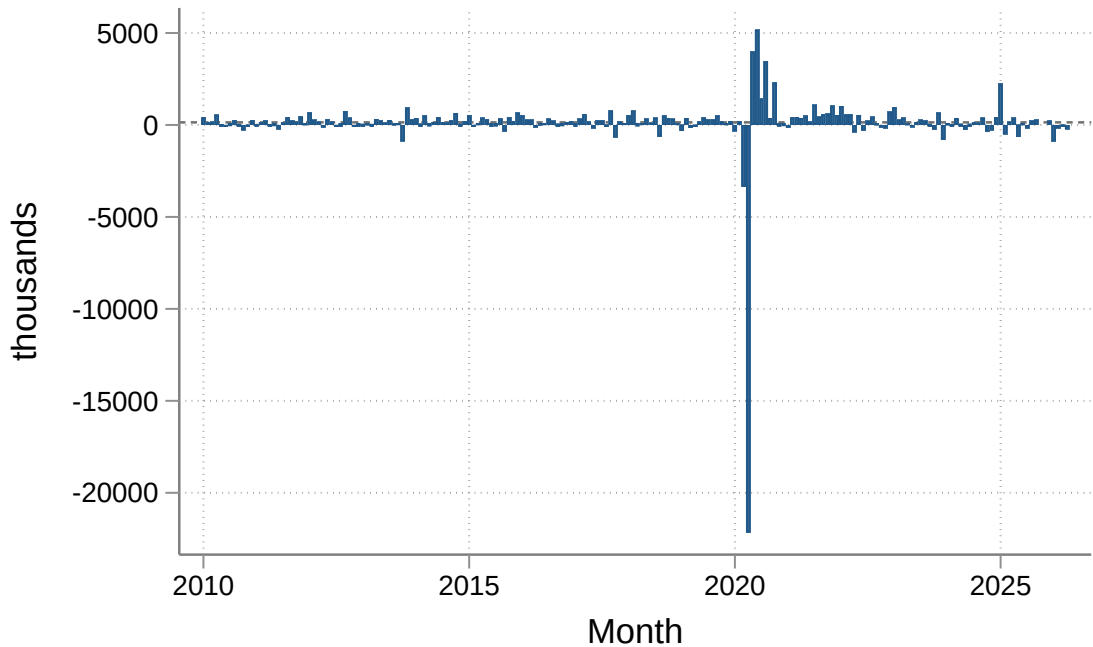
# Monthly Survey Employment Change



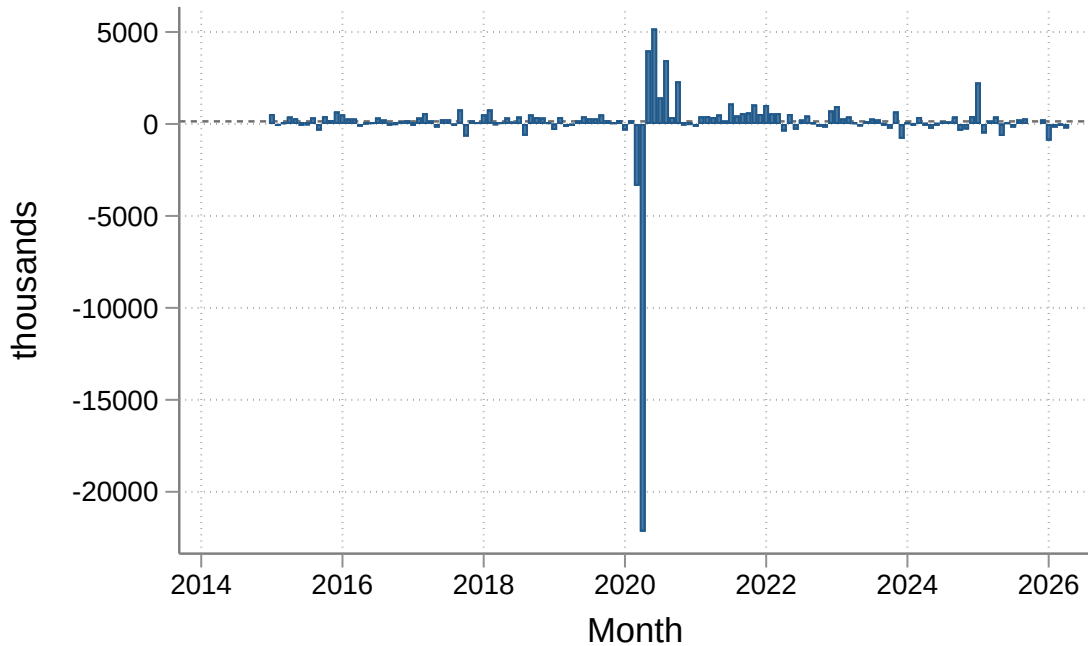
# Monthly Survey Employment Change



# Monthly Survey Employment Change

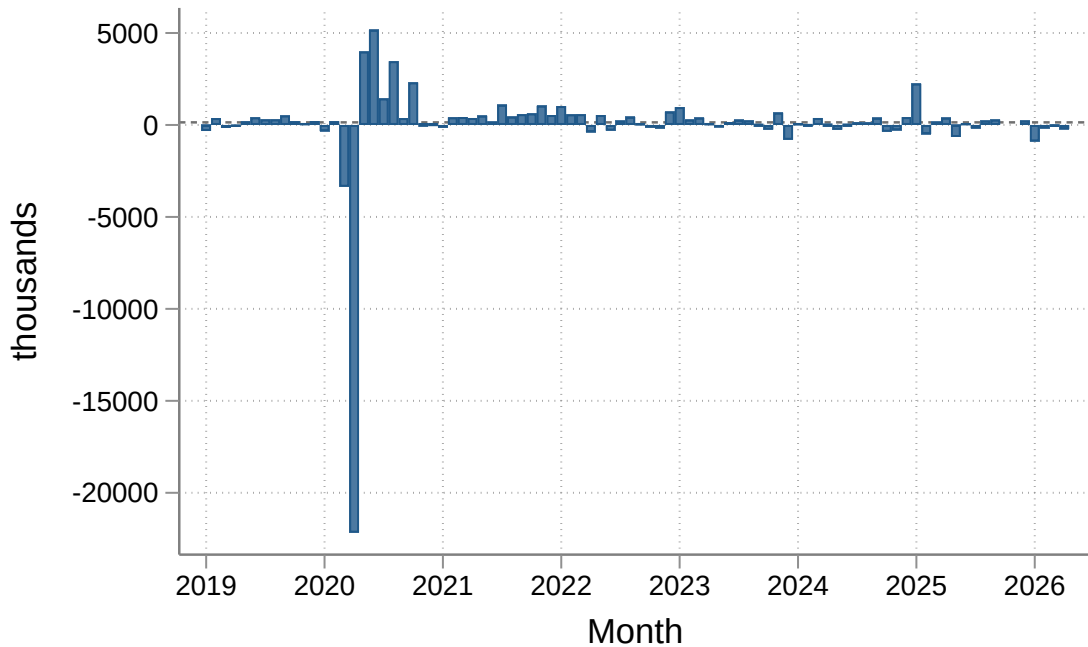


# Monthly Survey Employment Change

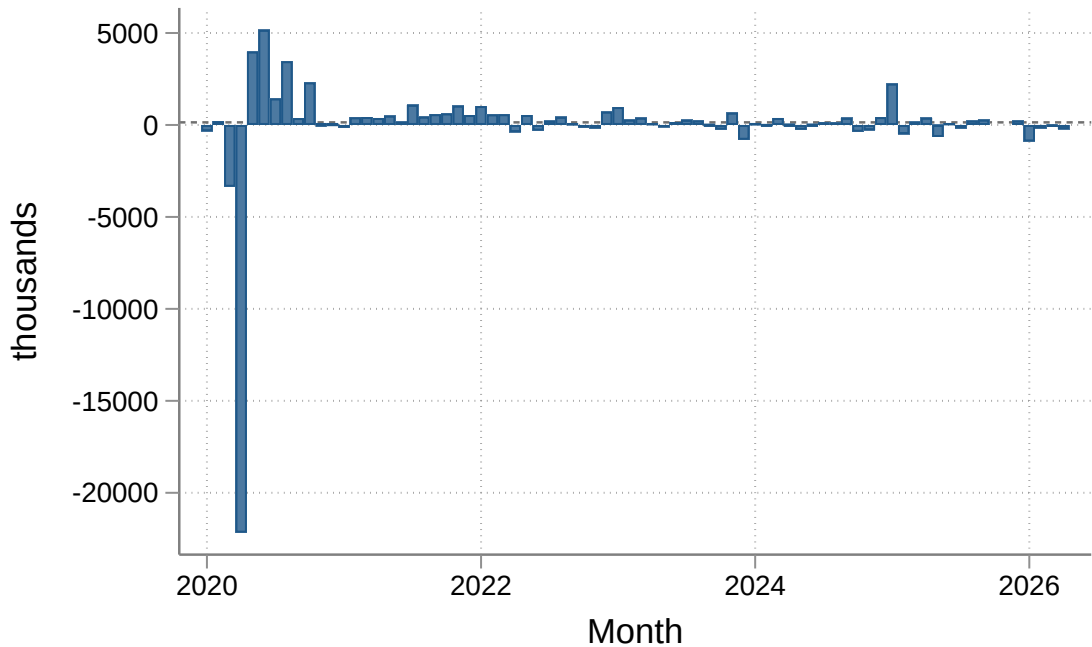




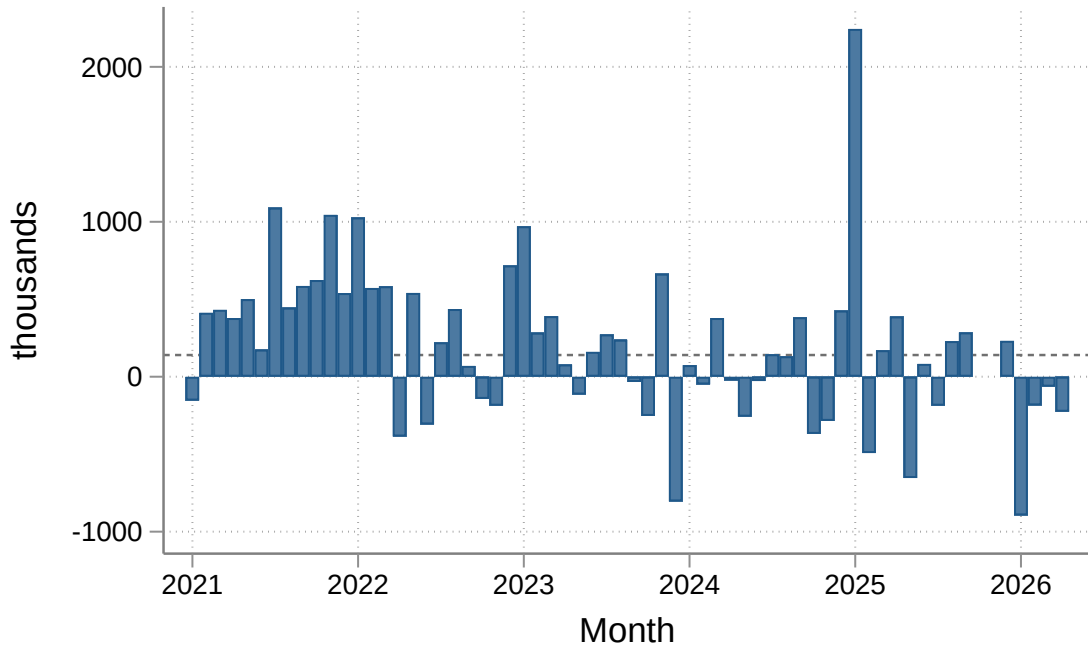
# Monthly Survey Employment Change



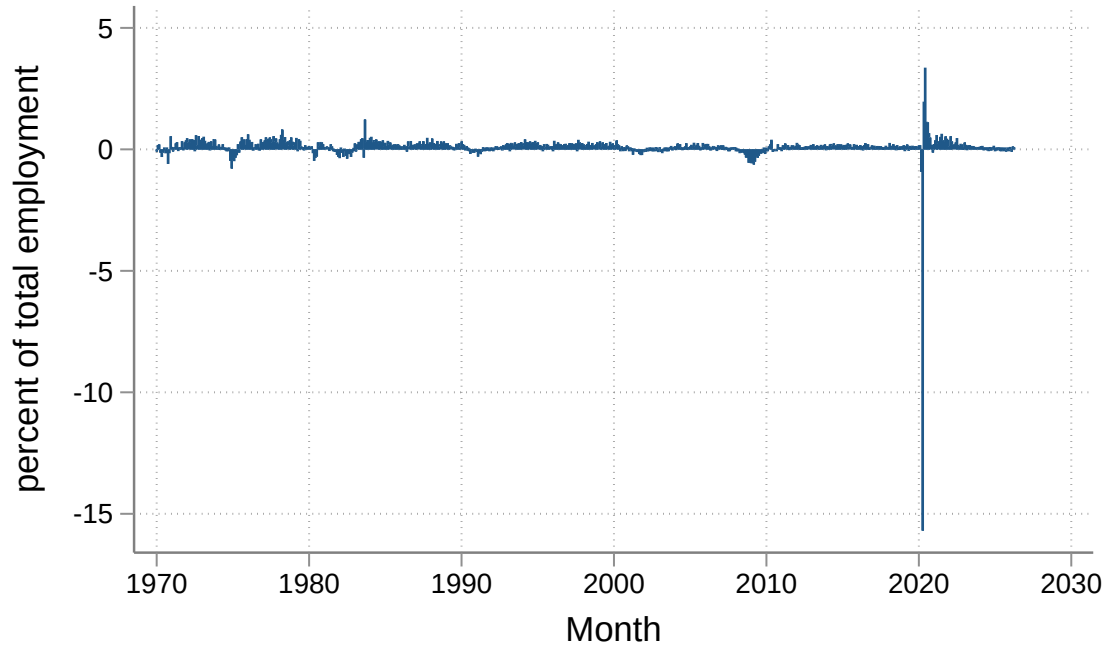
# Monthly Survey Employment Change



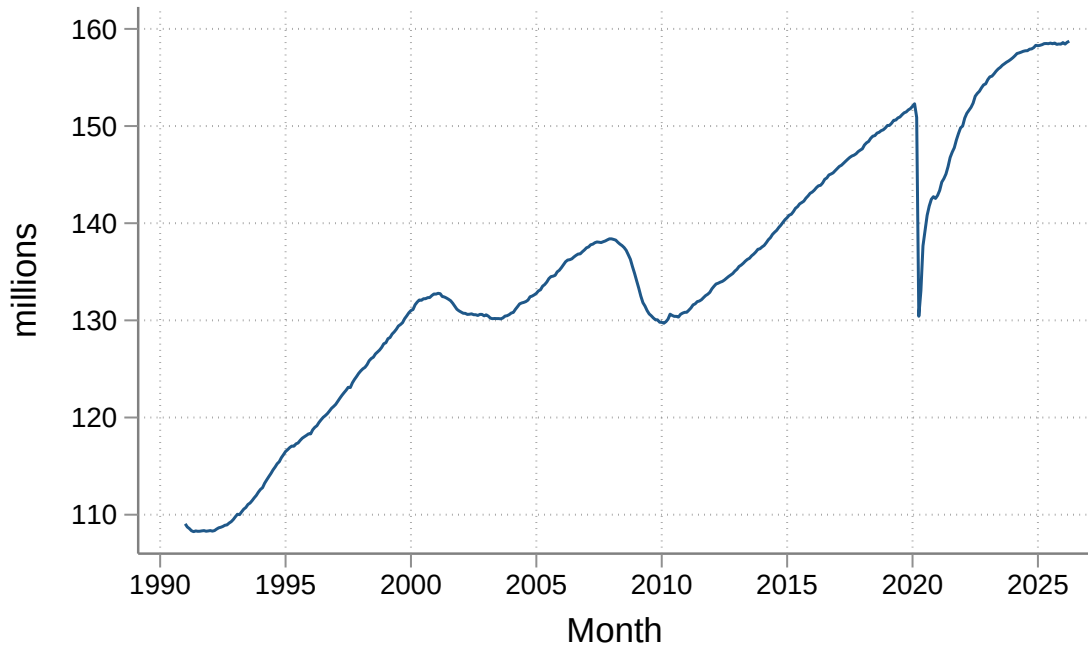
# Monthly Survey Employment Change



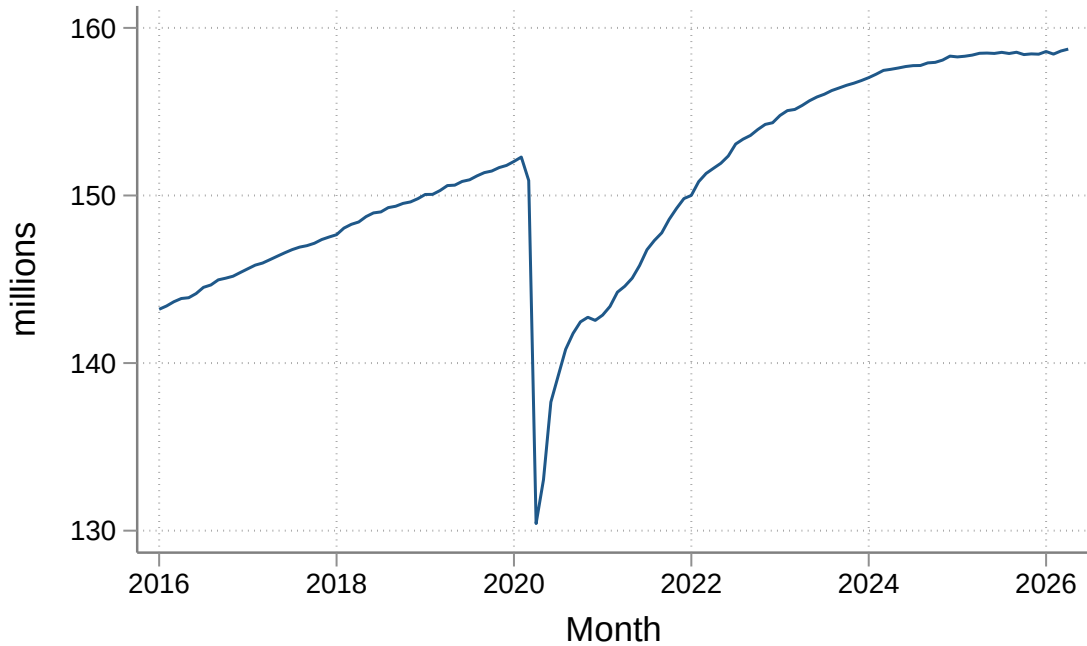
# Monthly Payroll Employment Change



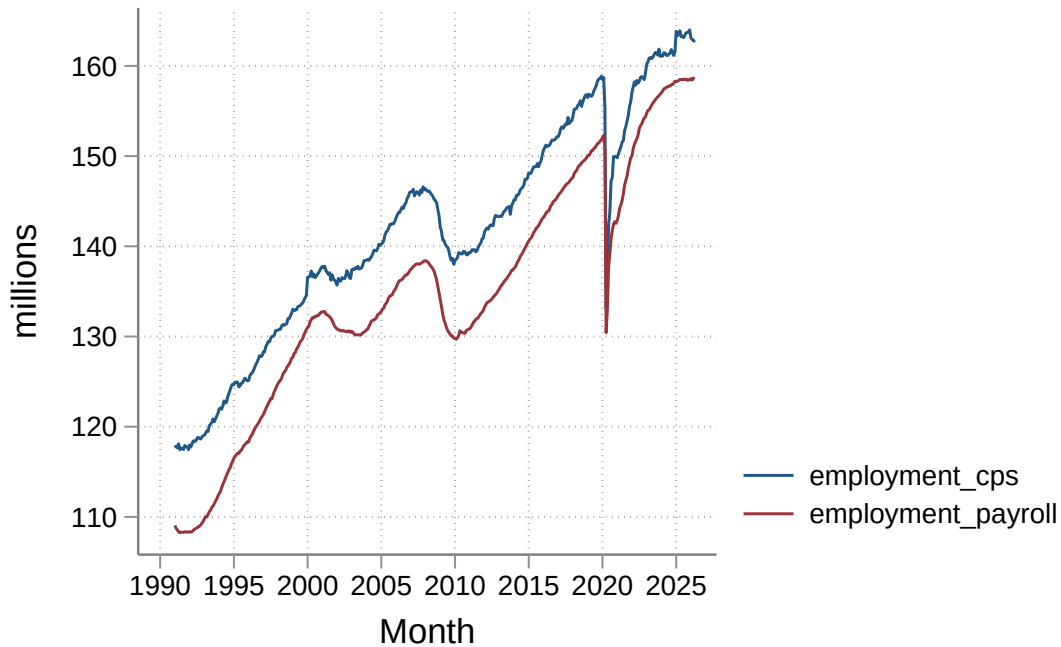
# Number of Employed Persons



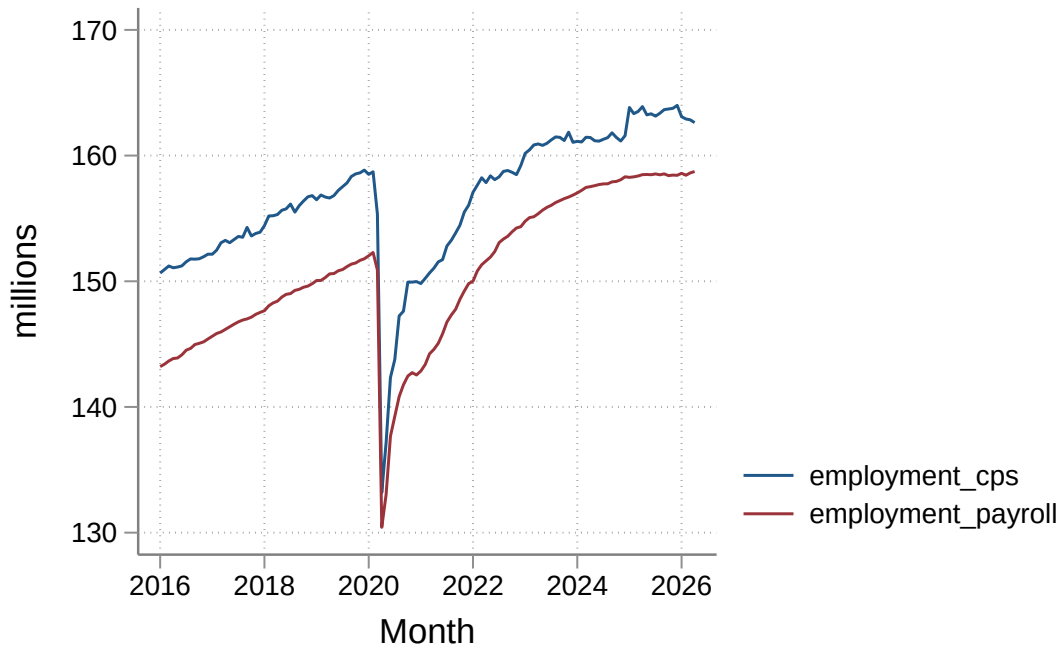
# Number of Employed Persons



## Number of Employed Persons

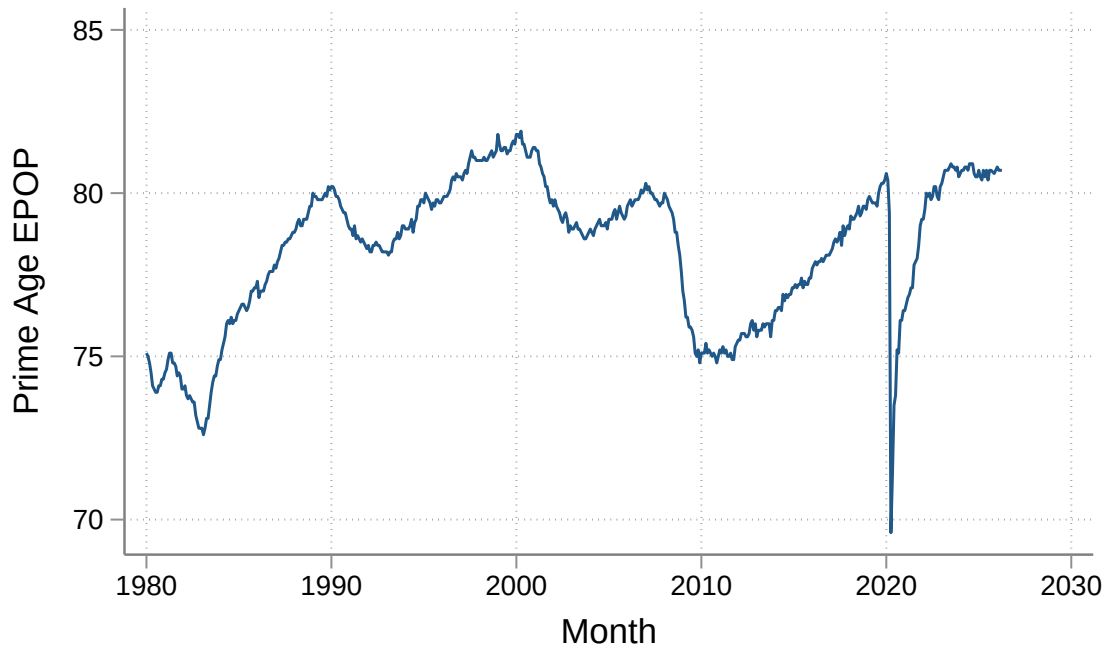


## Number of Employed Persons

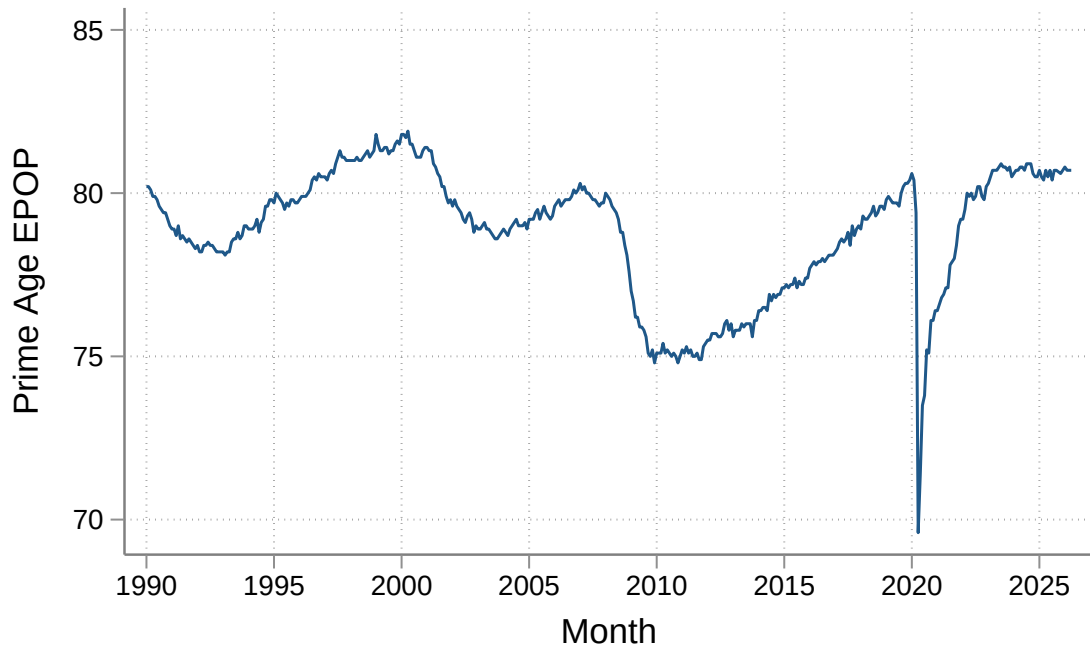




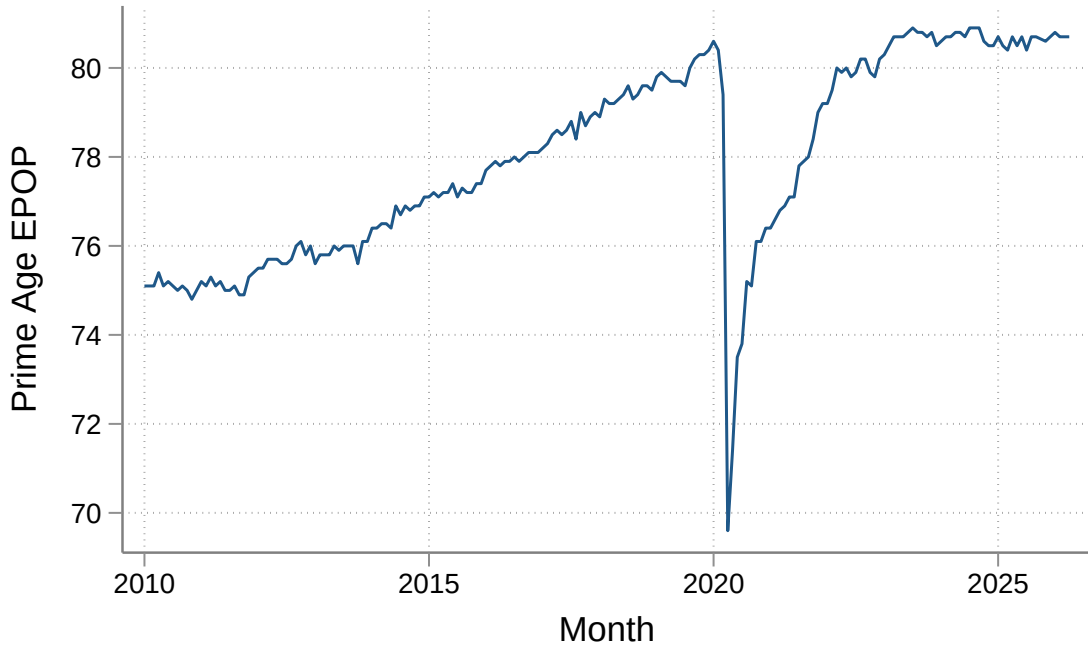
## Prime Age (25-54) EPOP



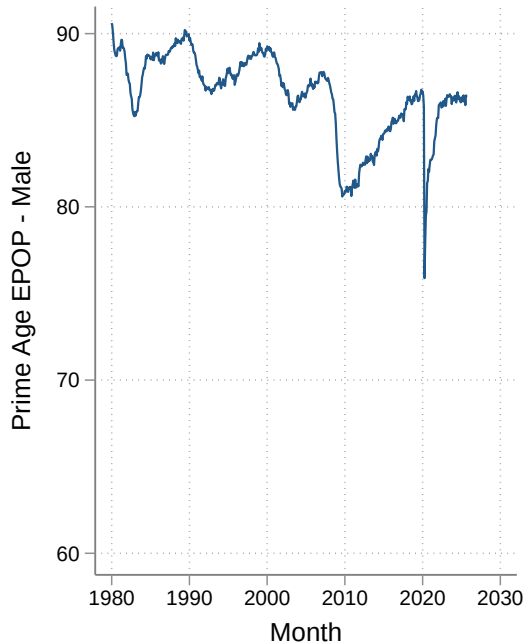
## Prime Age (25-54) EPOP



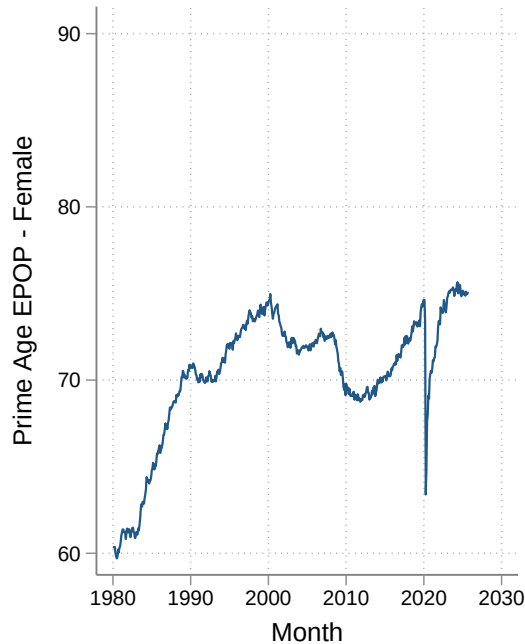
## Prime Age (25-54) EPOP



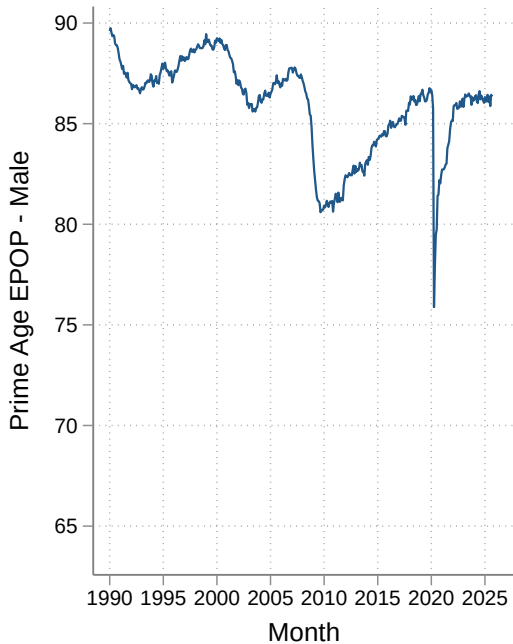
Prime Age (25-54) EPOP



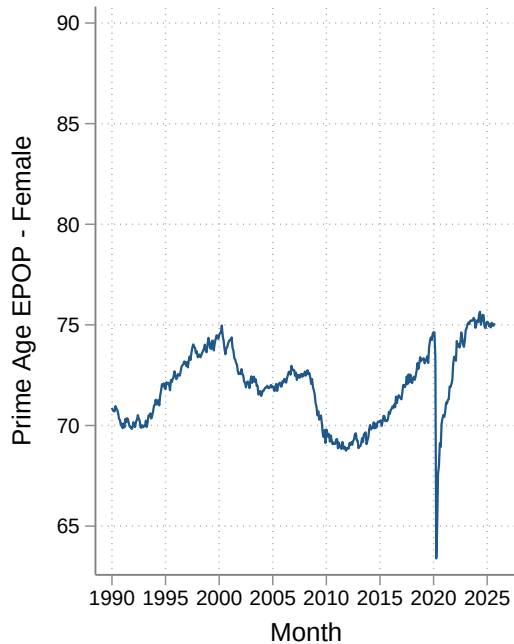
Prime Age (25-54) EPOP



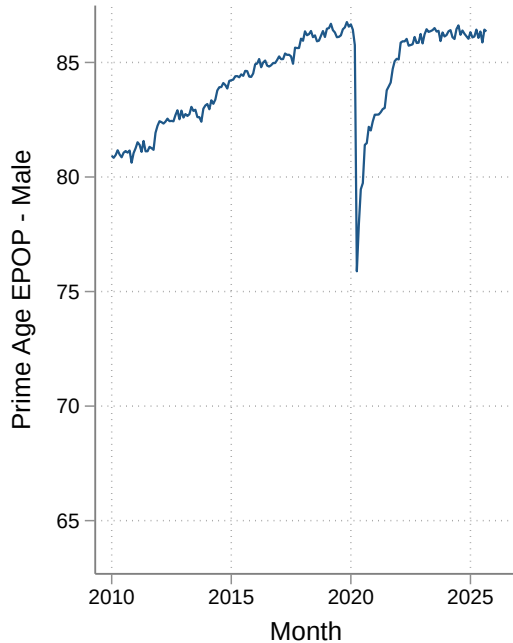
# Prime Age (25-54) EPOP



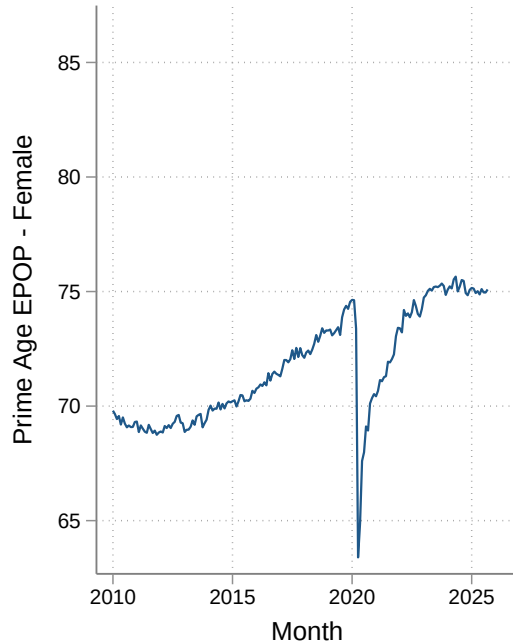
# Prime Age (25-54) EPOP



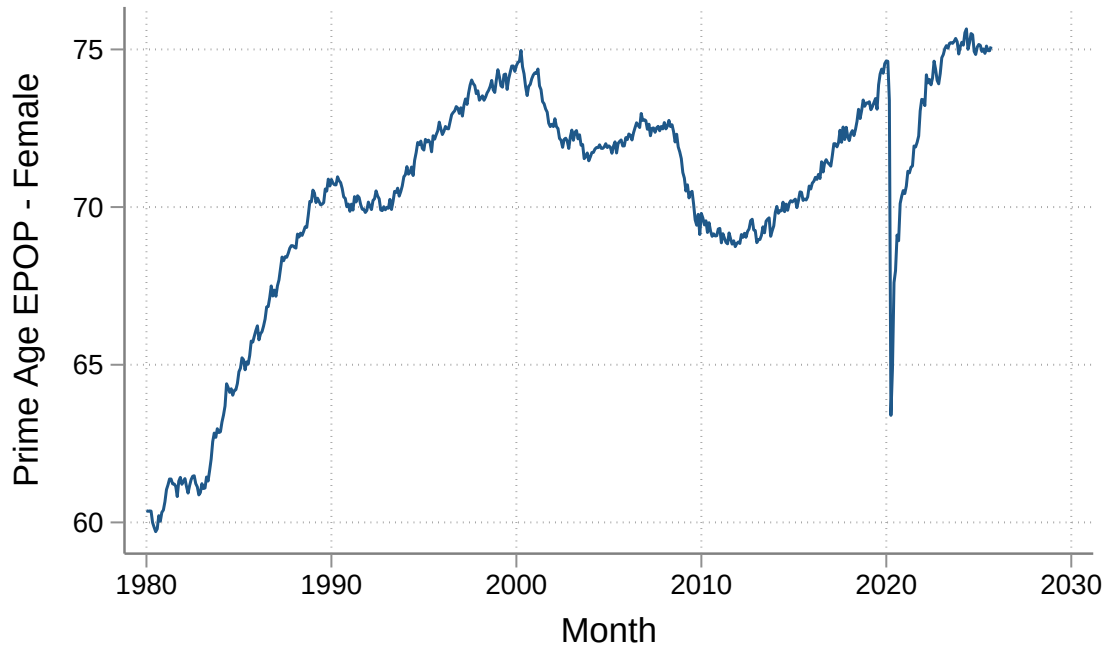
# Prime Age (25-54) EPOP



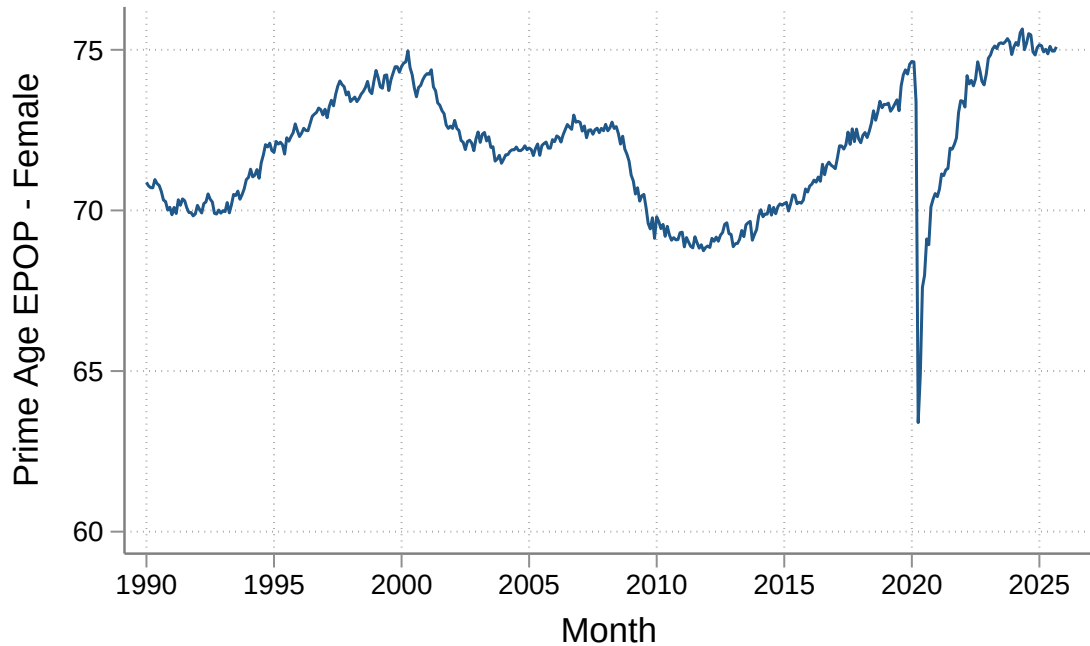
# Prime Age (25-54) EPOP



## Prime Age (25-54) EPOP

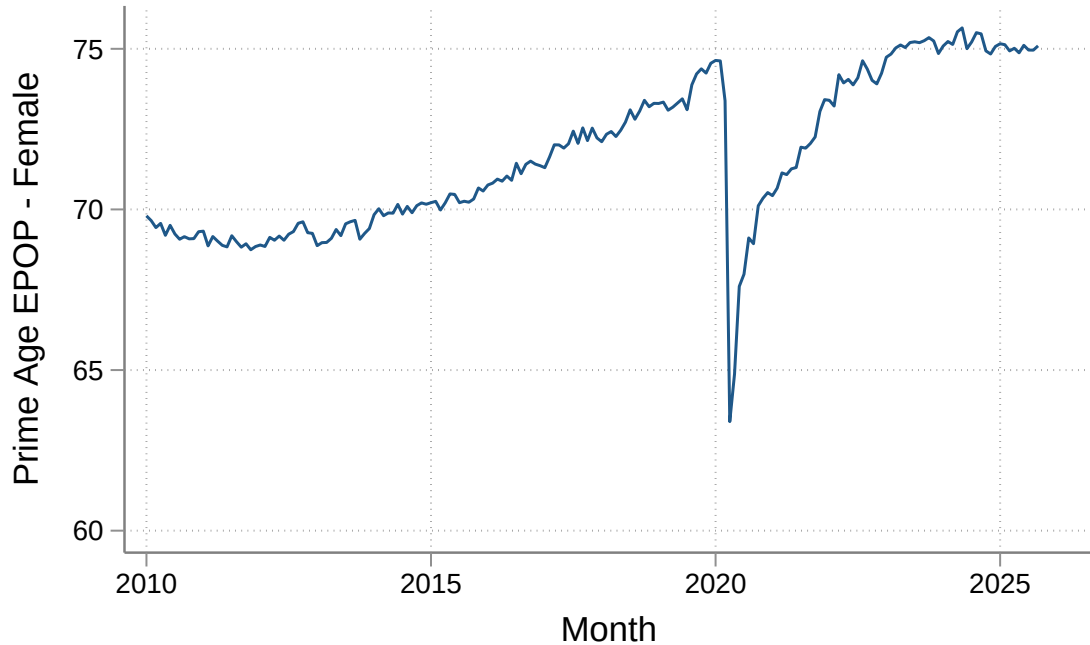


## Prime Age (25-54) EPOP

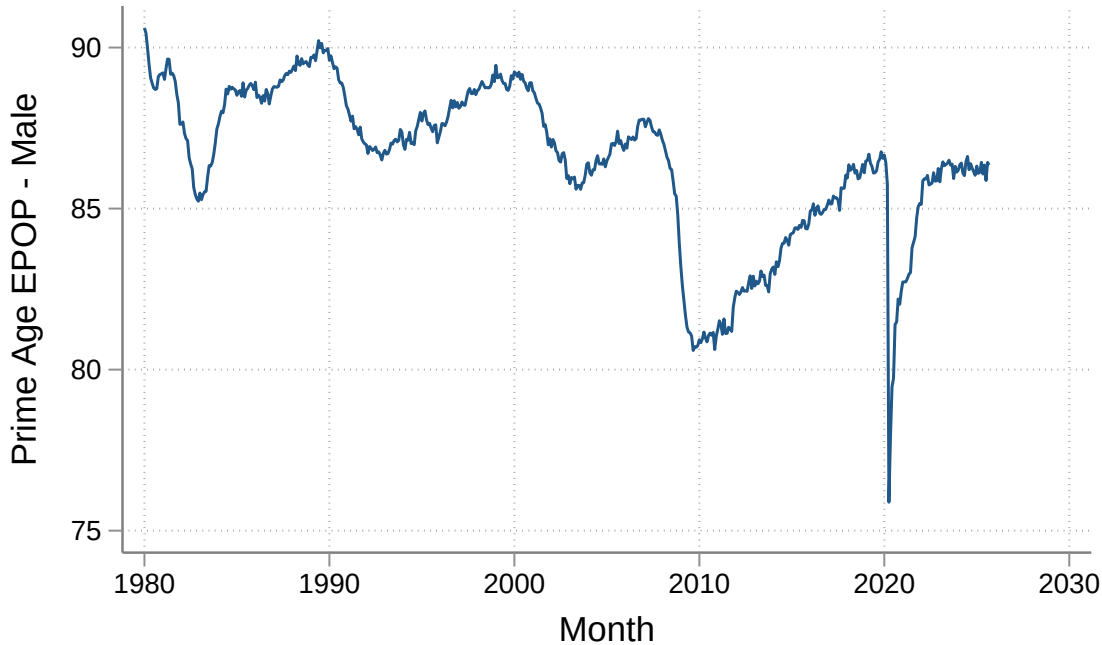




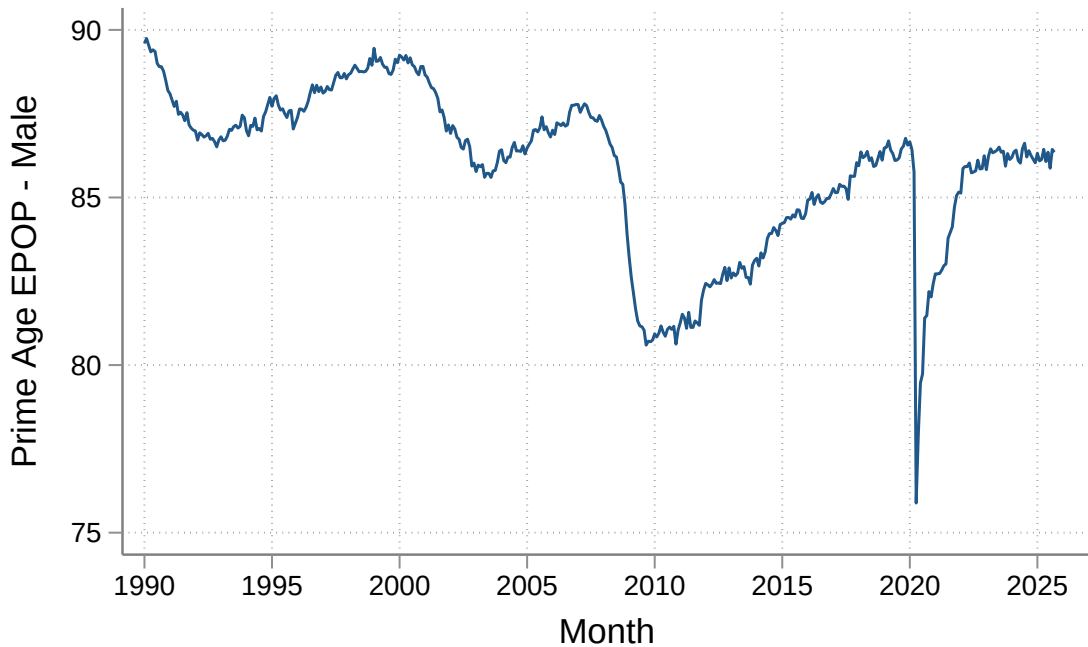
## Prime Age (25-54) EPOP



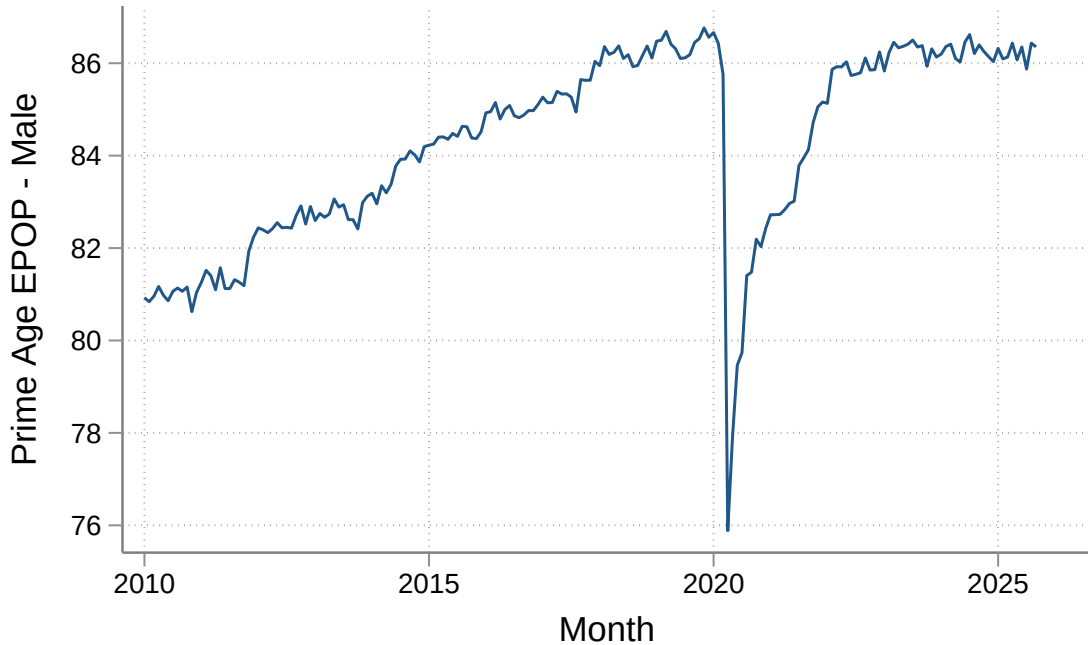
## Prime Age (25-54) EPOP



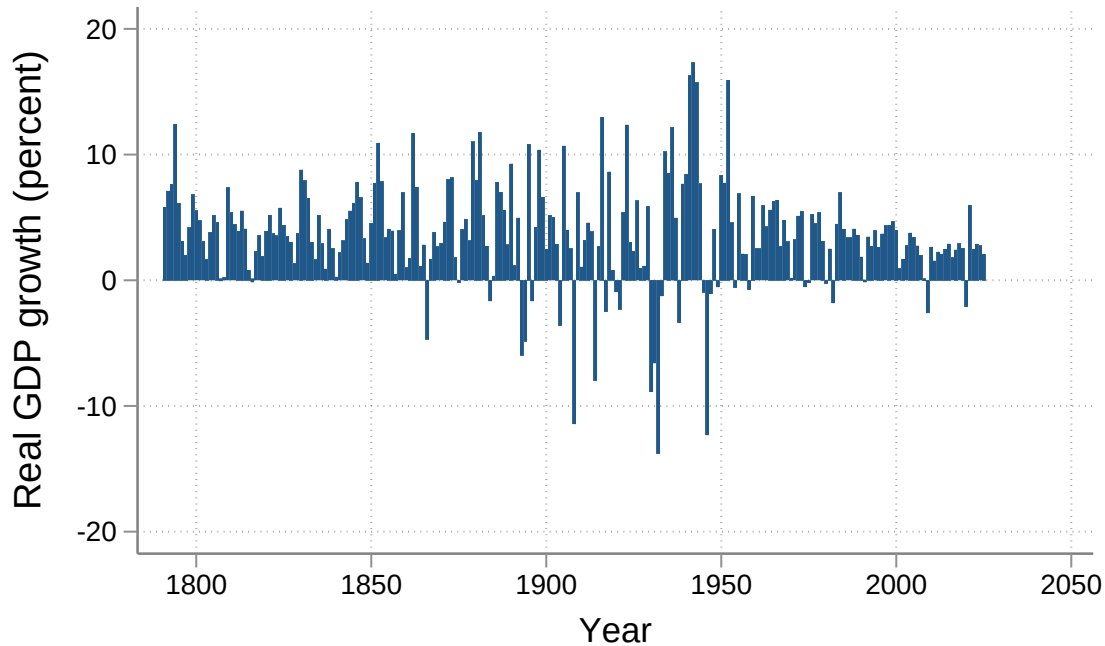
## Prime Age (25-54) EPOP



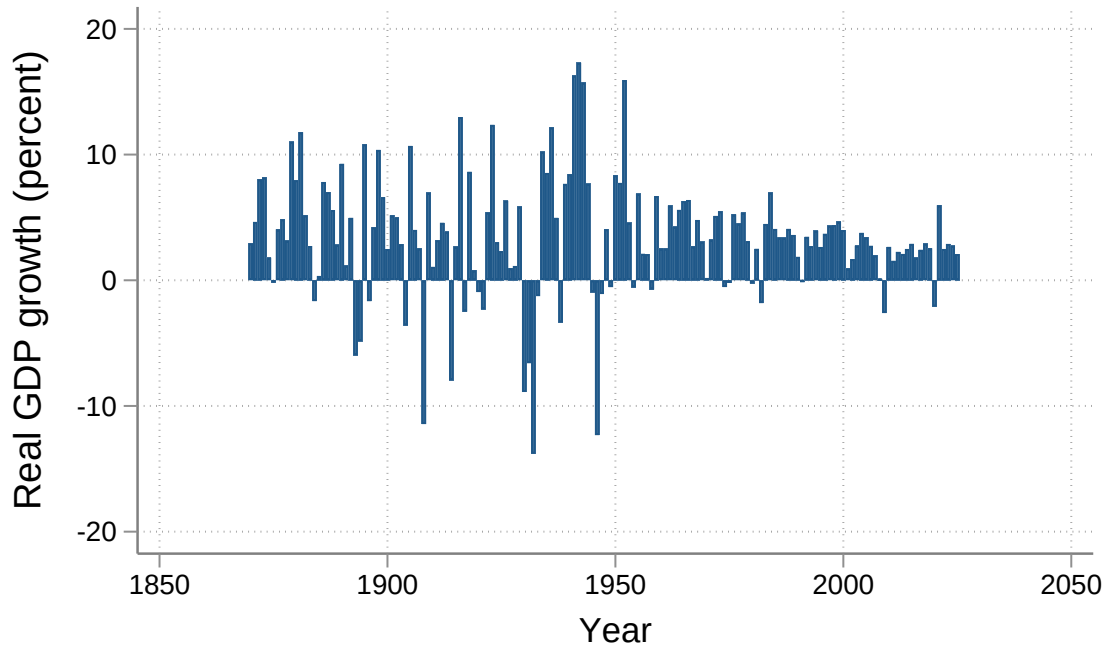
## Prime Age (25-54) EPOP



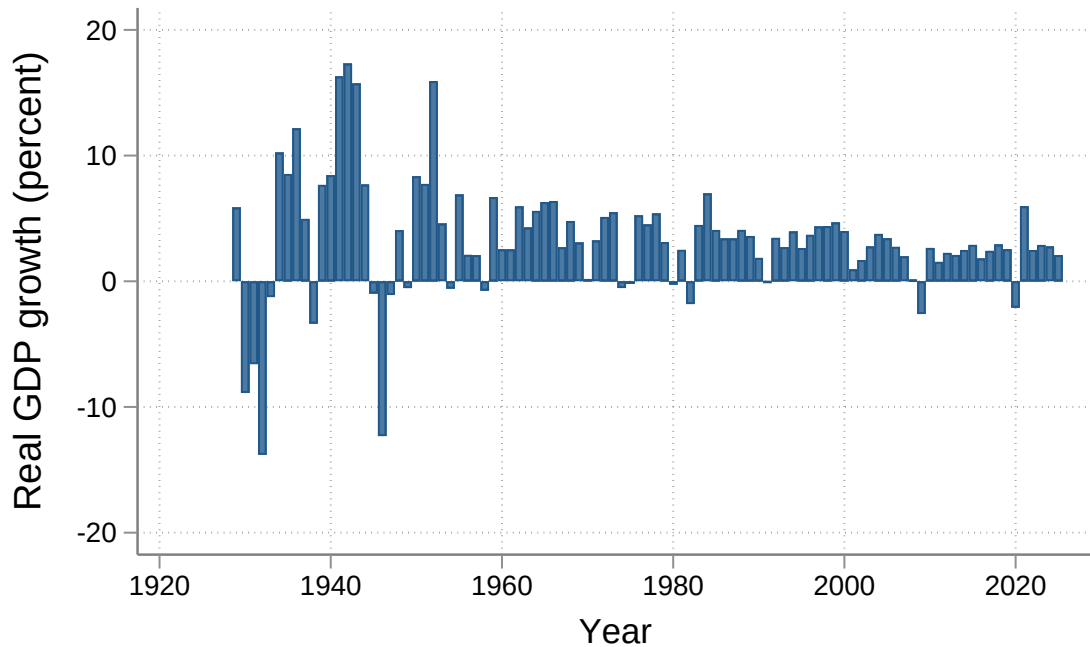
## US GDP growth



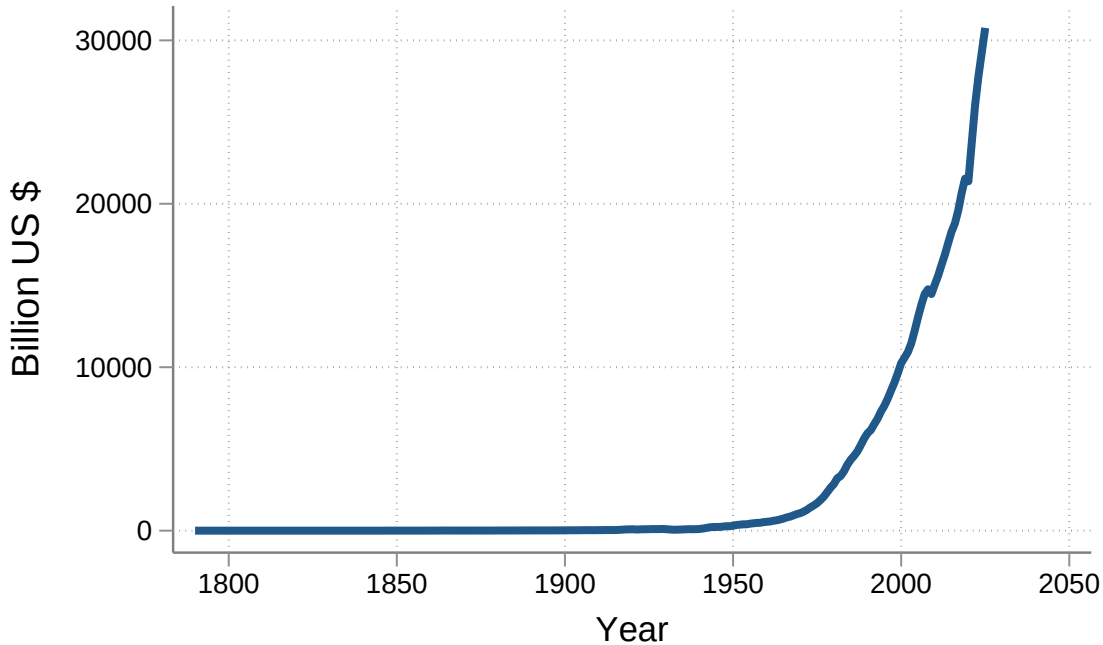
## US GDP growth



## US GDP growth

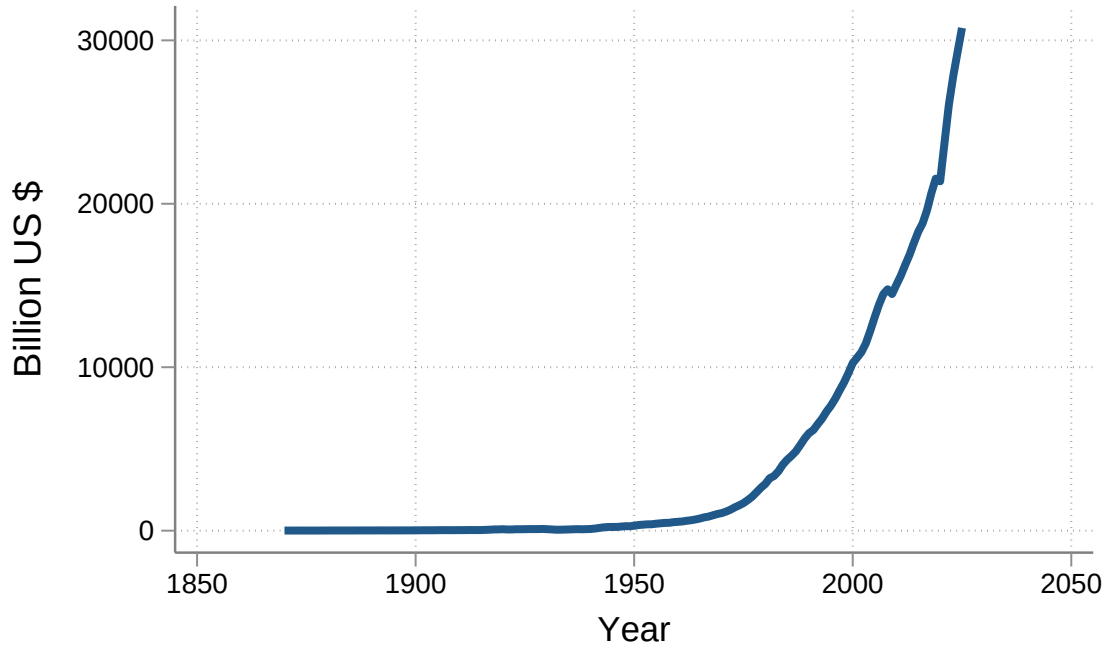


# Aggregate US GDP

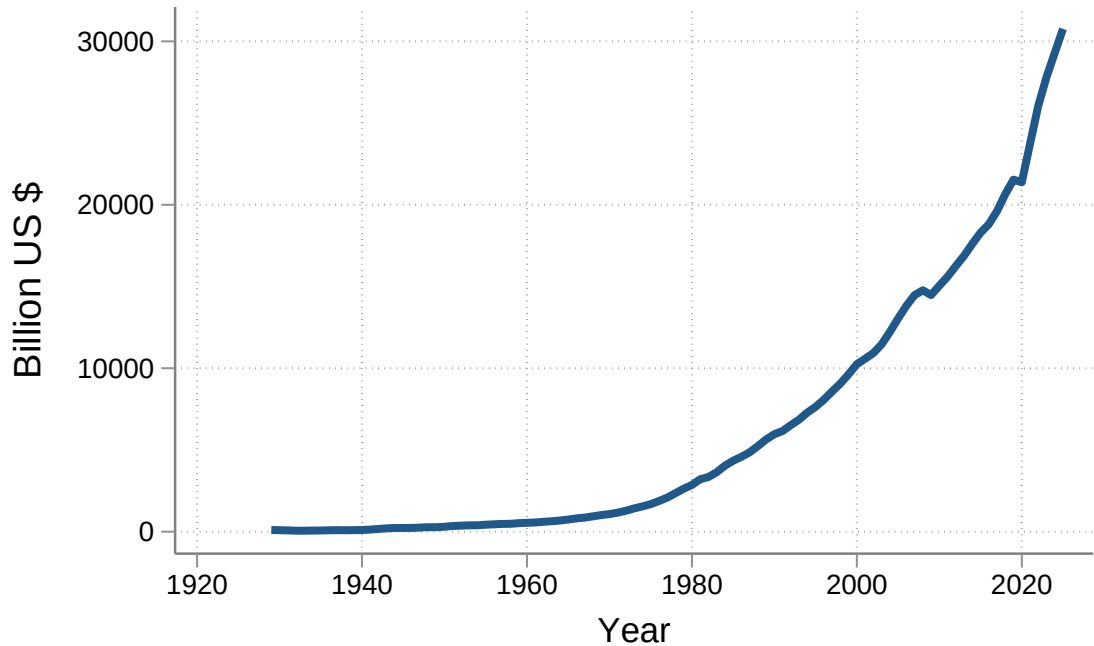




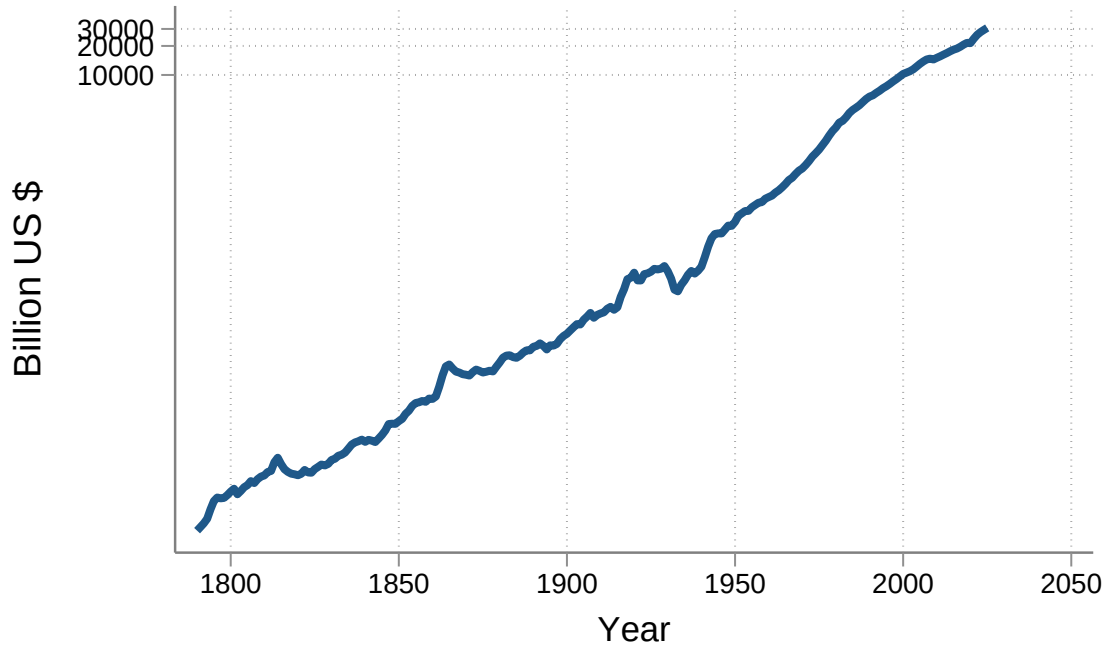
## Aggregate US GDP



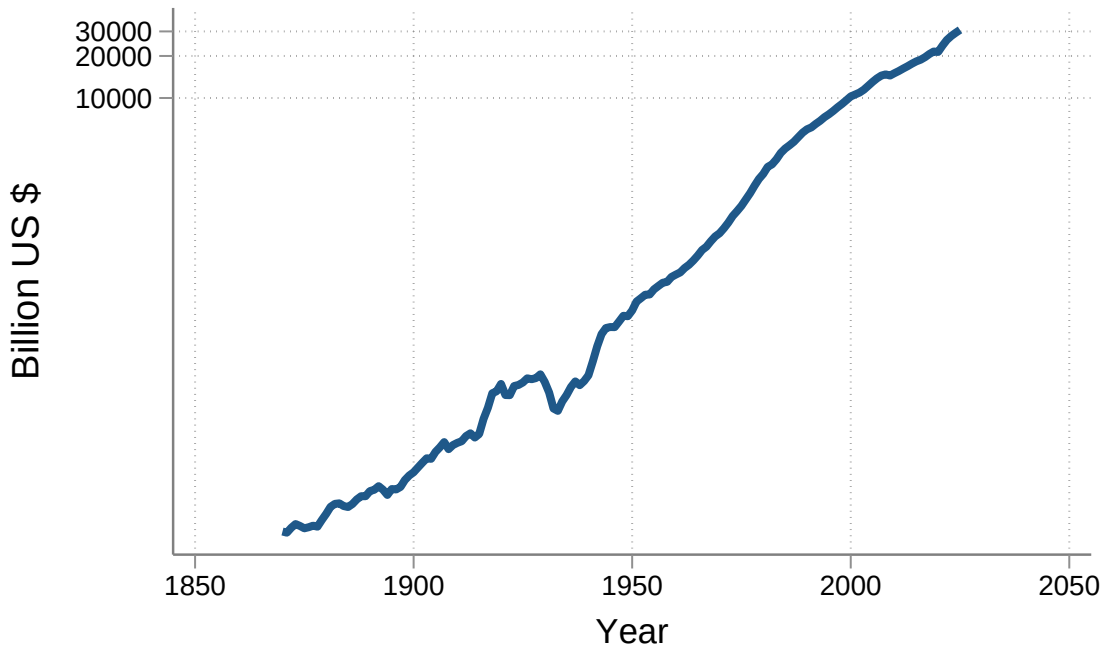
# Aggregate US GDP



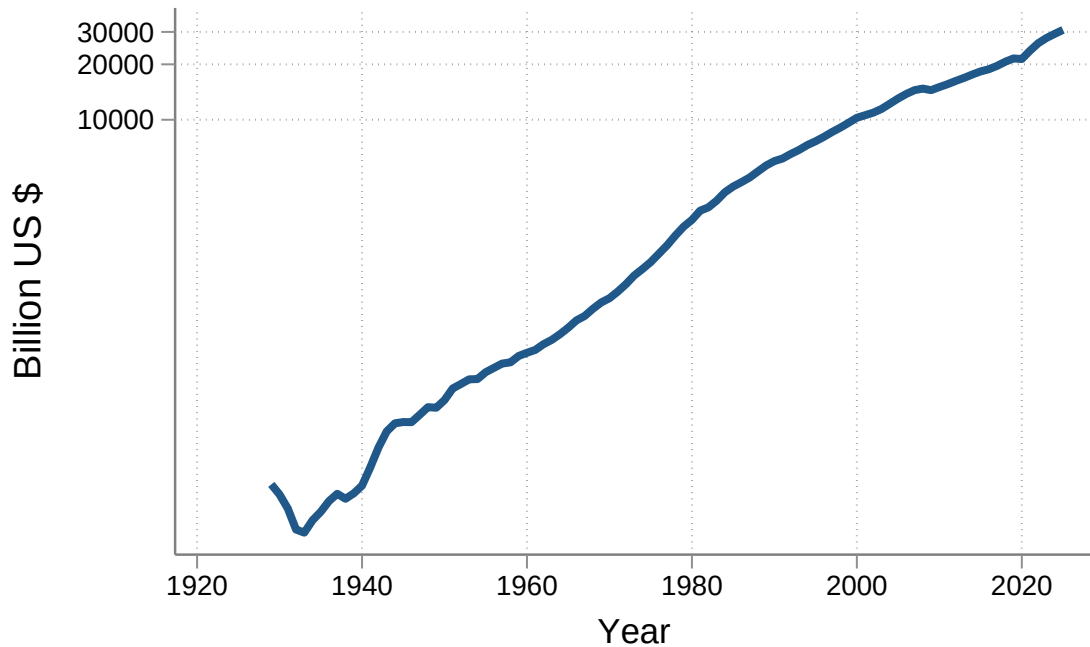
# Aggregate US GDP



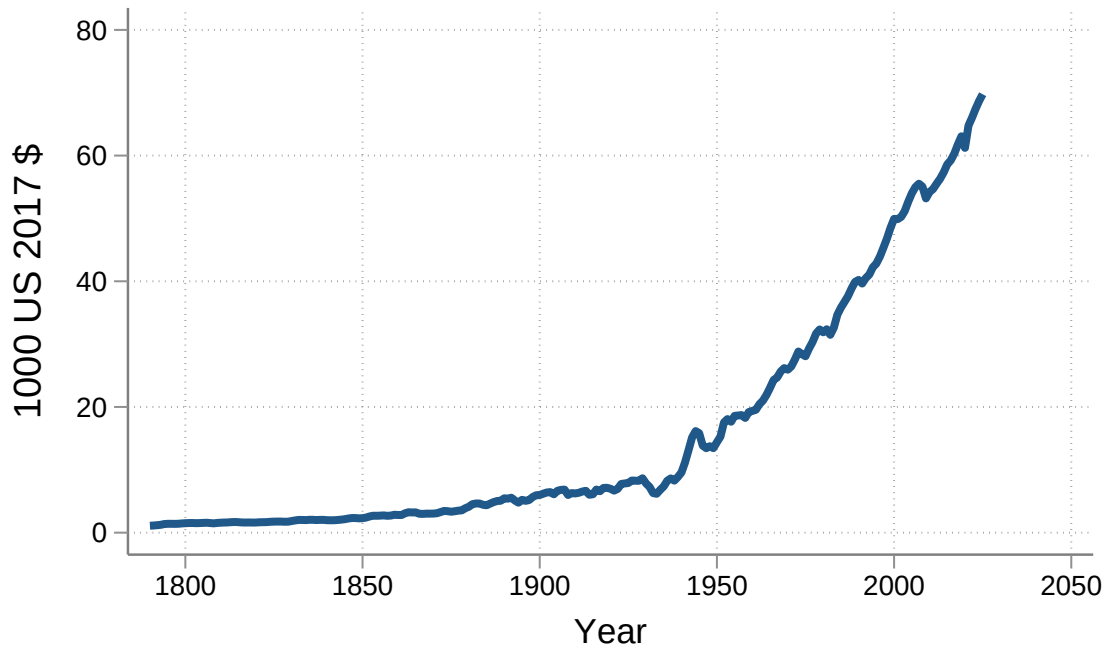
# Aggregate US GDP



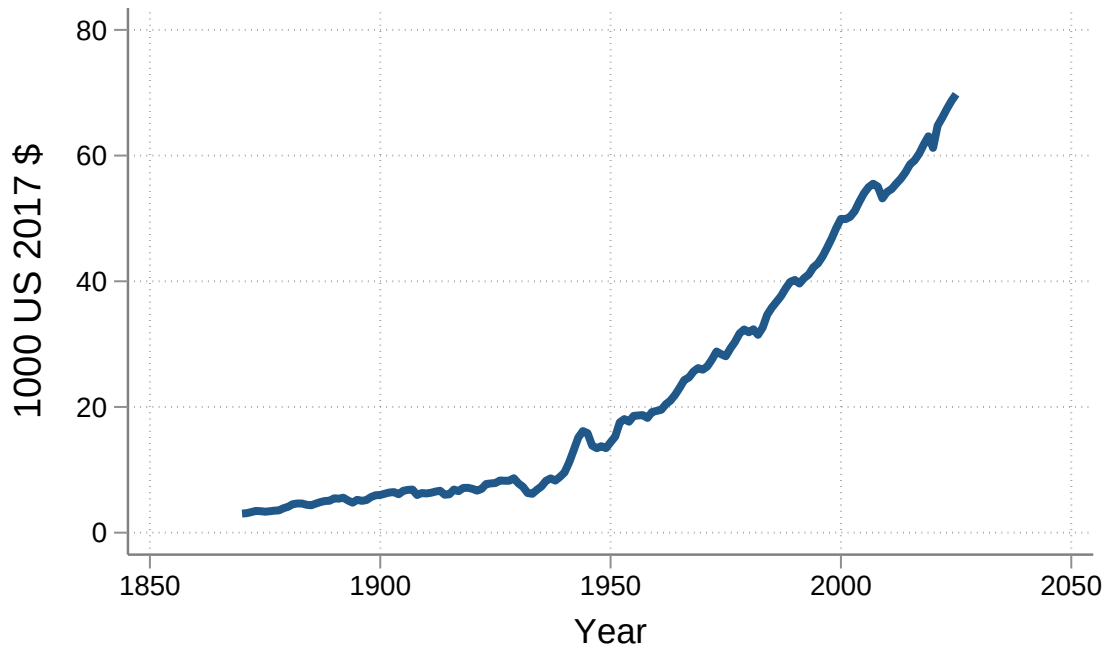
## Aggregate US GDP



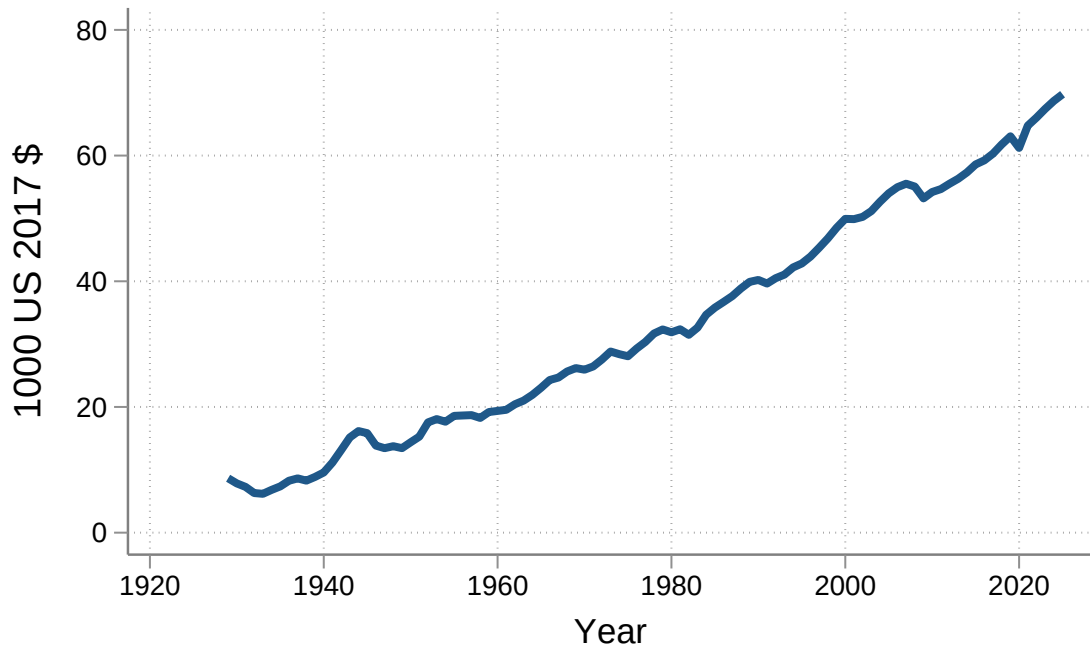
# Real US GDP per capita



Real US GDP per capita

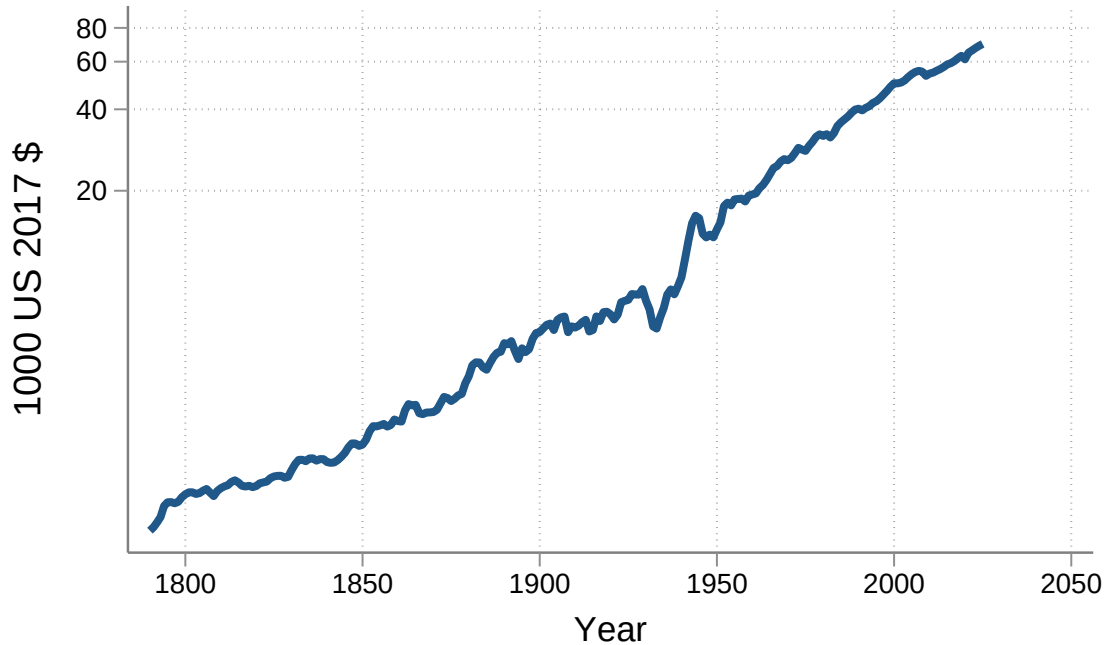


# Real US GDP per capita

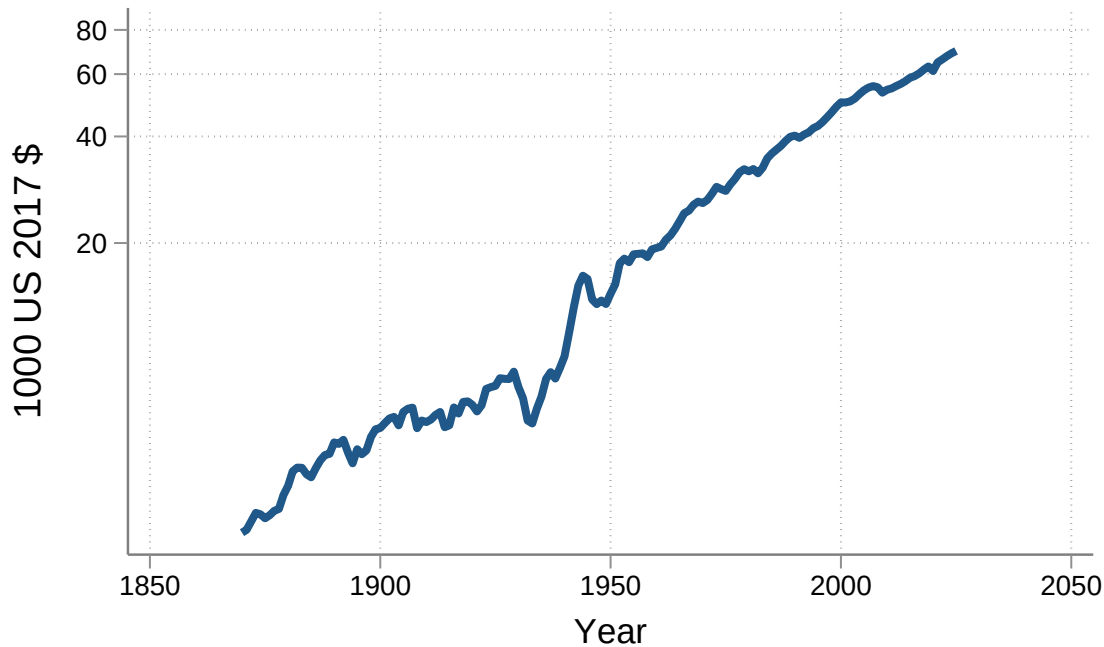




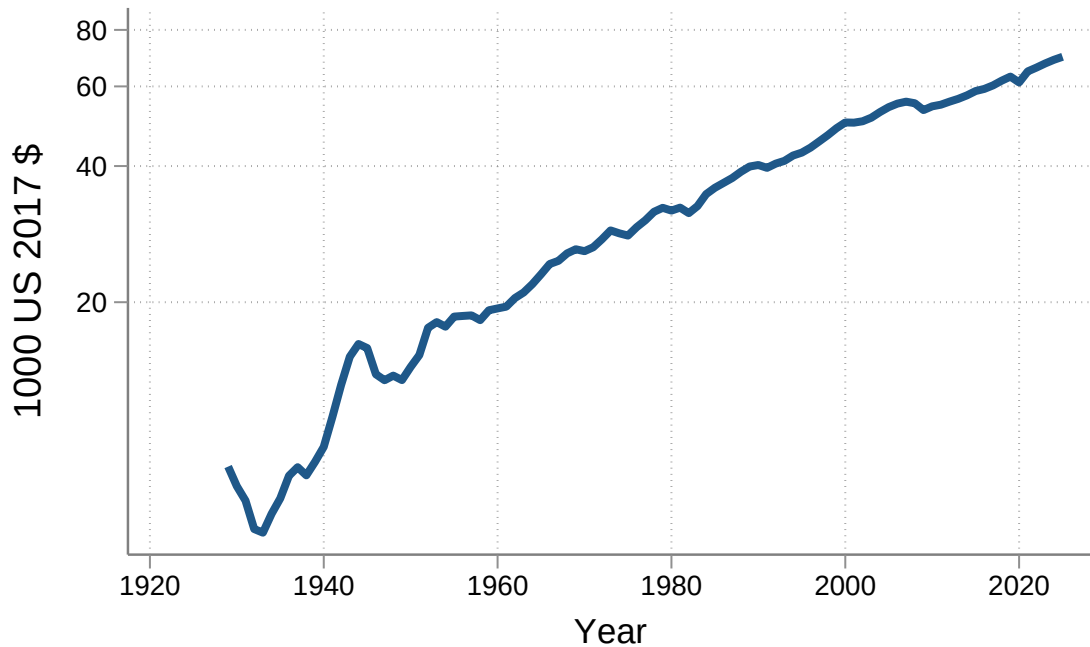
## Real US GDP per capita



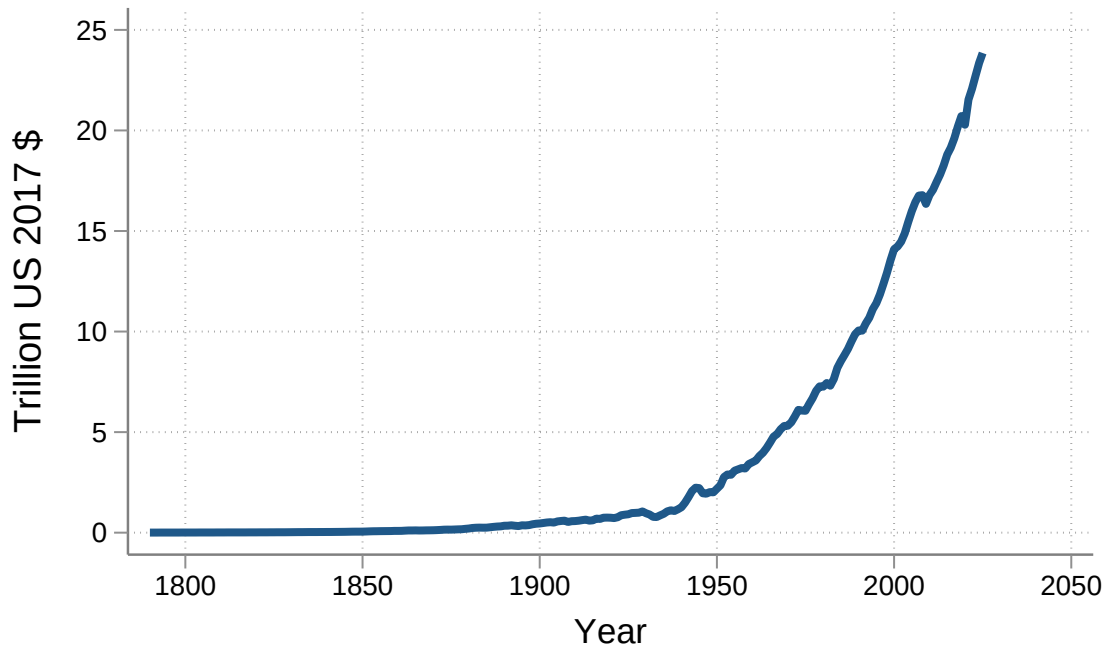
Real US GDP per capita



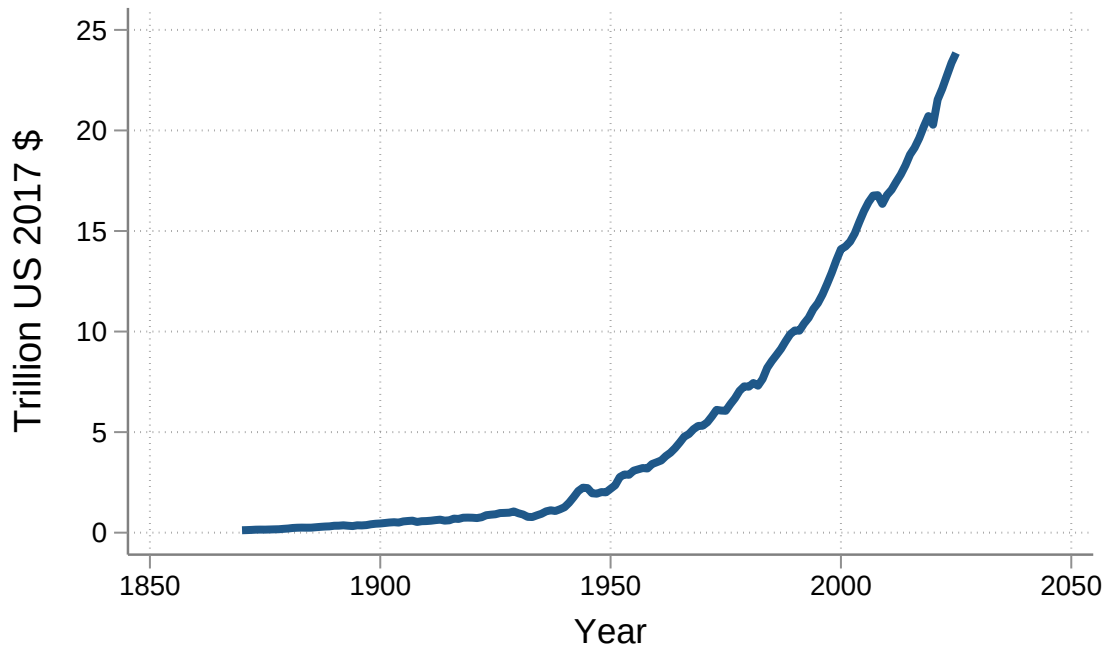
Real US GDP per capita



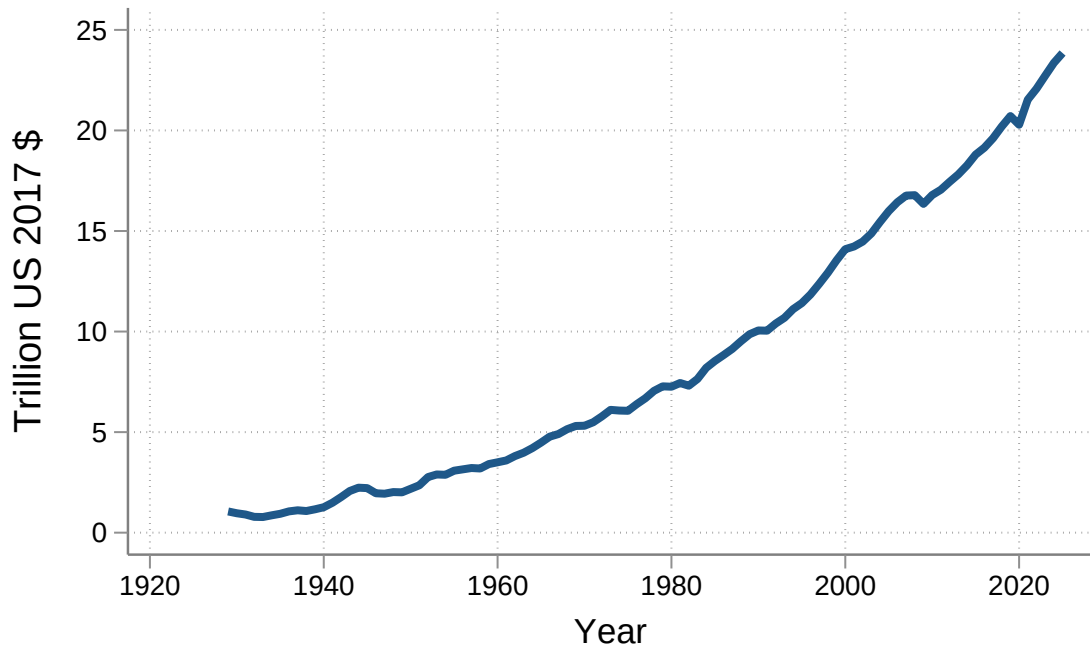
## Aggregate real US GDP



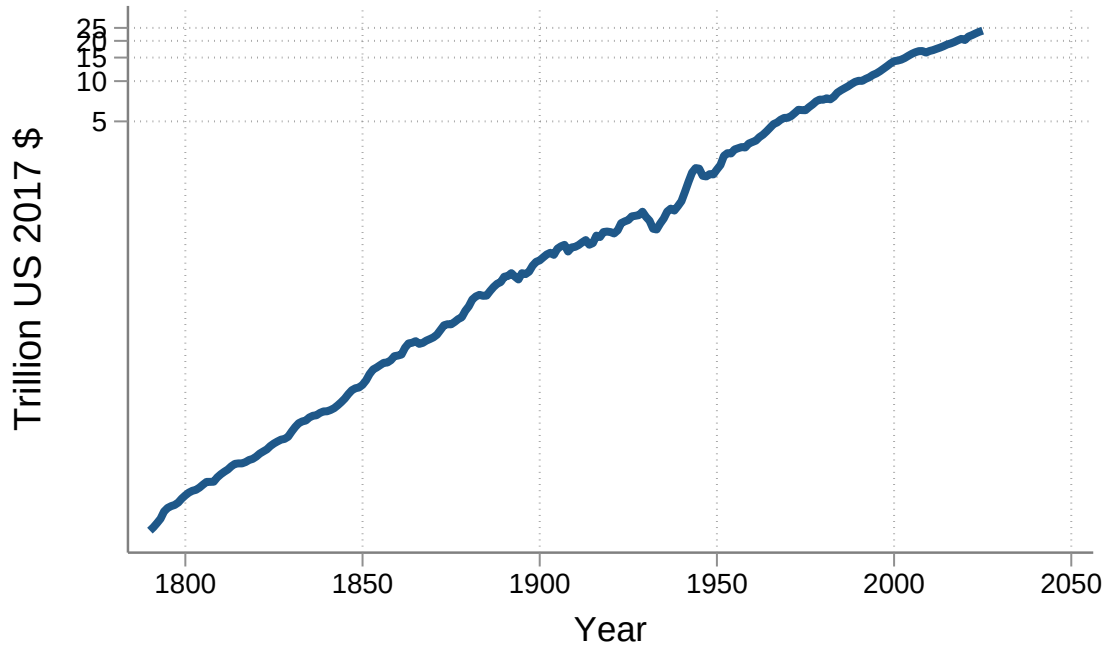
## Aggregate real US GDP



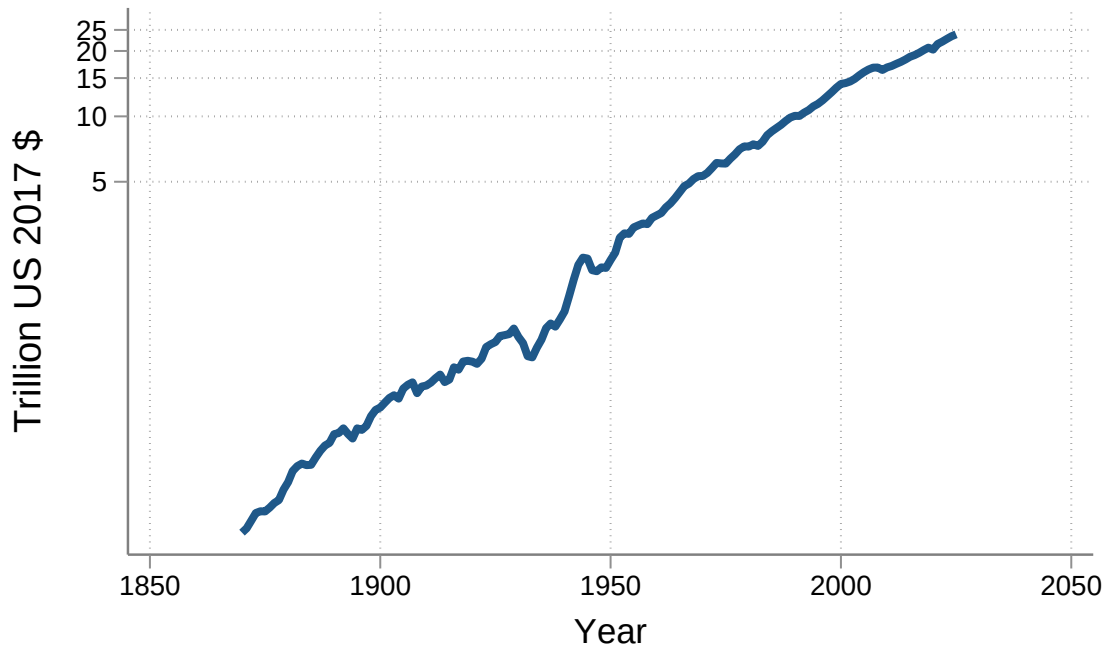
## Aggregate real US GDP



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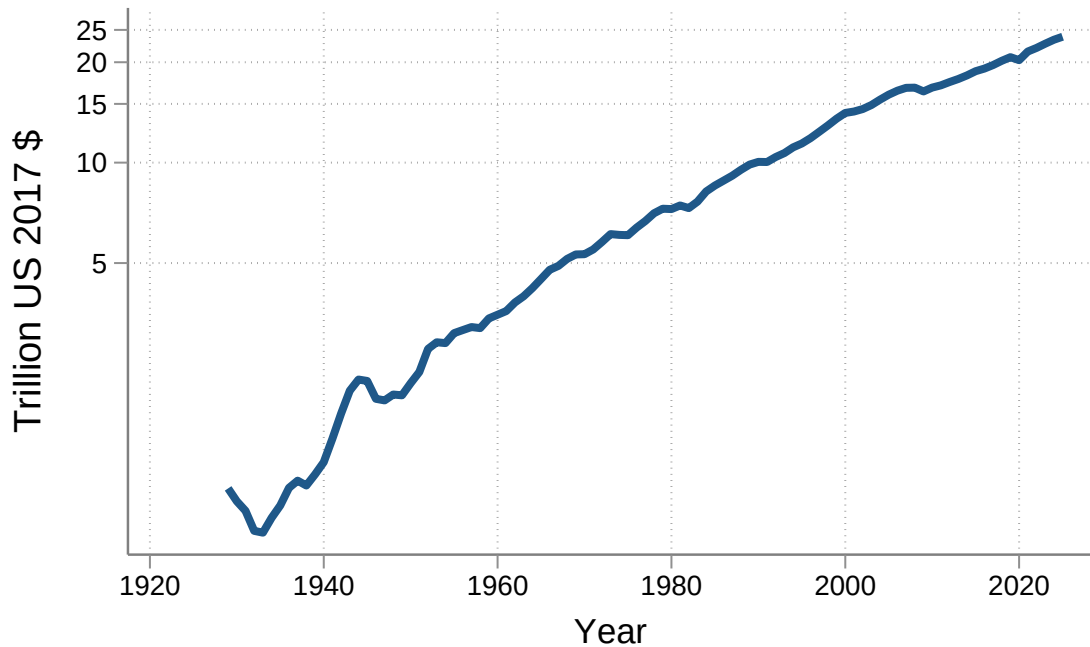


## Aggregate real US GDP

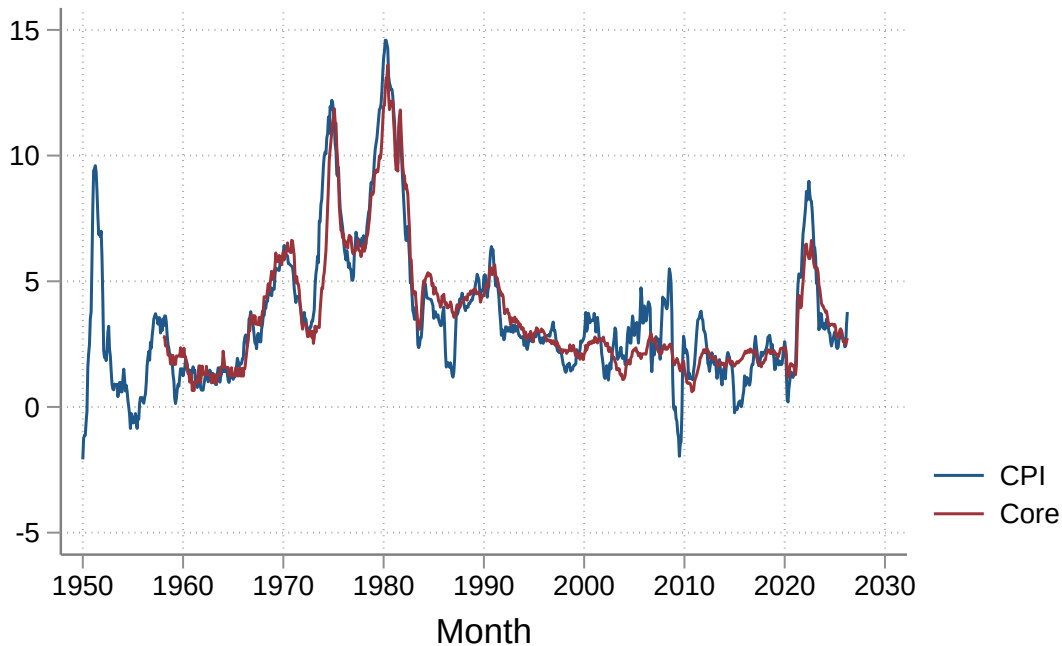




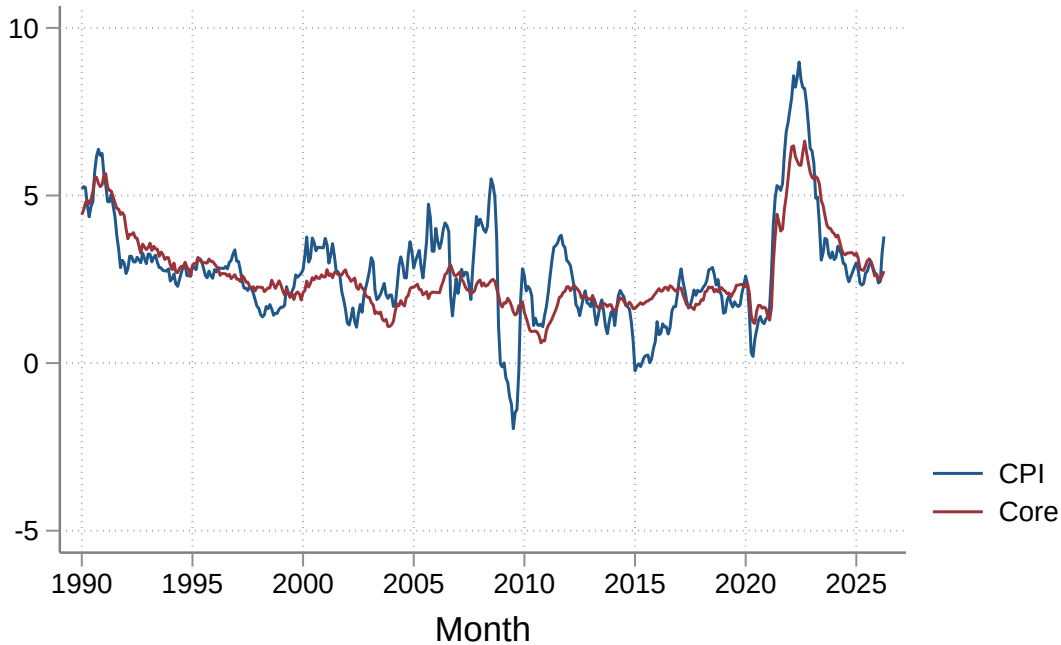
## Aggregate real US GDP



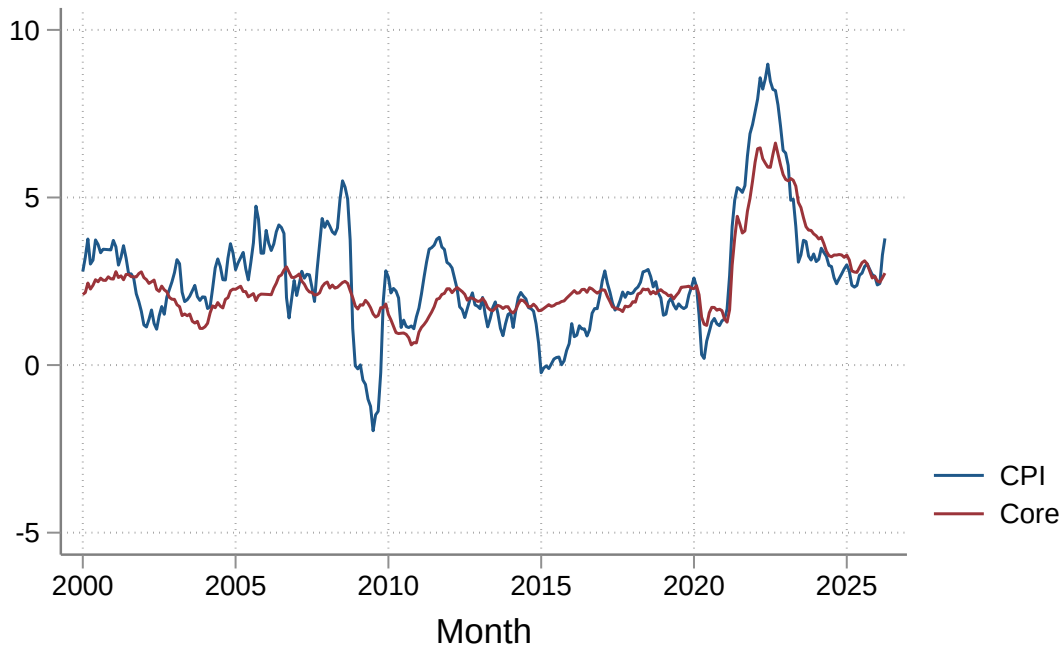
# US Inflation - 12 month price change



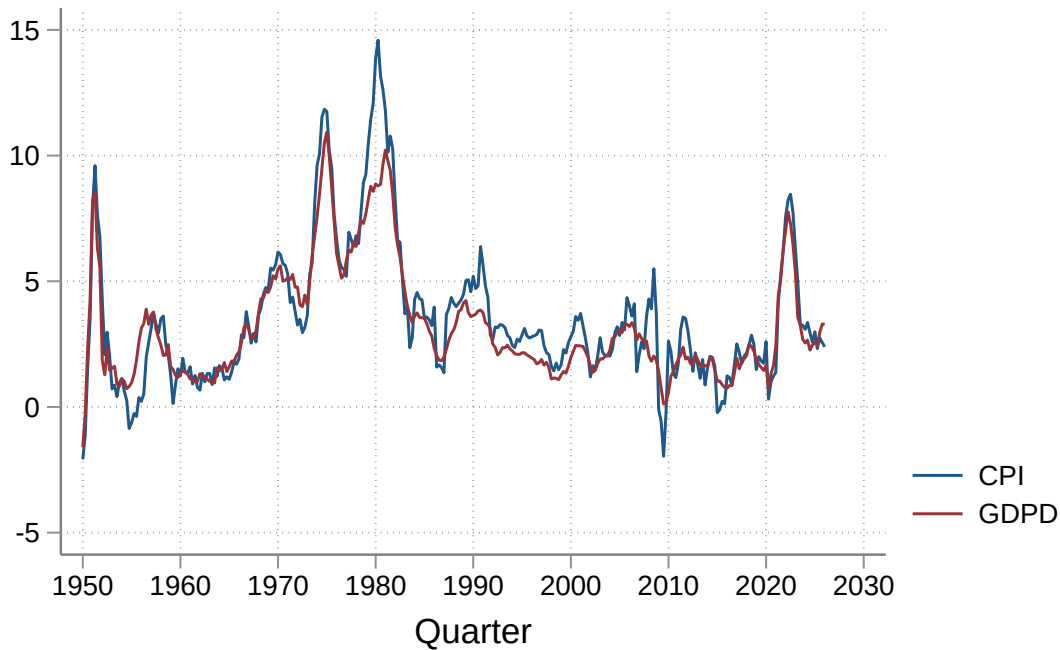
## US Inflation - 12 month price change



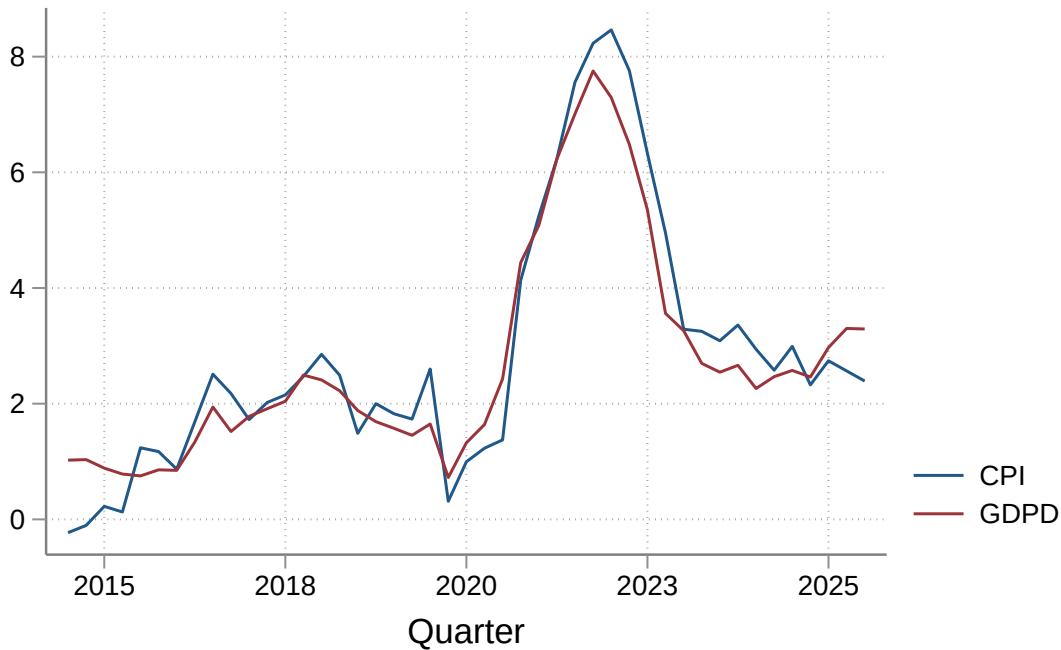
# US Inflation - 12 month price change



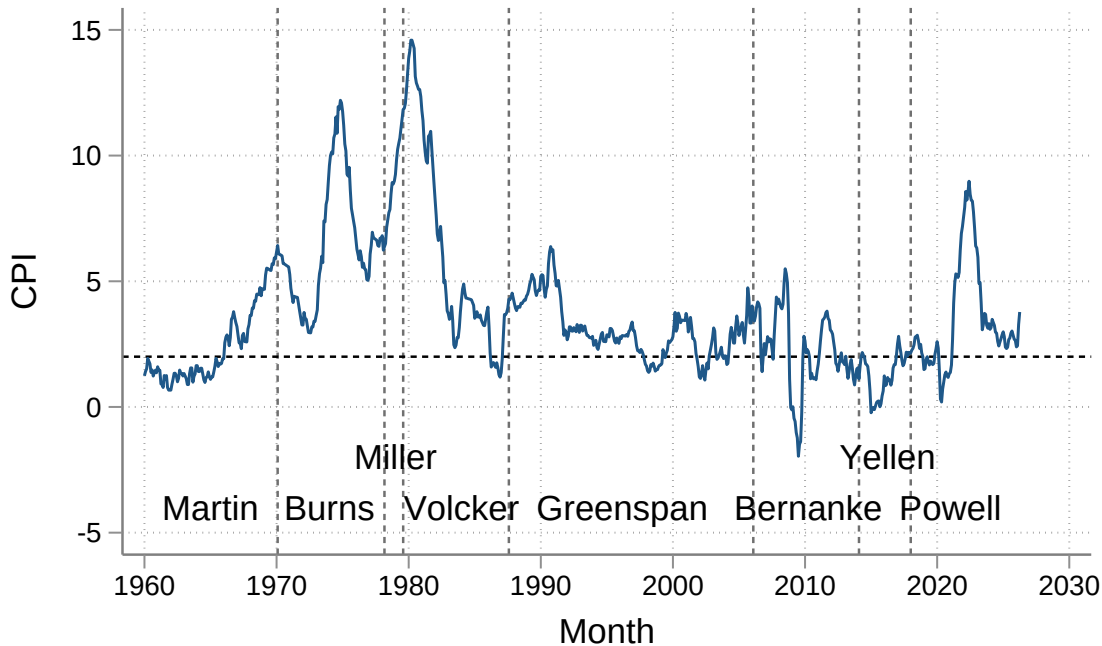
# US Inflation - 12 month price change



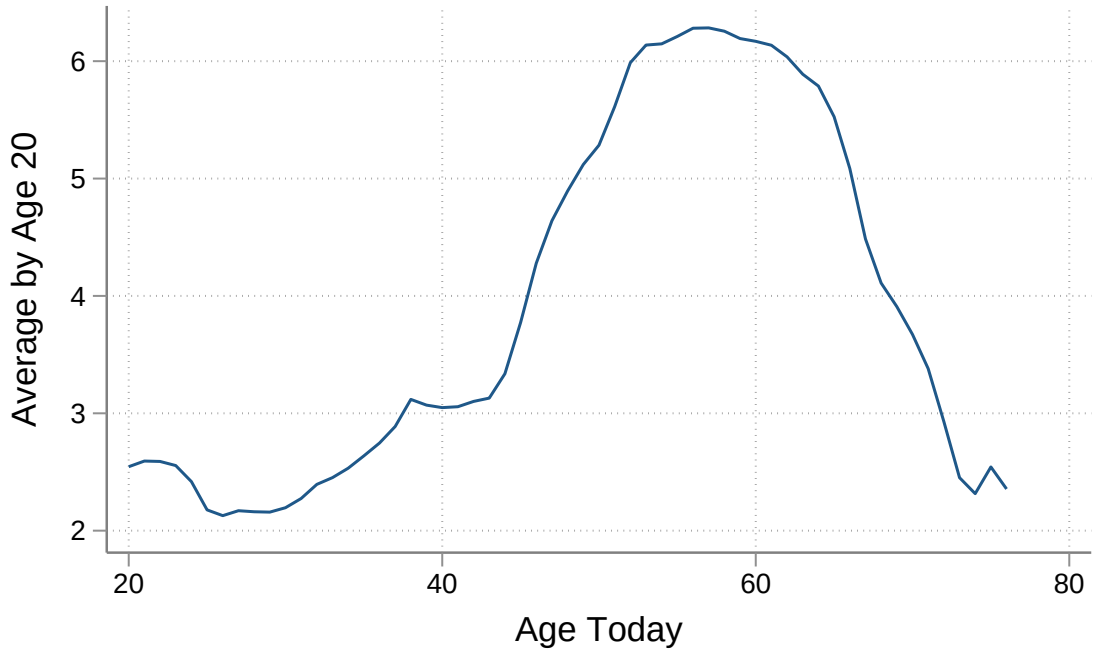
## US Inflation - 12 month price change



# US Inflation - 12 month price change

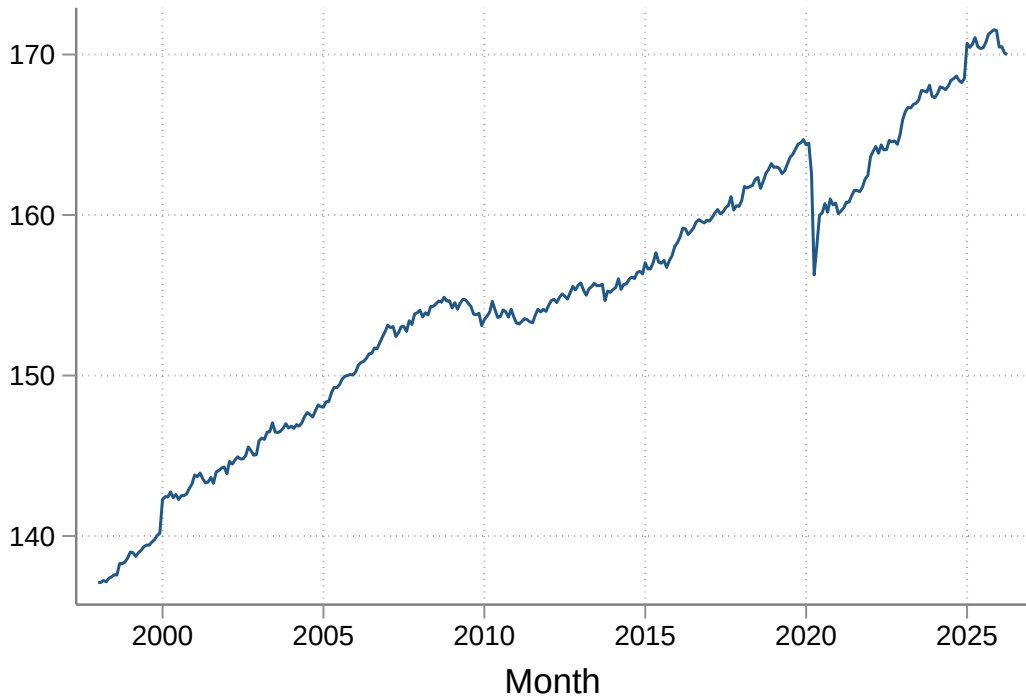


## Average Inflation Experienced by Age 20

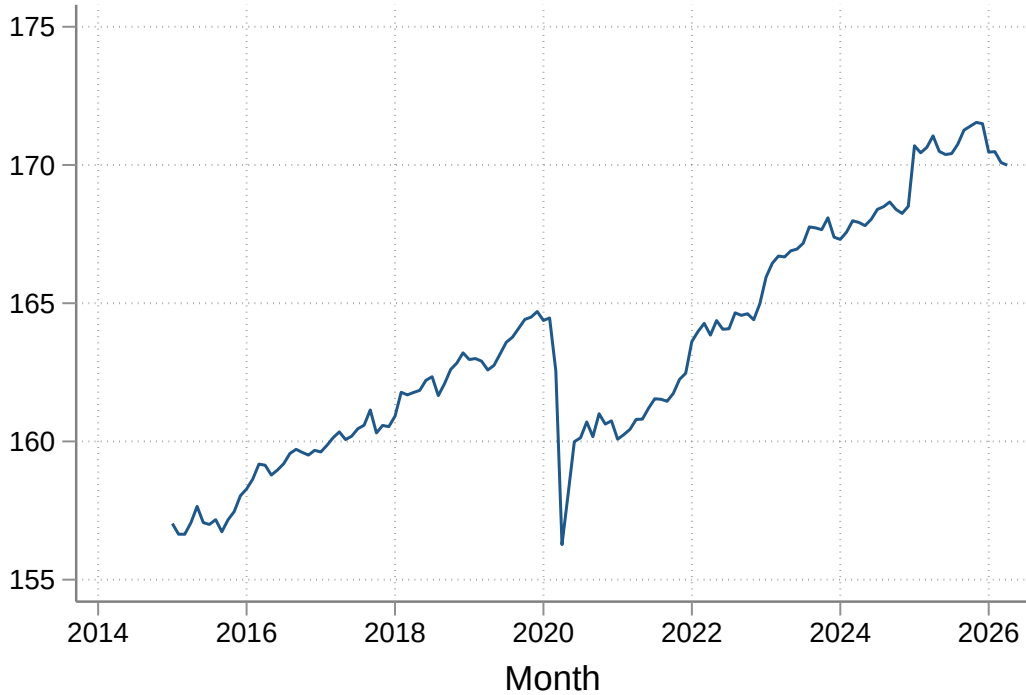




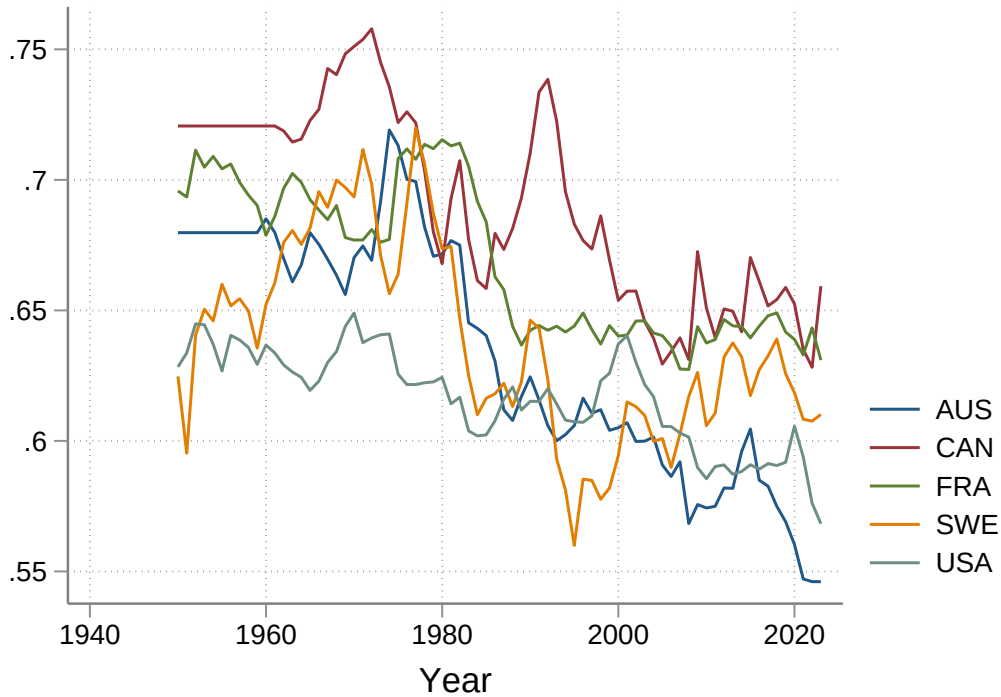
Labor Force (millions)



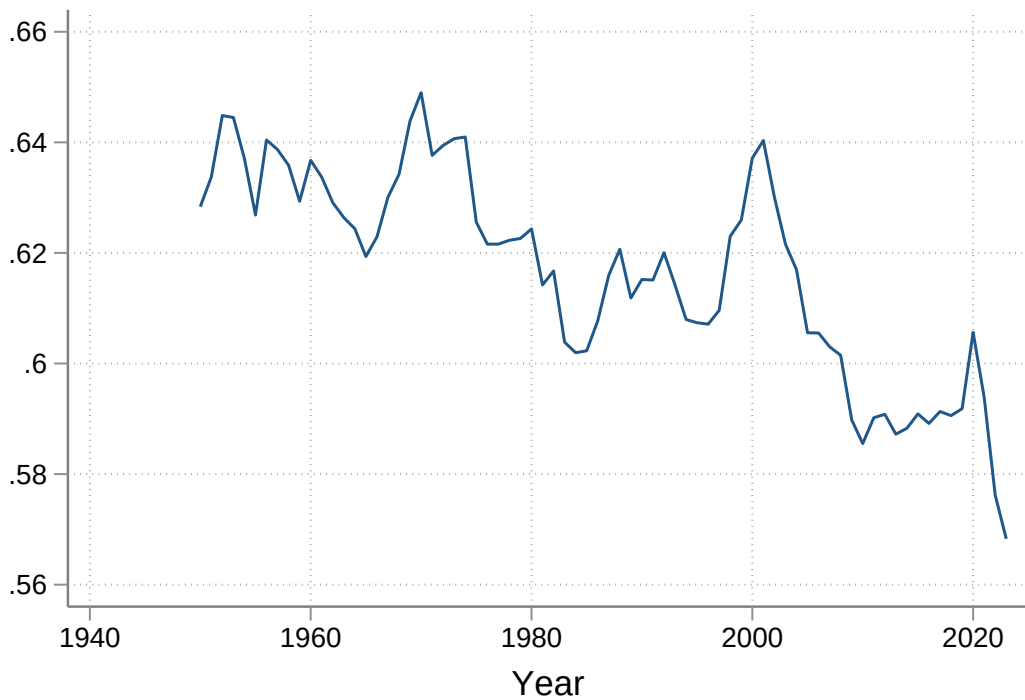
Labor Force (millions)



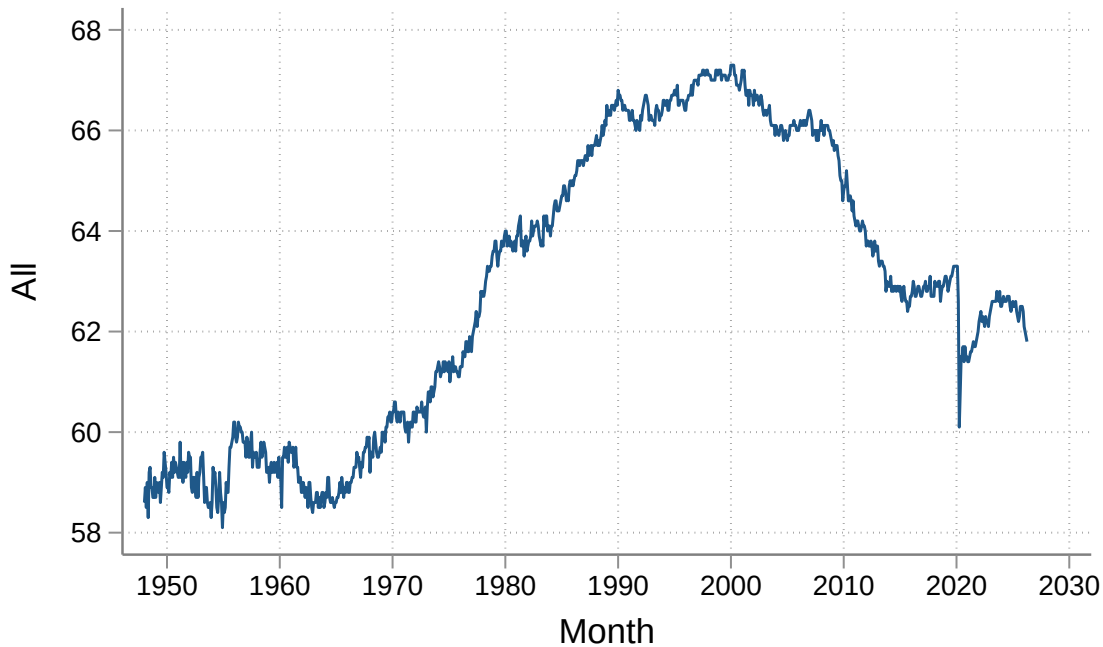
Labor's Share of Income



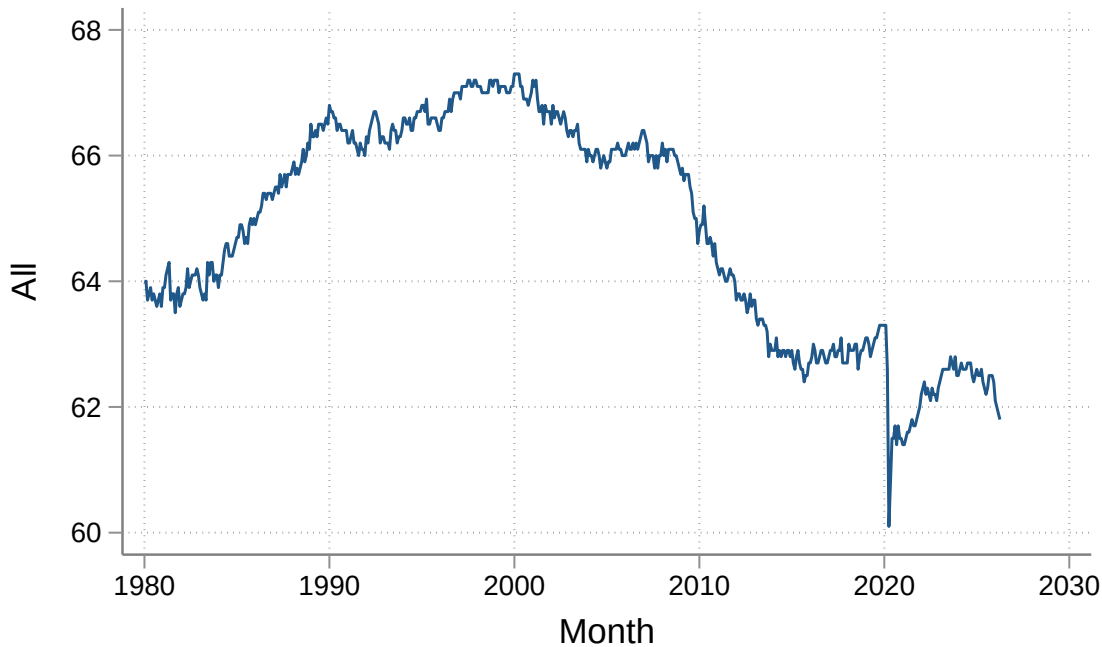
Labor's Share of Income



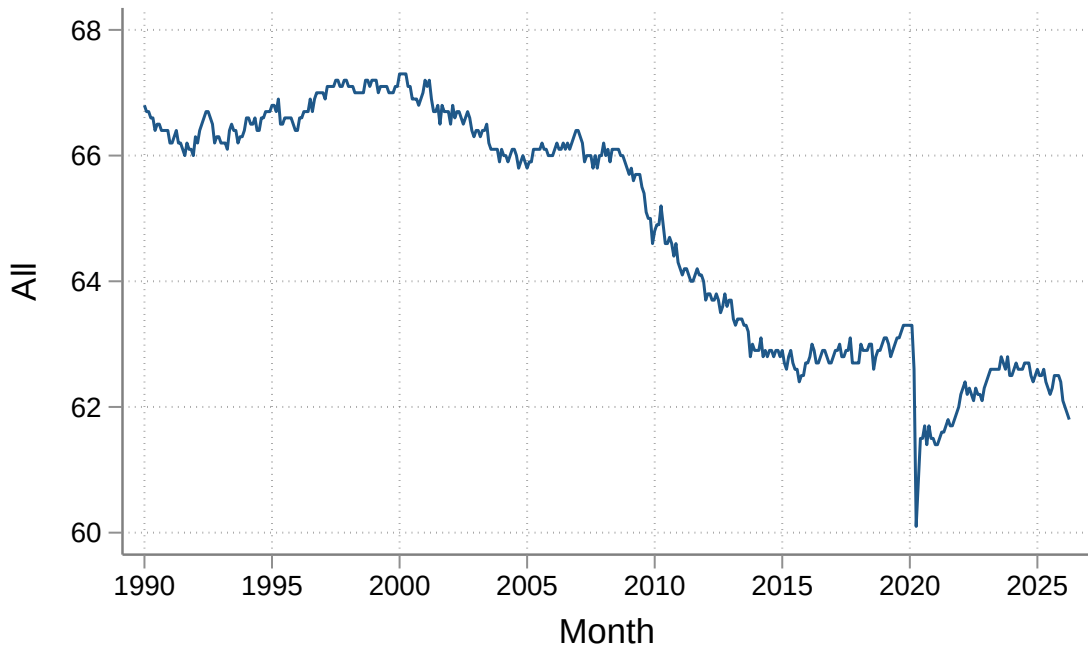
# Labor Force Participation Rate



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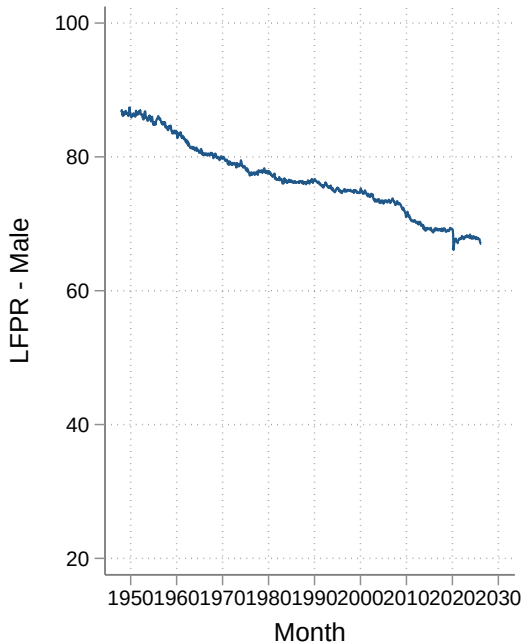


# Labor Force Participation Rate

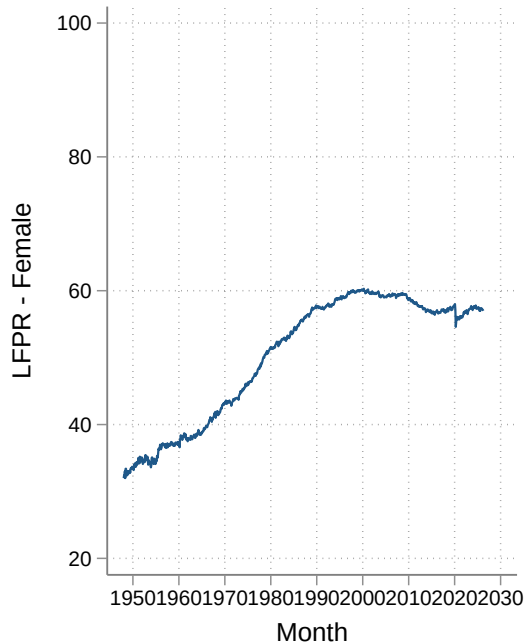




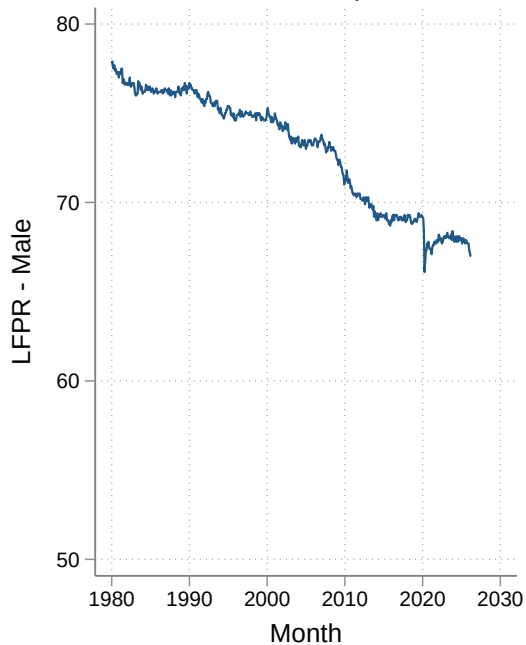
### Labor Force Participation Rate



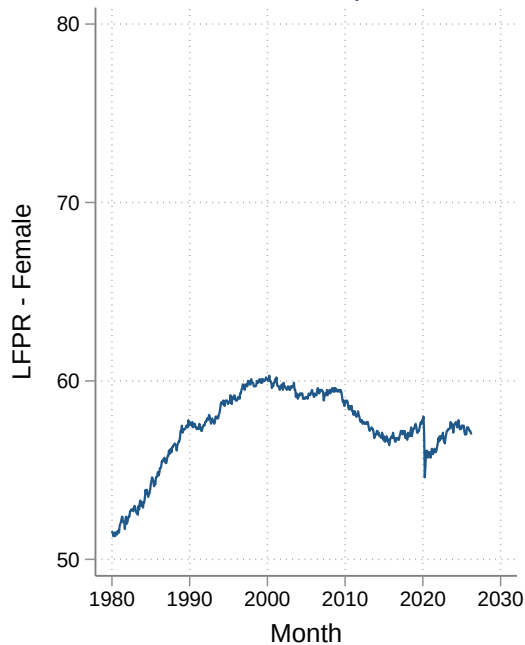
### Labor Force Participation Rate



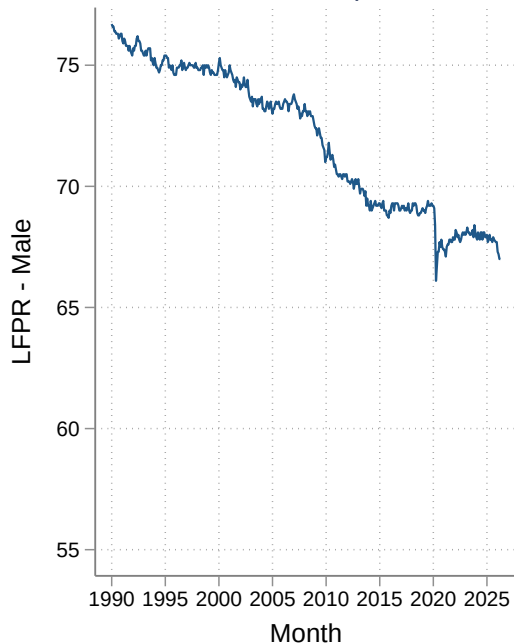
Labor Force Participation Rate



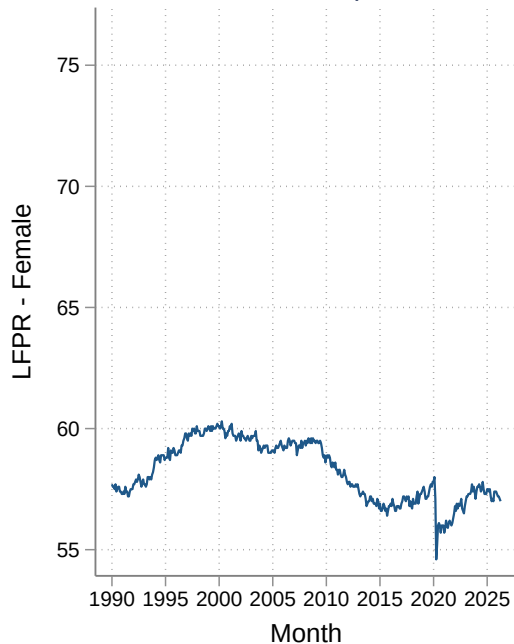
Labor Force Participation Rate



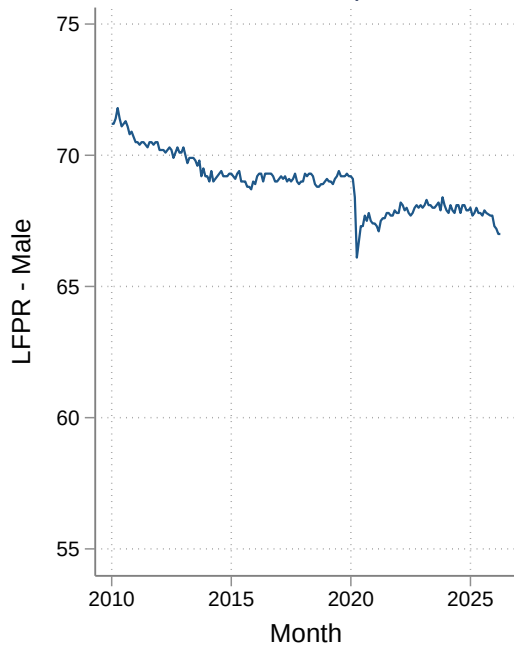
Labor Force Participation Rate



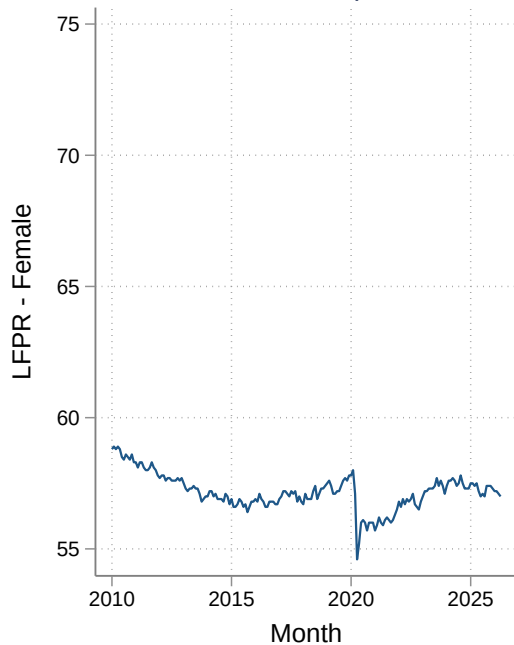
Labor Force Participation Rate



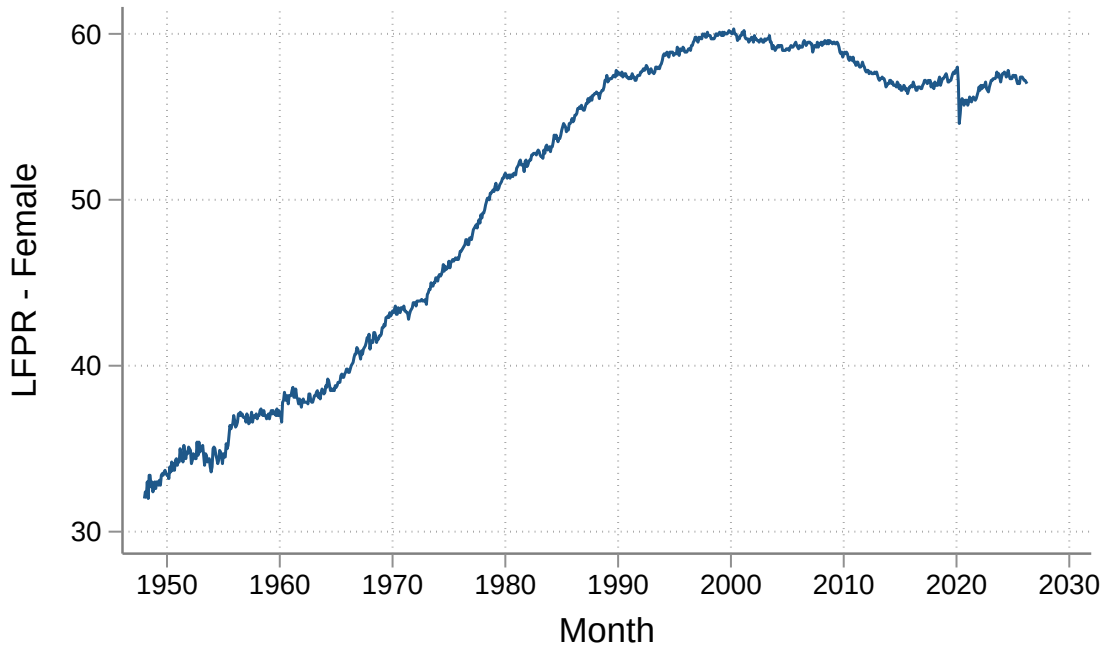
Labor Force Participation Rate



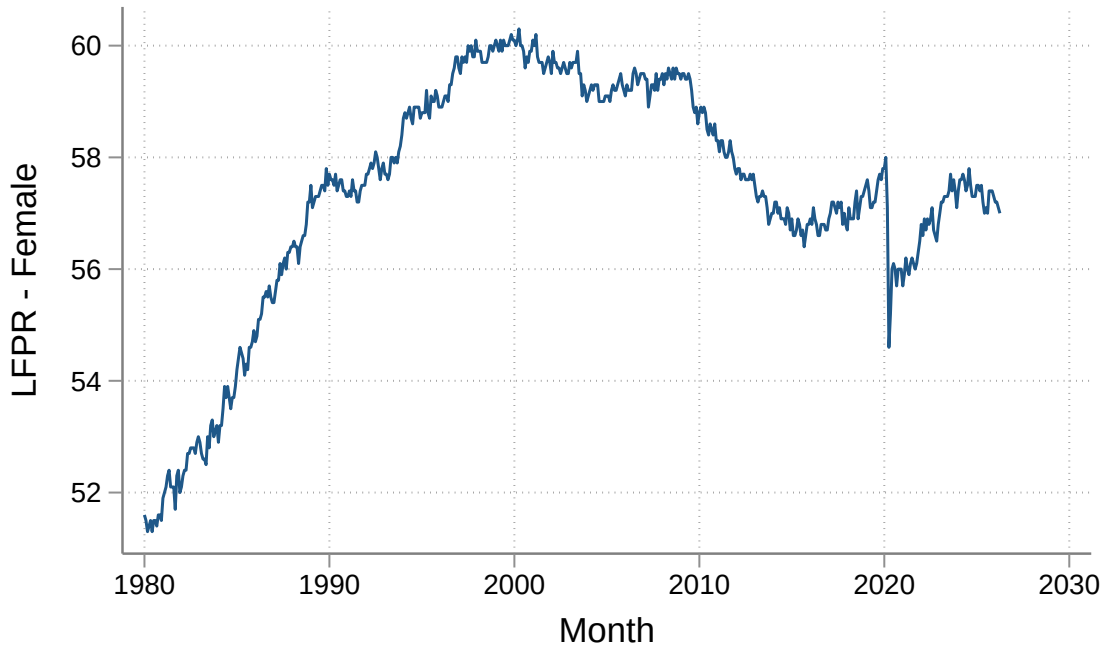
Labor Force Participation Rate



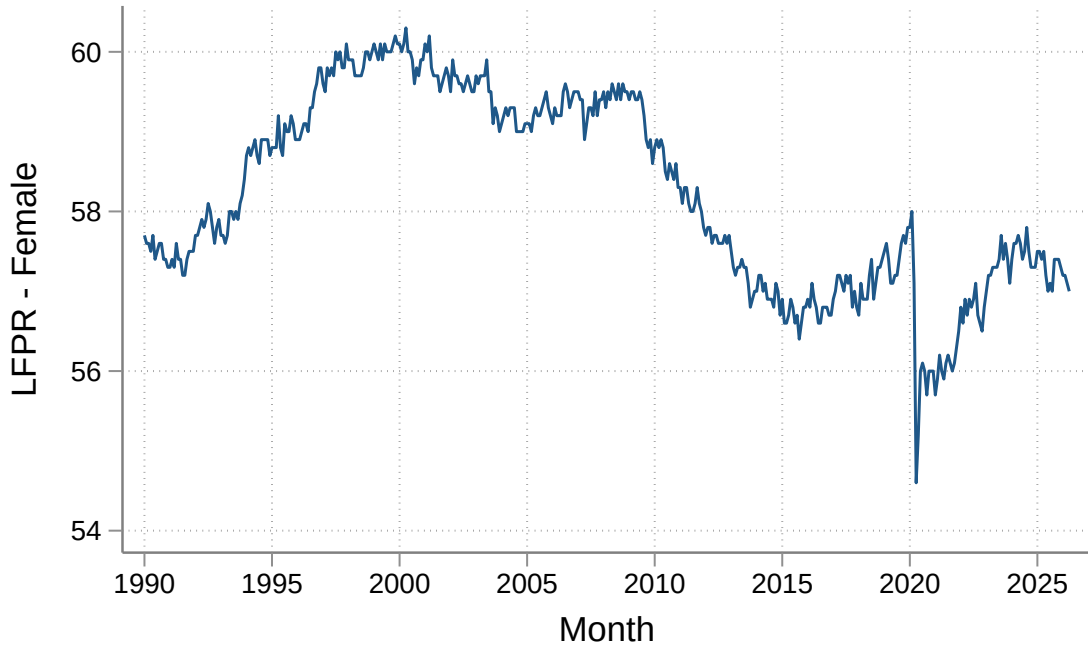
# Labor Force Participation Rate



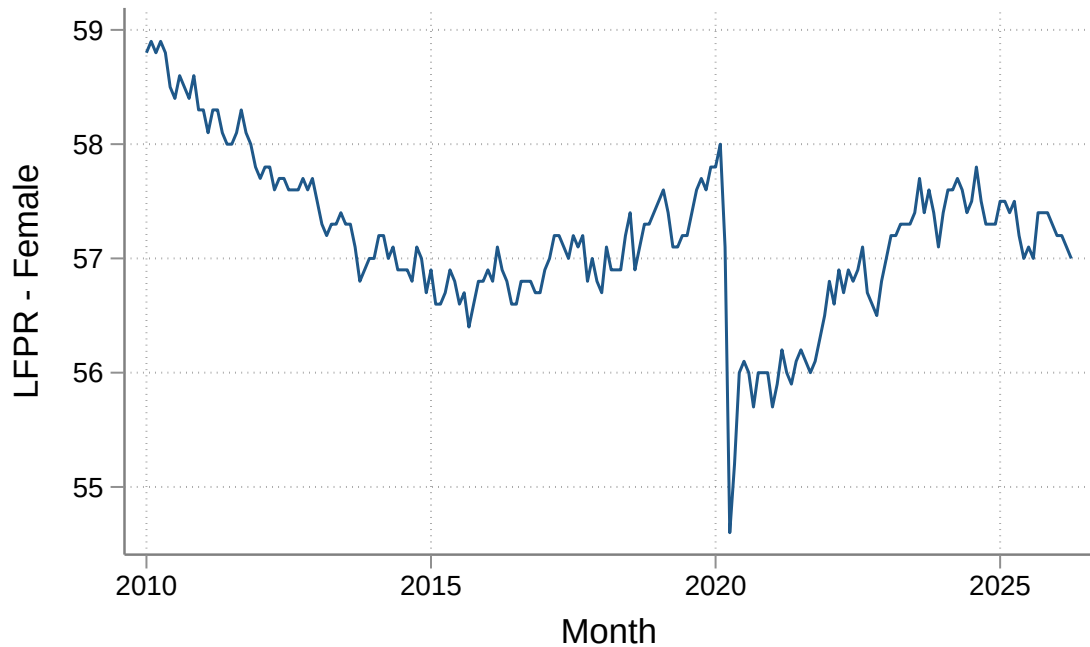
# Labor Force Participation Rate



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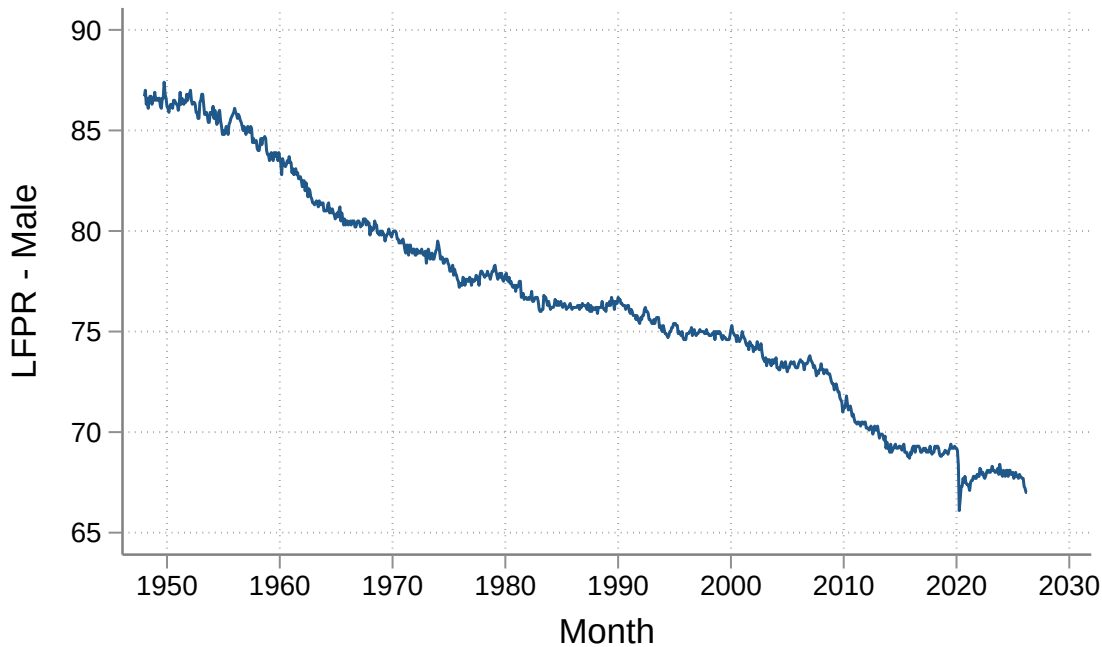


# Labor Force Participation Rate

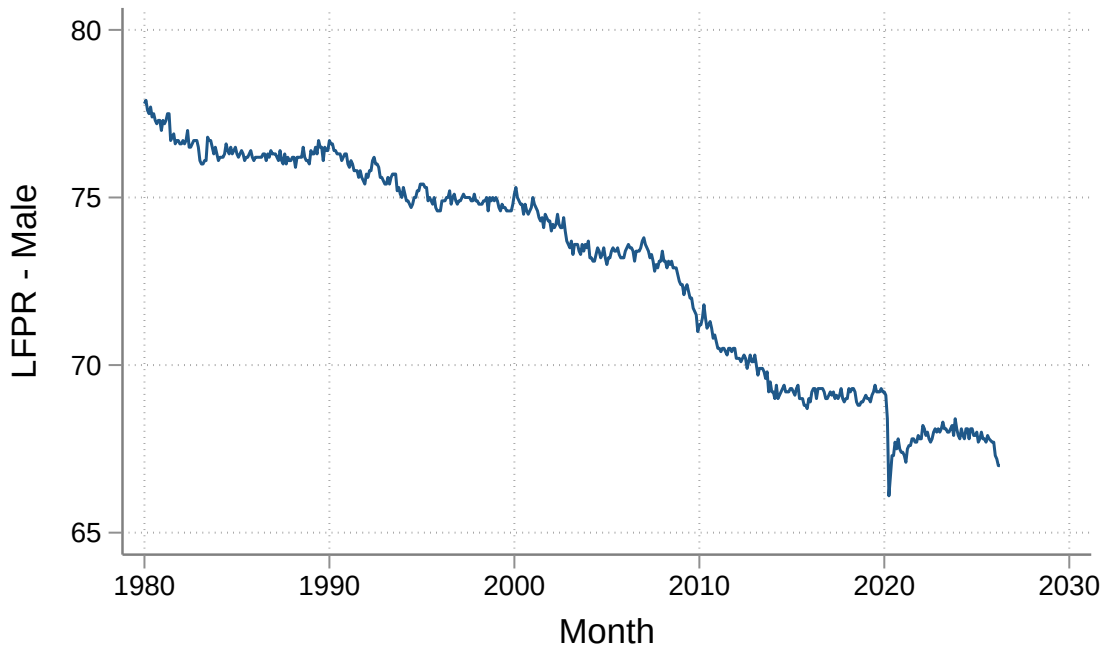




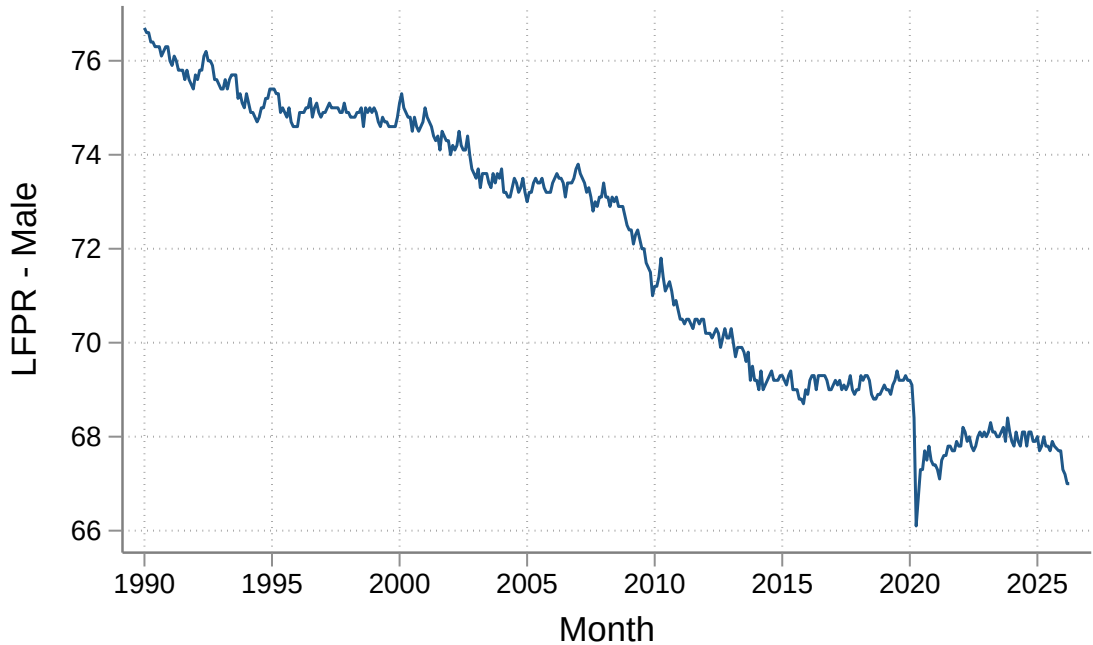
# Labor Force Participation Rate



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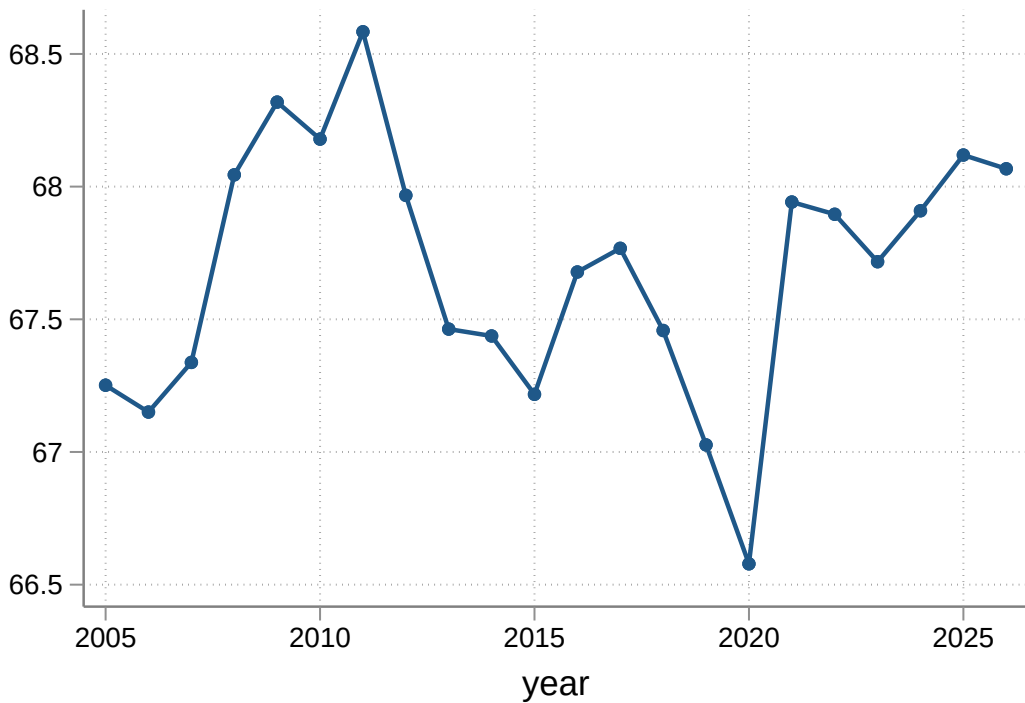
# Labor Force Participation Rate



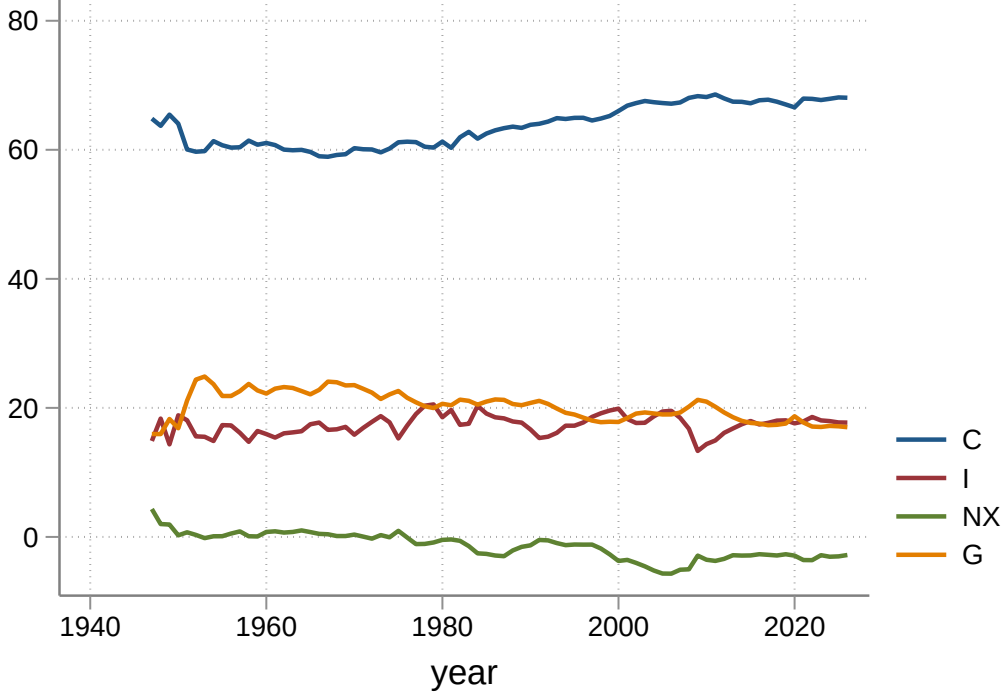
# Labor Force Participation Rate



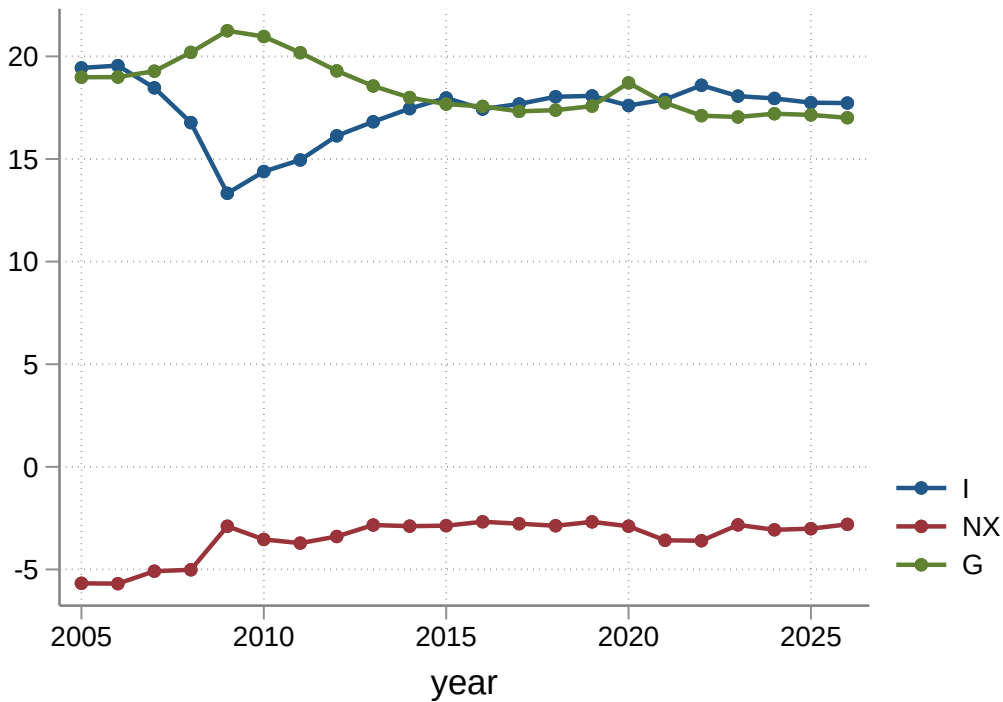
Percentage of GDP



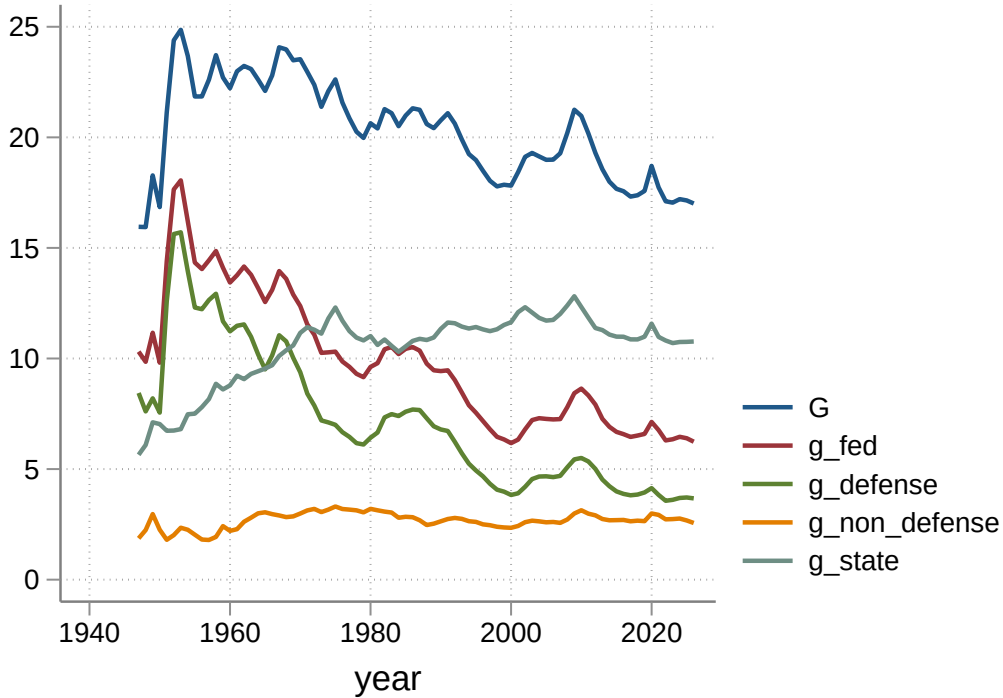
Percentage of GDP



Percentage of GDP

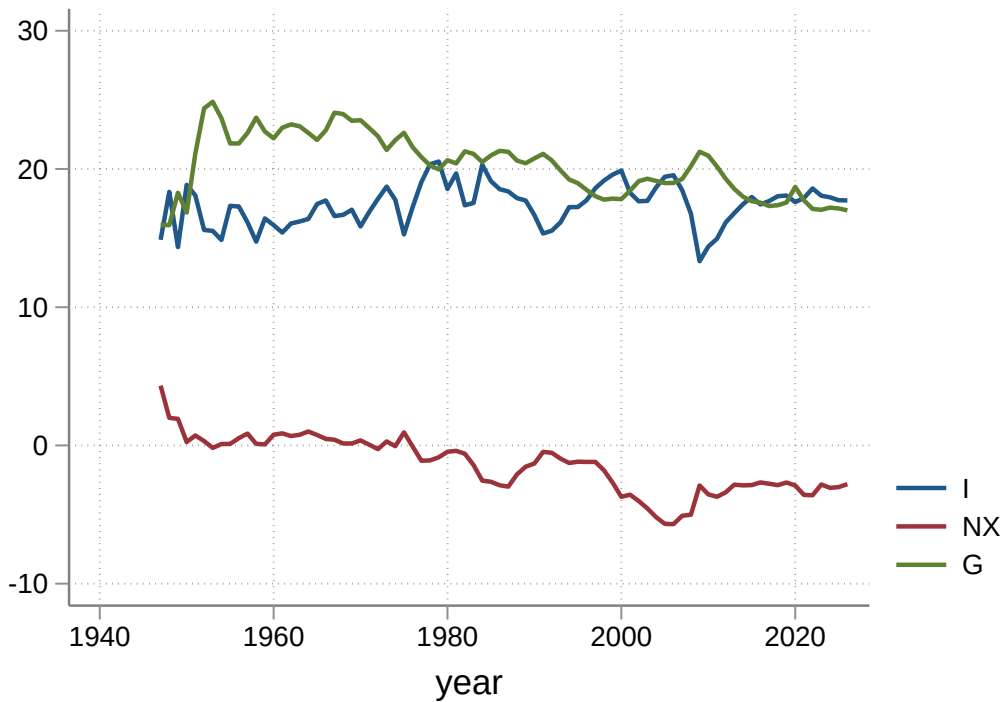


Percentage of GDP

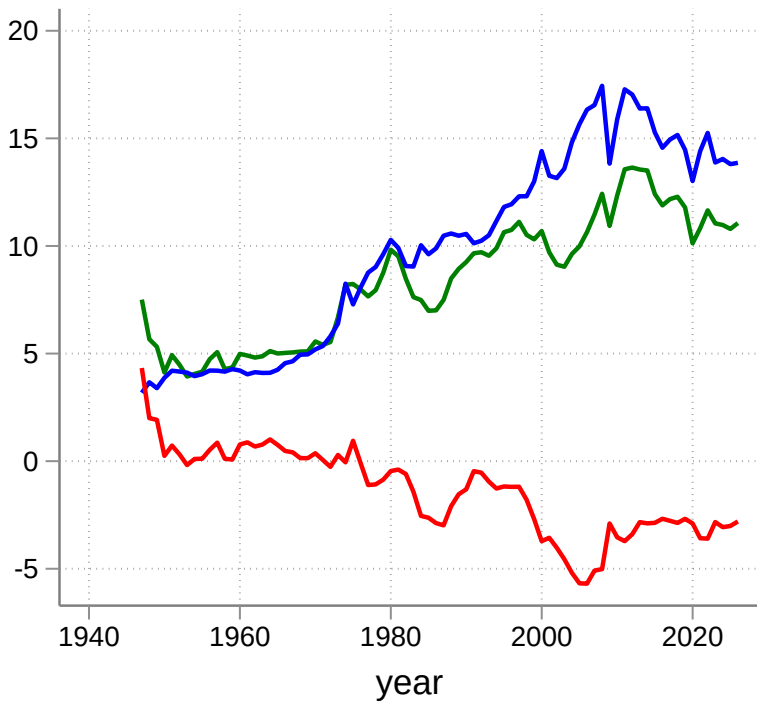




Percentage of GDP

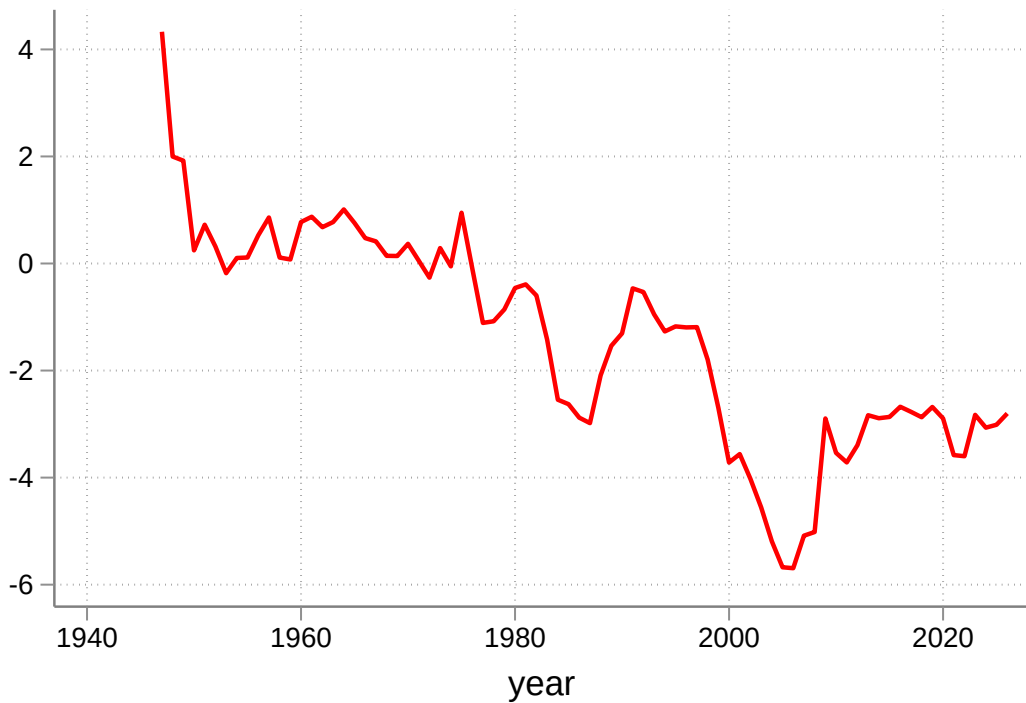


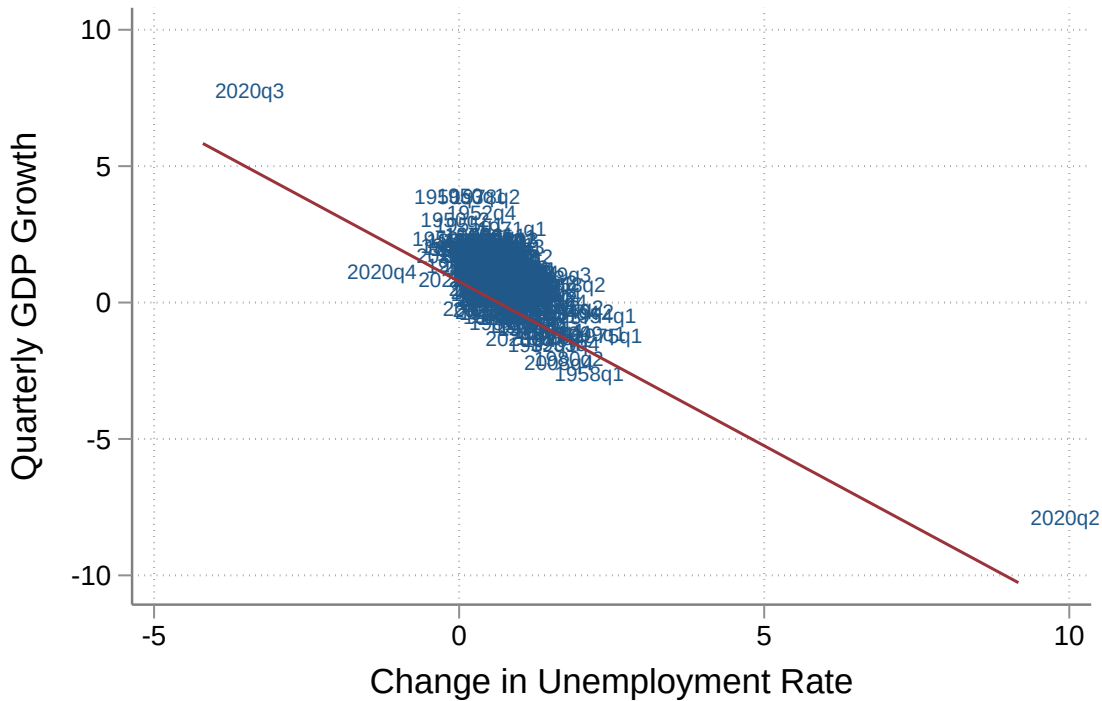
Percentage of GDP



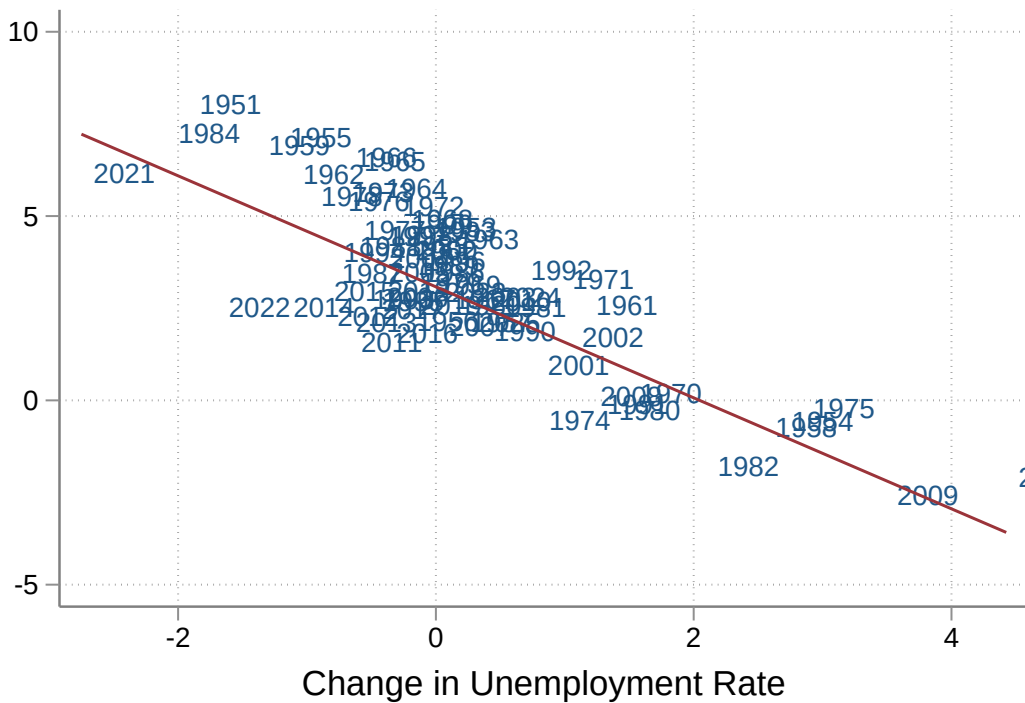
- Exports
- Imports
- Net Exports

Percentage of GDP

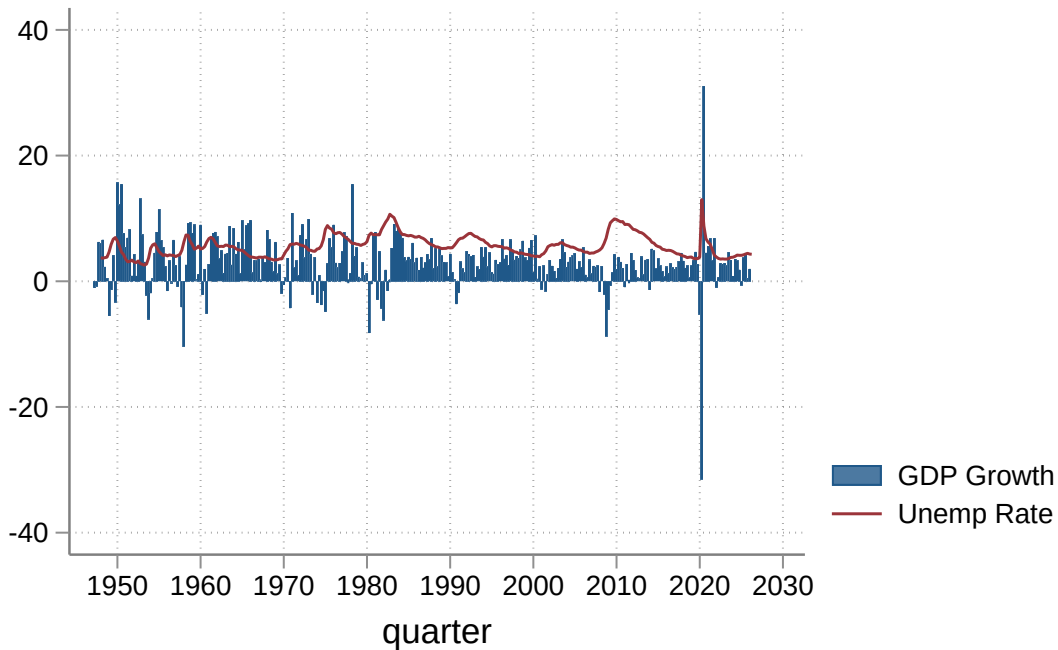


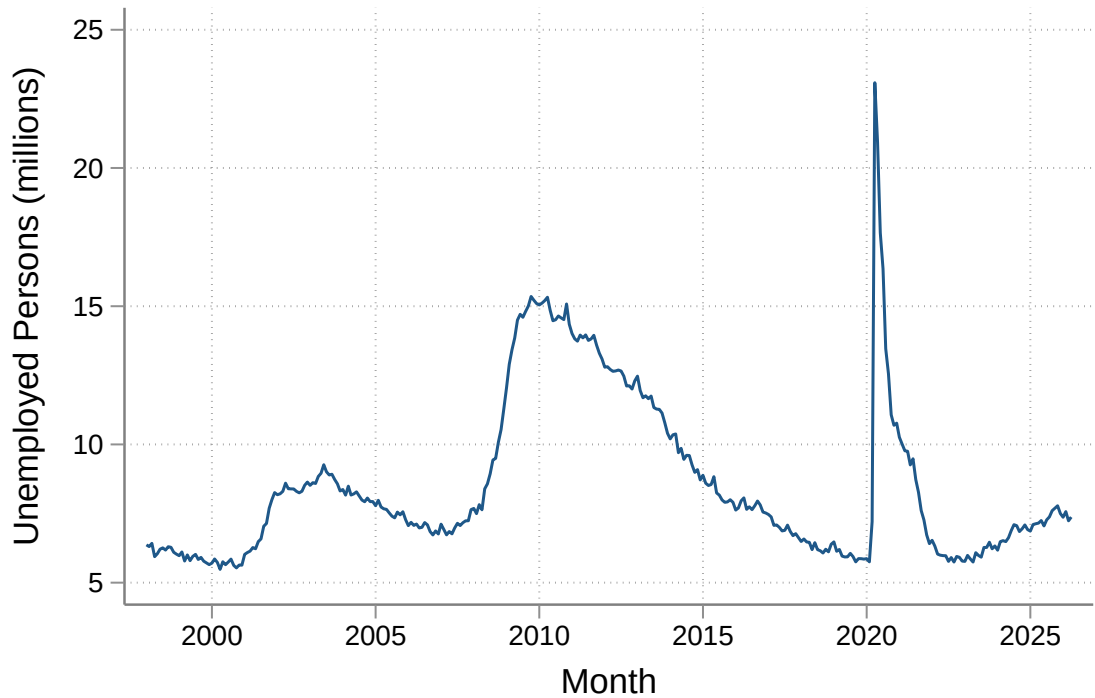


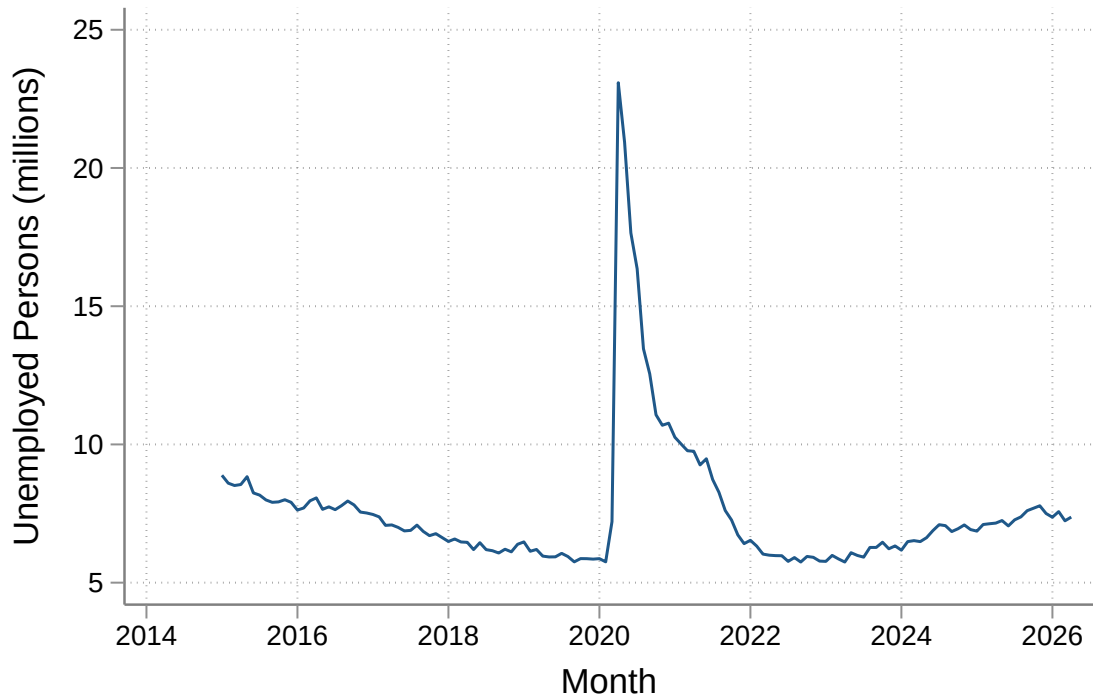
Yearly GDP Growth



# Quarterly GDP Growth (Annualized)

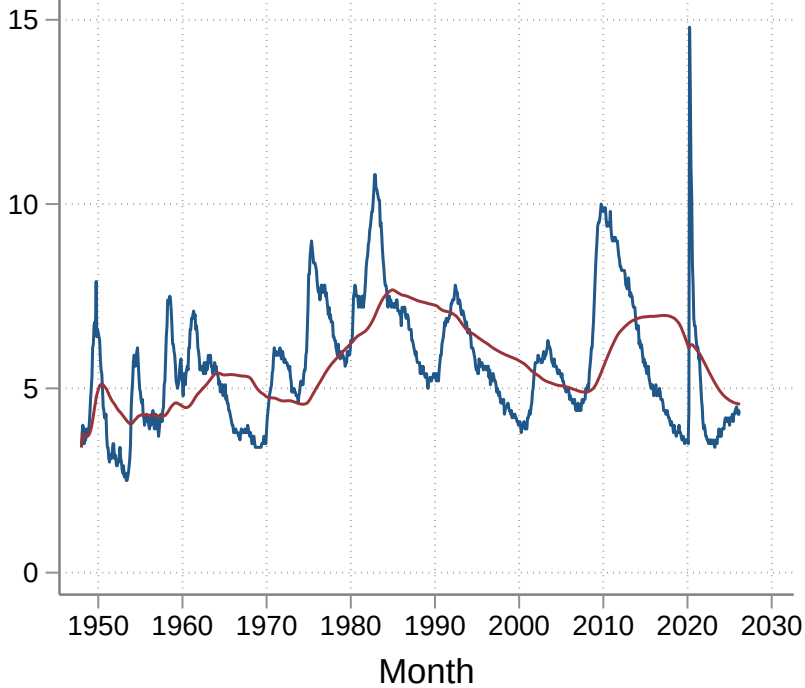






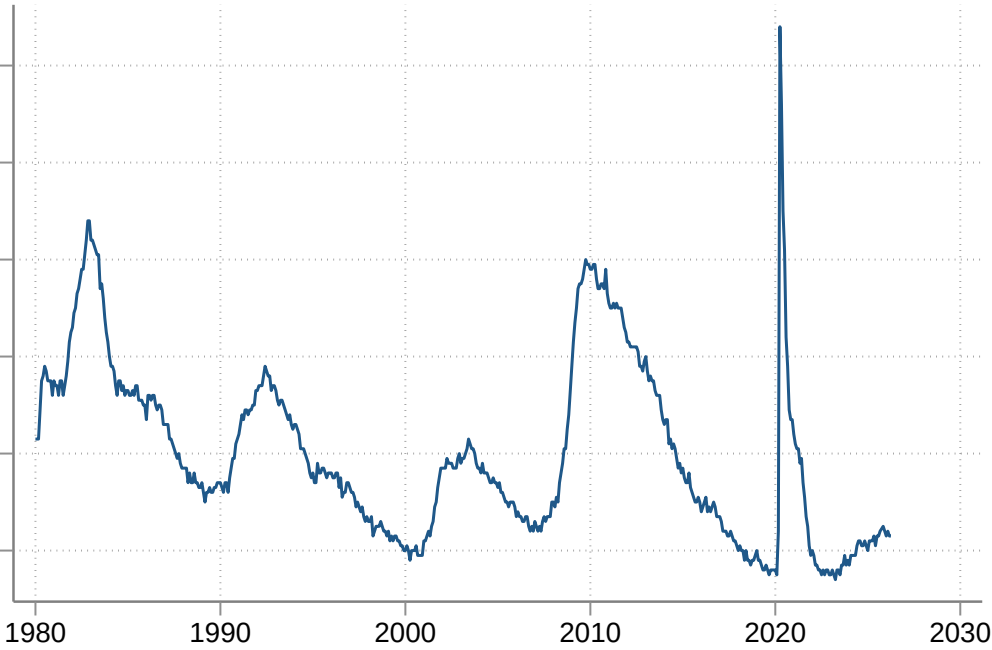


Unemployment Rate



Unemployment Rate

14  
12  
10  
8  
6  
4

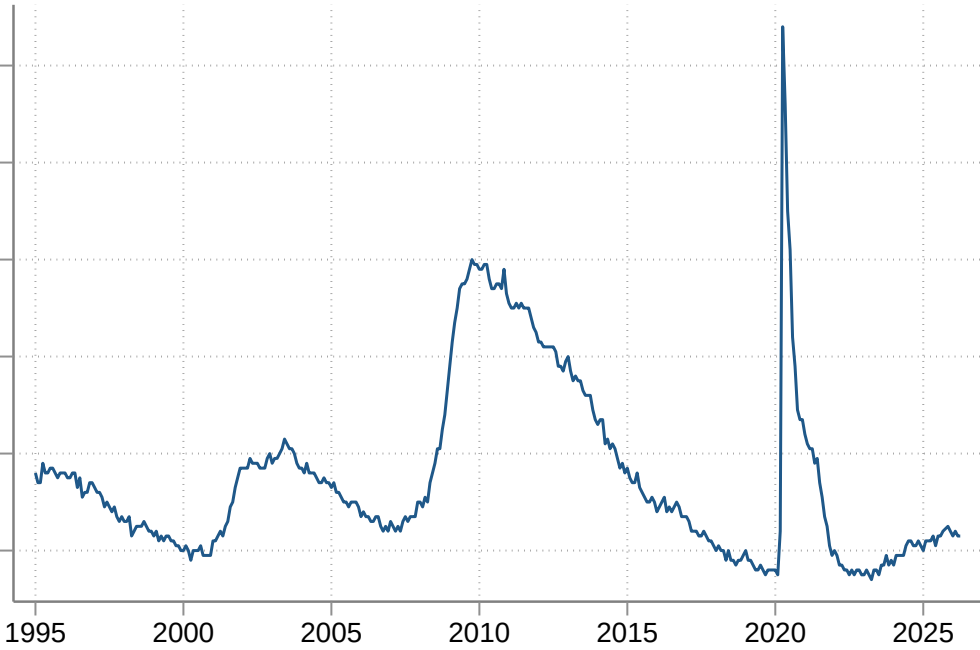


Month

2030

Unemployment Rate

14  
12  
10  
8  
6  
4



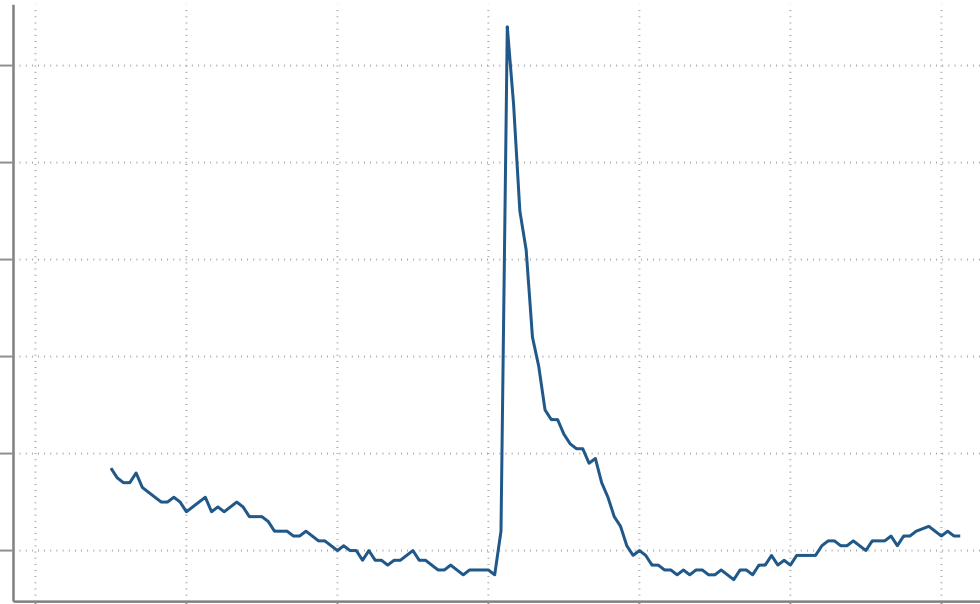
Month

Unemployment Rate

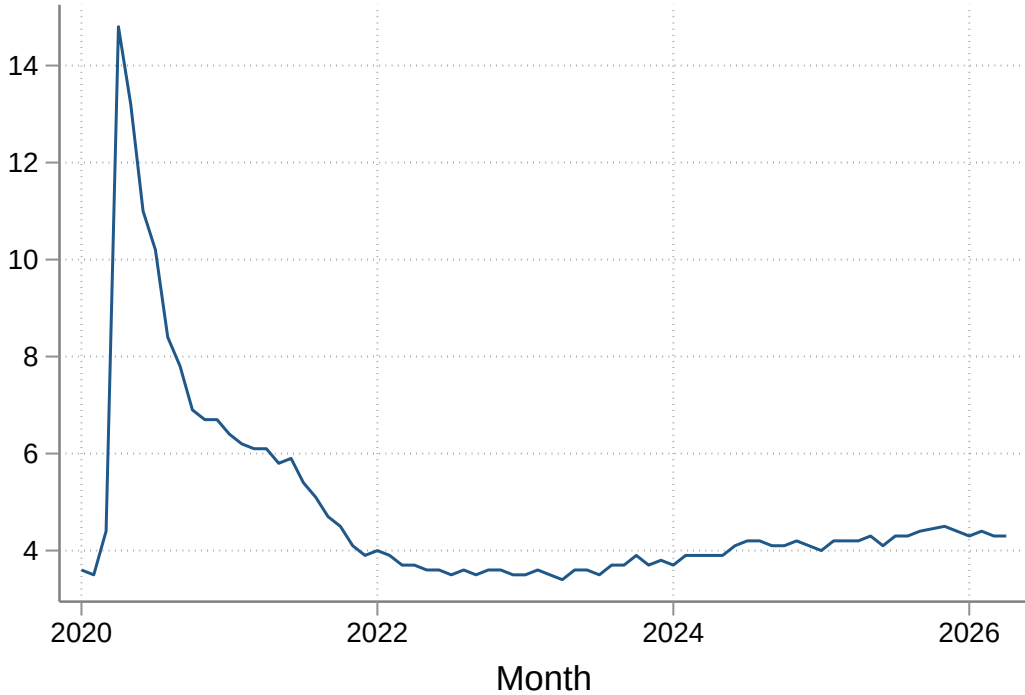
14  
12  
10  
8  
6  
4

2014 2016 2018 2020 2022 2024 2026

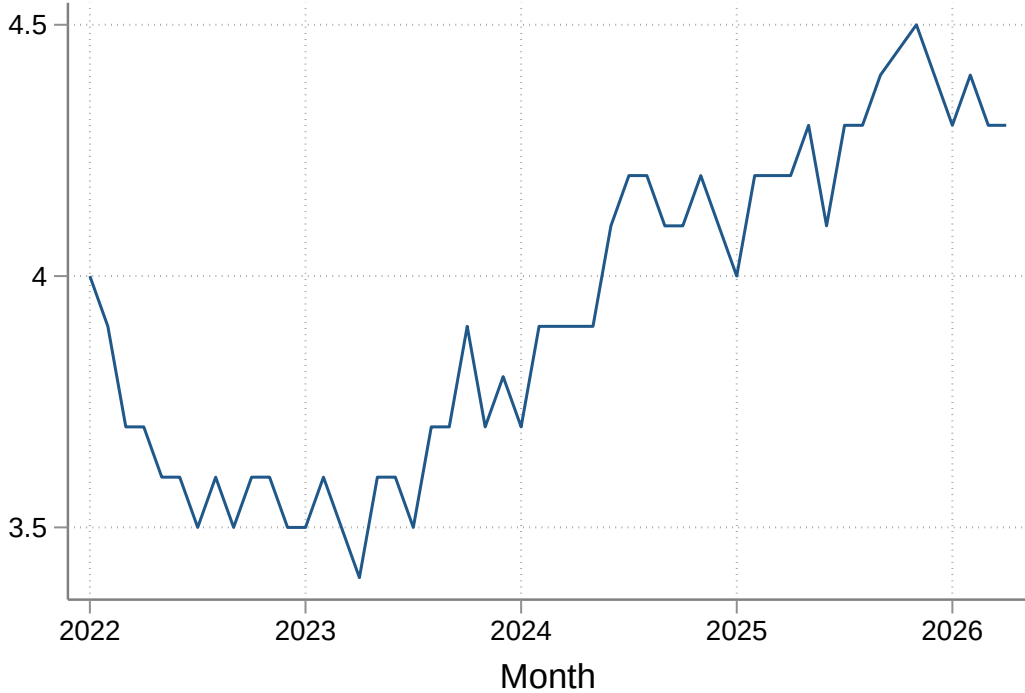
Month



Unemployment Rate



Unemployment Rate



Unemployment Rate

15

10

5

0

jan05

jan10

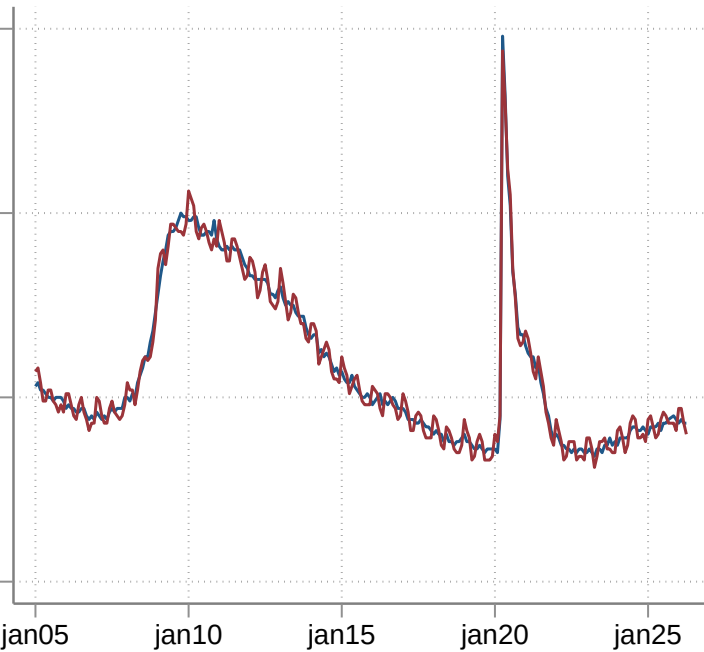
jan15

jan20

jan25

Month

UR  
not seas adj



Unemployment Rate

15

10

5

0

jan12

jan14

jan16

jan18

jan20

jan22

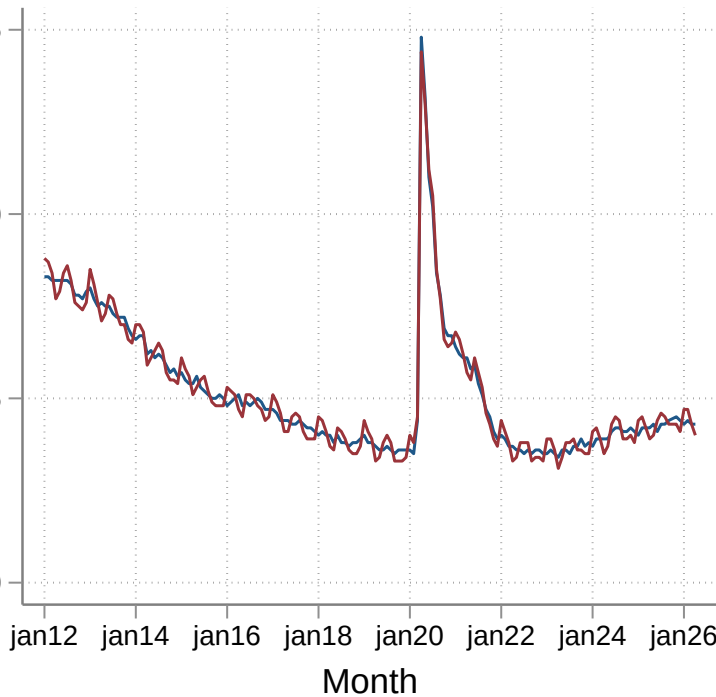
jan24

jan26

Month

— UR

— not seas adj





Unemployment Rate

0.5  
0  
-0.5  
-1

jan05

jan10

jan15

jan20

jan25

Month

